

GE Healthcare

Mobile C-Arm X-Ray Product

Operator Manual

Brivo OEC 715/Brivo OEC 785/Brivo OEC 865



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For USA, OEC Brivo Prime/OEC Brivo Essential/OEC Brivo Plus are respectively described as Brivo OEC 715/Brivo OEC 785/Brivo OEC 865 in the manual.

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U.S. federal law restricts this device to sale by or on the order of a physician, veterinarian, or other designated licensed practitioner as appropriate for its clinical use.

IMPORTANT SAVE THESE INSTRUCTIONS. PLEASE READ THIS MANUAL BEFORE USING Brivo OEC 715/Brivo OEC 785/Brivo OEC 865 Mobile C-Arm X-Ray Product.

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The contents of this document are accurate at the time of publication. However, changes in design and additional features can, at any time, be incorporated in the hardware and software and may not be reflected in this version of the document. Contact GE Healthcare Technical Support for clarification, if discrepancies arise.

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REGULATORY REQUIREMENTS

International Electrotechnical Commission (IEC), international standards organization, when applicable.

This product conforms with the Medical Device Directive, Council Directive 93/42/EEC: June 14, 1993, as amended by 2007/47/EC following the provisions of Annex II, when it bears the following CE Mark of Conformity.

CE 0459

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Contents

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Chapter1. Introduction and Safety	1-1
1.1. Overview	1-2
1.2. Owner Responsibilities.....	1-2
1.2.1. System Compatibility	1-2
1.2.2. Operator Qualifications	1-2
1.2.3. Continued Compliance	1-2
1.2.4. Unauthorized Modifications.....	1-3
1.3. Manufacturer's Responsibilities.....	1-4
1.3.1. X-Ray Equipment Certification.....	1-4
1.3.2. After-sale Operating and Safety Practices	1-4
1.3.3. Essential Performance	1-4
1.4. Communication Center Telephone Numbers.....	1-4
1.5. Safety Hazards.....	1-5
1.5.1. Safety Hazard Alerts	1-5
1.5.2. Explosion.....	1-5
1.5.3. Equipment Stability and Positioning	1-6
1.5.4. Motorized Mechanical Movement.....	1-6
1.5.5. Improperly Attached Equipment.....	1-6
1.5.6. Electrical Fire.....	1-7
1.5.7. Ground Fault.....	1-8
1.5.8. Improper Access	1-8
1.6. Radiation Exposure.....	1-8
1.6.1. General Protection	1-8
1.6.2. Source-to-Skin Distance	1-8
1.6.3. Dose to Patient	1-9
1.7. Ingress of Fluids.....	1-16
1.8. Cooling Efficiency	1-16
1.9. Burns.....	1-16
1.10. Electromagnetic Compatibility Statement.....	1-17
1.11. Equipment Malfunction	1-24
1.12. Mains Isolation.....	1-24

1.13.	External Devices	1-24
1.14.	Patient Environment	1-25
1.14.1.	Within the United States	1-25
1.14.2.	Outside the United States	1-25
Chapter2.	System Overview	2-1
2.1.	General Description.....	2-2
2.2.	C-Arm Components	2-3
2.3.	Workstation Components	2-5
Chapter3.	Mechanical Positioning	3-1
3.1.	Overview.....	3-2
3.2.	Positioning	3-2
3.2.1.	C-Arm Orbital Rotation	3-2
3.2.2.	Lateral Rotation	3-3
3.2.3.	Wig-Wag	3-4
3.2.4.	Vertical Column Operation	3-5
3.2.5.	Horizontal Cross-Arm.....	3-6
3.2.6.	C-Arm Brakes	3-7
3.2.7.	C-Arm Steering Handle.....	3-8
3.2.8.	Workstation Monitor Adjustment.....	3-8
3.2.9.	Workstation Brakes.....	3-9
3.3.	Moving the C-Arm and Workstation	3-10
3.3.1.	Overview.....	3-10
3.3.2.	Moving the C-Arm	3-11
3.3.3.	Moving the Workstation.....	3-12
Chapter4.	Start Up and Operating	4-1
Control	4-1	
4.1.	Overview.....	4-2
4.2.	Start Up	4-2
4.2.1.	Power On.....	4-2
4.2.2.	Standby or Power Off.....	4-5
4.3.	C-Arm controls.....	4-6
4.3.1.	C-Arm Control Panel	4-6
4.3.2.	Footswitch and Handswitch.....	4-14

4.3.3.	Emergency Switch	4-16
4.4.	Workstation Controls.....	4-17
4.4.1.	Workstation Membrane Keyboard	4-17
4.4.2.	Touch Screen.....	4-23
4.5.	Image Quality Control	4-23
Chapter5. Usual Operations.....		5-1
5.1.	Exam Management.....	5-2
5.1.1.	Create a New Exam.....	5-2
5.1.2.	Scheduled Exams Feature	5-3
5.1.3.	Resume a Performed Exam.....	5-5
5.1.4.	Edit Patient Information	5-6
5.2.	Imaging Mode	5-7
5.2.1.	Normal Fluoro Imaging.....	5-9
5.2.2.	High Level Fluoro Imaging	5-9
5.2.3.	Digital Spot Imaging.....	5-10
5.2.4.	Pulsed Fluoro Imaging	5-10
5.2.5.	Low Dose Imaging	5-11
5.2.6.	Imaging Mode List.....	5-11
5.2.7.	Modes of Operation: Engaging and Disengaging.....	5-13
5.3.	Image Review, Print and Archive	5-13
5.3.1.	Review Images.....	5-13
5.3.2.	Review Dose Information	5-18
5.3.3.	Print, Export and Import.....	5-25
5.3.4.	Retrieve a Performed Exam.....	5-30
5.4.	Pediatric Use	5-30
5.4.1.	General Instructions for Small or Pediatric Patients.....	5-30
5.4.2.	General Dose Planning	5-31
5.5.	Emergency Mode.....	5-32
Chapter7. Radiographic Film.....		7-1
7.1.	Overview	7-2
7.2.	Setup and Make a Film Exposure	7-2
Chapter8. System Setup		8-1
8.1.	System Information.....	8-2

8.2.	Date Time & Region.....	8-3
8.3.	C-Arm System	8-4
8.4.	Security.....	8-5
8.5.	Maintenance.....	8-7
Chapter9.	Maintenance.....	9-1
9.1.	Overview.....	9-2
9.2.	Performance Checks	9-3
9.2.1.	Mechanical Performance Check.....	9-3
9.2.2.	Electrical Performance Check.....	9-3
9.2.3.	Emergency Switch Performance Check.....	9-4
9.2.4.	Fluoro Mode Performance Check.....	9-4
9.2.5.	Film Mode Performance Check	9-5
9.3.	Cleaning and Disinfections	9-6
9.3.1.	Cleaning	9-6
9.3.2.	Disinfections	9-7
9.4.	Storage	9-8
9.4.1.	Temporary Storage (less than 60 days).....	9-8
9.4.2.	Long Term Storage or Shipment (60 days or more)	9-8
9.5.	Maintenance Schedule.....	9-9
9.5.1.	User Routine Checks Items.....	9-9
9.5.2.	Periodic Maintenance	9-10
Chapter10.	Display Messages.....	10-1
10.1.	Overview.....	10-2
10.2.	Error Recovery Steps	10-2
10.3.	Messages.....	10-2
Chapter11.	Labels and Symbols	11-1
11.1.	Overview.....	11-2
11.2.	Labels and Locations.....	11-2
11.2.1.	Labels.....	11-2
11.2.2.	Locations of the Labels	11-15
11.3.	Symbols.....	11-17
Chapter12.	Technical Reference.....	12-1
12.1.	Overview.....	12-2

12.2.	System Specifications.....	12-2
12.2.1.	Classification Type	12-2
12.2.2.	Environmental Requirements.....	12-2
12.2.3.	Power Requirements.....	12-3
12.2.4.	Physical Specifications.....	12-4
12.3.	Specification of Key Components.....	12-6
12.3.1.	X-Ray Generator	12-6
12.3.2.	X-Ray Source Assembly.....	12-8
12.3.3.	Grid	12-11
12.3.4.	Image Intensifier	12-11
12.3.5.	Camera.....	12-11
12.3.6.	Others.....	12-11
12.4.	Tube Rating Chart.....	12-13
12.4.1.	Filament Emission Characteristics	12-13
12.4.2.	Single Load Ratings	12-14
12.4.3.	Thermal Characteristic	12-15
12.5.	Scatter Radiation	12-16
12.6.	Room Interface (RIF)	12-23
12.7.	Options.....	12-24
12.7.1.	Thermal/ Film Printers.....	12-24
12.7.2.	Laser Aimer.....	12-25
12.7.3.	30cm Skin Spacer.....	12-26
12.7.4.	Film Cassette Holder	12-27
12.7.5.	Video Distributor	12-27
12.7.6.	Removable Filter	12-27
12.7.7.	Removable Grid.....	12-27
12.7.8.	Wi-Fi Internet Adapter.....	12-28
12.8.	Material Safety Data Sheets.....	12-28
12.9.	Material Recycling.....	12-28
12.10.	Replacement Parts.....	12-29
12.11.	System Block.....	12-30

Chapter1. Introduction and Safety

1.1. Overview

This manual describes operation for Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product only. It is intended for qualified medical personnel who have been trained in the use of medical imaging equipment. It is not designed to replace or substitute for certified training in the radiological or medical field.

The Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product is designed to provide digital spot-film imaging and fluoroscopic image guidance across all adult and pediatric populations for orientations between patient anatomy and surgical instruments. The product is used for general surgical applications and musculoskeletal procedures to visualize, for example, implant localization/s or needle positions for aspirations, injections or biopsy. Not for interventional use.

Contraindications: pregnant or lactating women, persons who can't tolerate surgery, persons who have mental disorders and can't cooperate in surgery, etc.

1.2. Owner Responsibilities

The owner has the responsibility to ensure system compatibility, operator qualifications and the continued compliance of equipment and operating specifications. The owner has the responsibility to ensure that only properly trained, qualified personnel who have obtained credentials from the appropriate authorities operate the system. Systems should only be used in designated use areas with approved AC receptacles. Unauthorized changes or modifications to any part of the system could have hazardous consequences. Changes or modifications must not be made unless specifically authorized by GE HUALUN Medical Systems Co, Ltd.

1.2.1. System Compatibility

Damage may result to the system if incompatible components are connected. Read your operator manual thoroughly prior to connecting components that you are uncertain it's compatible.

1.2.2. Operator Qualifications

It is the responsibility of the owner to ensure that the system is operated only by properly trained, qualified personnel who have obtained credentials from the appropriate authorities.

1.2.3. Continued Compliance

The owner is responsible for verifying continued compliance with all applicable regulations and standards. Consult local, state, federal and/or international agencies regarding specific requirements and regulations applicable to the use of this type of medical electronic equipment.

International standards include the following but not limited:

- US Federal Performance standard 21CFR 1020.30; 21CFR 1020.31; 21CFR1020.32
- IEC 60601-1 1988 (2nd Edition) +Amd. 1 & Amd. 2/EN 60601-1 (1990) + Amd. 1 & Amd. 2, Medical electric equipment Part 1: General requirements for safety
- IEC 60601-1-1 2000 2nd Edition, Medical electrical equipment Part 1-1 - Collateral Standard: Safety Requirements for Medical Electrical Systems

- IEC 60601-1-2 (2001) & Amd. 1 (2004)/EN 60601-1-2 (2001) & Amd. 1 (2006), Medical electrical equipment Part 1-2: Collateral standard: Electromagnetic compatibility Requirements and tests
- IEC 60601-1-3 (1994)/EN 60601-1-3 (1994), MEDICAL ELECTRICAL EQUIPMENT Part 1-3: Collateral Standard: General Requirements For Radiation Protection In Diagnostic X-Ray Equipment
- IEC 60601-1-4 (1996) & Amd. 1 (1999)/EN 60601-1-4 (1996) & Amd. 1 (1999), Medical electrical equipment Part 1-4 - Collateral Standard: Programmable electrical medical systems
- IEC 60601-1-6 (2006)/EN 60601-1-6 (2007), Medical electrical equipment Part 1-6: General requirements for safety - Collateral Standard: Usability
- IEC 60601-2-7 (1998)/EN 60601-2-7 (1998), Medical Electrical Equipment Part 2: Particular requirements for the safety of high-voltage generators of diagnostic X-Ray generator
- IEC 60601-2-28 (1993)/EN 60601-2-28 (1993), Medical electrical equipment Part 2: Particular requirements for the safety of X-Ray source assemblies and X-Ray tube assemblies for medical diagnosis
- IEC 60601-2-32 (1994)/EN 60601-2-32 (1994), Medical electrical equipment Part 2: Particular requirements for the safety of associated equipment of X-Ray equipment
- IEC 60601-1 (2005)/EN 60601-1 (2006), Medical electrical equipment Part 1: General requirements for basic safety and essential performance
- IEC 60601-1-2 (2007)/EN 60601-1-2 (2007), Medical electrical equipment Part 1-2: Collateral standard: Electromagnetic compatibility Requirements and tests
- IEC 60601-1-3 (2008)/EN 60601-1-3 (2008), Medical electrical equipment Part 1-3: Collateral Standard: Radiation protection in diagnostic X-Ray equipment
- IEC 60601-1-6 (2010)/EN 60601-1-6 (2010), Medical electrical equipment Part 1-6: Collateral standard: Usability
- IEC 60601-2-43 (2010)/EN 60601-2-43 (2010), Medical electrical equipment Part 2-43: Particular requirements for basic safety and essential performance of X-Ray equipment for interventional procedures
- IEC 60601-2-28 (2010)/EN 60601-2-28 (2010), Medical electrical equipment Part 2-28: Particular requirements for the basic safety and essential performance of X-Ray tube assemblies for medical diagnosis

1.2.4. Unauthorized Modifications

This equipment meets US Federal regulations and International standards. Unauthorized modifications to the equipment may impact adherence to these standards and make the equipment unsafe to operate. Never make any modifications or adjustments to the equipment unless directed by a qualified GE Healthcare representative.



U.S. federal law restricts this device to sale by or on the order of a physician, veterinarian, or other designated licensed practitioner as appropriate for its clinical use.

1.3. Manufacturer's Responsibilities

GE HUALUN Medical Systems, Co. Ltd. certifies each system and X-Ray source assembly. After-sale operating practices and safety are the responsibility of the owner/operator.

1.3.1. X-Ray Equipment Certification

GE HUALUN Medical Systems, Co. Ltd. certifies that when assembled according to manufacturer's instructions, the X-Ray equipment complies with applicable international standards.

1.3.2. After-sale Operating and Safety Practices

GE HUALUN Medical Systems, Co. Ltd. assumes no responsibility or liability for personal injury or damage resulting from misuse of its systems.

1.3.3. Essential Performance

In order to meet its intended use and comply with regulatory standards, the system provides the following essential performances:

1. Patient/Image database integrity;
2. Acceptable fluoroscopic image quality;
3. Display & Indication of Last Image Hold (LIH) following termination of X-ray exposure;
4. Maintain X-ray tube voltage accuracy within 10%;
5. Ensure increment/decrement between two settings is within 50% to 150% of indicated change;
6. Maintain X-ray tube current accuracy within 20%;
7. Maintain current time product at $\pm (10\% + 0.2\text{mAs})$;
8. The coefficient of variation of MEASURED VALUES of AIR KERMA is not greater than 0.05 for any combination of LOADING FACTORS;
9. Linearity of AIR KERMA in RADIOGRAPHY is not greater than 0.2;
10. Automatic Brightness Stabilization (ABS) function implementing the Automatic Exposure Rate Control feature;
11. Provide radiation dose structured report (RDSR) as per IEC/PAS 61910-1:2007.

1.4. Communication Center Telephone Numbers

If the system does not operate properly or fails to respond to the controls as described in your operator's manual, call GE HUALUN Medical Systems, Co. Ltd. to request service. The communication center's telephone and Fax numbers are listed below:

GE HUALUN MEDICAL SYSTEMS CO. Ltd.

No 1 Yong Chang North Road

Beijing Economic-Technological Development Zone

Beijing CHINA.

P.O.: 100176

Tel: 8610-58068888

Call center: 800-810-8188

European Representative:

GE Medical Systems S.C.S.

283 rue de la Minière

78530 BUC France

Tel: +33 1 30 70 40 40

You may also call or fax these numbers to order circuit diagrams, component part lists, calibration instructions or other information which will assist qualified service engineers to repair the system.

1.5. Safety Hazards

Potential hazards exist in the use of medical electronic devices and X-Ray systems. Operators using the equipment should understand the safety issues, emergency procedures, and the operating instructions provided.

The following pages describe hazardous and potentially hazardous conditions, and how to adequately protect yourself and others from possible injury.

1.5.1. Safety Hazard Alerts

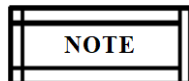
There are three hazard classifications, which are denoted and prioritized by the alert words:



WARNING indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.



CAUTION indicates a potentially hazardous situation that, if not avoided, may result in moderate to minor injury, equipment damage or loss of data.



NOTE is given in situation requiring special attention.

1.5.2. Explosion



The system is not designed for use in explosive atmosphere (e.g. anesthetic gases).

If an abnormal condition occurs, such as the room fills with flammable gas, steps must be taken to prevent the gas from coming in contact with the equipment. Follow these guidelines:

Do not turn the system off or unplug it from the AC receptacle.

Do not operate any other electrically powered equipment.

Evacuate all personnel from the area and ventilate with fresh air. Avoid operating any automated (electrically operated) doors or windows.

Contact your local fire department as soon as possible.

1.5.3. Equipment Stability and Positioning

Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product is mounted on wheels. If it is moved or operated improperly it could roll out of control. Follow these guidelines:

Two people should maintain control of the equipment when moving up or down an incline.

Place all mechanical assemblies in their most compact (transport) position and lock brake handles prior to moving the equipment.

Use the handles designed for moving the equipment and mechanical assemblies.

Never attempt to move the system up or down steps.

Do not operate the equipment on unlevelled floors.

Do not lock the wheel brakes and leave the equipment unattended on unlevelled floors.

Always apply the wheel locks when the system is in its final position.

Do not move the equipment if the casters or wheels are not functioning properly.

Mechanical shocks to the equipment while disk drives are accessing information may cause damage to the disk drive.

1.5.4. Motorized Mechanical Movement

Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product has motorized mechanical assemblies. Please follow these guidelines.

Always observe mechanical assemblies when operating the motor to avoid pinching or collision with a person or object.

Be careful when working around equipment to avoid unintentional motor actuation. Do not carelessly place objects on the equipment or bump or lean against the equipment.

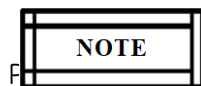
Observe and prevent articles of clothing from getting caught in moving parts.

1.5.5. Improperly Attached Equipment

If your equipment accommodates a film cassette holder or some other piece of equipment that can be attached or removed, follow these guidelines:

Use only equipment supplied by GE HUALUN Medical Systems Co. Ltd.

Attach the equipment properly. Incorrectly attached equipment could fall, causing injury to the patient or operator.



Refer to Chapter 12 "Technical Reference" for the correct dimensions of items that may be used in conjunction with this product, such as radiographic film cassettes.

WARNING

Electrical circuits inside the equipment may use voltages that are capable of causing serious injury or death from electric shock. To avoid this hazard, never remove any of the system covers.

WARNING

Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product is not waterproof. If water, soap or other liquids drip into the equipment, this can cause short circuit leading to electric shock and fire hazards. If liquids are accidentally spilled into the system, do not power or turn on the system until the liquids dry or evaporate completely.

WARNING

To avoid risk of electric shock, this equipment must only be connected to a supply mains with protective earth.

Observe the following safety procedures to avoid electric shock or serious injury to operators and patients and to avoid system malfunction.

Make all electrical connections to equipment while outside the patient environment. Do not touch a connector and the patient at the same time.

Do not bypass, jumper or otherwise disable the safety interlocks.

Do not remove any of the assembly covers.

Do not place food or beverage containers on any part of the equipment. If spilled they can cause short circuits.

Always remove power to the equipment before cleaning. Use a slightly damp cloth or sponge for cleaning.

Only qualified service engineers are allowed to service or repair a system.

1.5.6. Electrical Fire

In the event of electrical fire perform the following emergency procedure:

NOTE

Any emergency procedure developed by the owner, for the area in which the system is used, should include these safety measures:

Remove electrical power to the system by placing the power switch in the off position.

Unplug the power cord from the AC receptacle.

Evacuate personnel from the area.

Only use a fire extinguisher that is approved for use on electrical fires.

Call your local fire department for help if necessary.

WARNING

The use of the wrong type of fire extinguisher presents electrical shock and burn hazards. To avoid these hazards, a fire extinguisher, which meets applicable regulations and standards, must be available in the room where the equipment is being used.

1.5.7. Ground Fault

If the operating room has a ground fault alarm and the alarm is actuated:

Do not operate the system.

Call a qualified service technician.

1.5.8. Improper Access

WARNING

Improper access into the cabinet and covers of the system may cause injury to your hand or fingers. Do not put your hands or fingers into the cabinet or covers of the system.

1.6. Radiation Exposure

1.6.1. General Protection

WARNING

This equipment either produces or is used in the vicinity of ionizing radiation. Observe proper safety practices during operation.

The owner must designate areas suitable for safe operation and service of the equipment and ensure they are only used in those areas.

The owner must ensure that all personnel wear appropriate protective clothing and radiation monitoring devices while using the equipment.

Remain alert for visual indicators and audible alarms that are activated when ionizing radiation is being produced by equipment in the work area.

1.6.2. Source-to-Skin Distance

International regulations specify that a minimum source-skin distance be maintained, except for specific surgical applications. Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product has a skin spacer attached in order to meet this requirement. The system is prefabricated with a spacer for 30cm source-to-skin distance.

WARNING

Operator should keep as far as possible away from the X-Ray source.

Removing the skin spacer may result in increased radiation exposure to the patient. The skin spacer can only be removed by the physician.

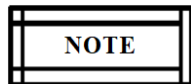
1.6.3. Dose to Patient

Skin Dose Levels

The skin dose levels in normal use may be high enough to cause deterministic effects. The availability of the different settings has a considerable effect on the radiation quality, the delivered dose rate and the image quality.

Patient Entrance Reference Point

Air Kerma (Kinetic Energy Released per unit Mass, i.e. Dose) is an indication of the radiation delivered to the patient entrance reference point, a point in space, and is measured in the units of Grays (Gy). The System determines the patient entrance reference point as a point along the central X-ray beam axis 70cm from the focal spot (30cm from the image intensifier entrance).



Selection of the patient entrance reference point is based on international convention and typical usage of a multi-use mobile fluoroscopy system. This point should represent a typical intersection of patient surface and the X-ray beam axis.

Radiation Data

The measurement is performed at patient entrance reference point. Following dosimetric data are typical values in normal use cases. There might be with some tolerance in actual use cases.

The following data are for Brivo OEC 715, 785 with I.I.(23XZ4ST/BS).

Operation Mode	Technique	Auto (PMMA 200mm)						Max	
	Pediatrics	With Grid, No Filter			With Filter, No Grid			With Grid, No Filter	With Filter, No Grid
	Mag mode	Mag0	Mag1	Mag2	Mag0	Mag1	Mag2	Mag0/1/2	Mag0/1/2
Normal Fluoro	kVp	80	81	80	72	73	74	110	110
	mA	2.1	3	4	1.7	2.5	3.6	4	4
	dose rate (mGy/min)	10.33	15.17	19.67	3.36	5.15	7.70	38.87	24.29
High Level Fluoro	kVp	81	81	80	72	73	74	110	110
	mA	3.2	4.6	6.4	2.5	3.7	5.4	6	6
	dose rate (mGy/min)	16.18	23.26	31.48	4.94	7.62	11.56	58.30	36.43

Low Dose	kVp	80	81	81	72	73	75	110	110
	mA	1	1.4	1.8	0.8	1.1	1.7	2	2
	dose rate (mGy/min)	4.92	7.08	9.10	1.58	2.26	3.78	19.43	12.14
Low Dose High Level Fluoro	kVp	81	81	81	72	73	74	110	110
	mA	1.5	2.1	3	1.2	1.7	2.5	2	2
	dose rate (mGy/min)	7.58	10.62	15.17	2.37	3.50	5.35	19.43	12.14
Pulsed Normal Fluoro 8pps	kVp	78	78	77	68	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	dose rate (mGy/min)	3.72	5.20	7.04	1.00	1.60	2.51	15.55	9.71
Pulsed High Level Fluoro 8pps	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6
	dose rate (mGy/min)	5.58	7.81	11.15	1.54	2.40	3.42	23.32	14.57
Pulsed Low Dose 8pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	dose rate (mGy/min)	1.62	2.10	2.89	0.37	0.60	0.98	7.77	4.86
Pulsed Low Dose High Level Fluoro 8pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	dose rate (mGy/min)	2.35	3.25	4.69	0.64	1.05	1.53	7.77	4.86
Pulse Normal Fluoro 4pps	kVp	78	78	77	68	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	dose rate (mGy/min)	1.86	2.60	3.52	0.50	0.80	1.25	7.77	4.86
Pulsed High Level	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6

Fluoro 4pps	dose rate (mGy/min)	2.79	3.90	5.58	0.77	1.20	1.71	11.66	7.29
Pulsed Low Dose 4pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	dose rate (mGy/min)	0.81	1.05	1.44	0.18	0.30	0.49	3.89	2.43
Pulsed Low Dose High Level Fluoro 4pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	dose rate (mGy/min)	1.17	1.62	2.35	0.32	0.52	0.76	3.89	2.43
Pulse Normal Fluoro 2pps	kVp	78	78	77	68	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	dose rate (mGy/min)	0.93	1.30	1.76	0.25	0.40	0.63	3.89	2.43
Pulsed High Level Fluoro 2pps	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6
	dose rate (mGy/min)	1.39	1.95	2.79	0.38	0.60	0.85	5.83	3.64
Pulsed Low Dose 2pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	dose rate (mGy/min)	0.41	0.53	0.72	0.09	0.15	0.24	1.94	1.21
Pulse Low Dose High Level Fluoro 2pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	dose rate (mGy/min)	0.59	0.81	1.17	0.16	0.26	0.38	1.94	1.21
Pulse Normal Fluoro 1pps	kVp	78	78	77	68	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	dose rate (mGy/min)	0.46	0.65	0.88	0.13	0.20	0.31	1.94	1.21
Pulsed High Level	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6

Fluoro 1pps	dose rate (mGy/min)	0.70	0.98	1.39	0.19	0.30	0.43	2.91	1.82
Pulsed Low Dose 1pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	dose rate (mGy/min)	0.20	0.26	0.36	0.05	0.08	0.12	0.97	0.61
Pulsed Low Dose High Level Fluoro 1pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	dose rate (mGy/min)	0.29	0.41	0.59	0.08	0.13	0.19	0.97	0.61
Digital Spot	kVp							110	110
	mA							16	16
	dose (mGy)							1.68	1.05
Film	kVp							110	110
	mAs							80	80
	dose (mGy)							12.96	8.10

The following data are for Brivo OEC 865 and Brivo OEC 715/785 with I.I.(E5830SD-P10A).

Operation Mode	Technique	Auto (PMMA 200mm)						Max	
	Pediatrics	With Grid, No Filter			With Filter, No Grid			With Grid, No Filter	With Filter, No Grid
	Mag mode	Mag0	Mag1	Mag2	Mag0	Mag1	Mag2	Mag0-2	Mag0-2
Normal Fluoro	kVp	79	79	79	71	73	73	110	110
	mA	2	2.9	4	1.6	2.4	3.5	4	4
	Dose Rate (mGy/min)	9.56	13.87	19.13	3.04	4.94	7.20	38.87	24.29
High Level Fluoro	kVp	80	79	78	71	72	73	110	110
	mA	3.1	4.4	6	2.5	3.6	5.2	6	6
	Dose Rate (mGy/min)	15.25	21.04	27.88	4.75	7.12	10.70	58.30	36.43
Low Dose	kVp	79	79	79	71	72	73	110	110
	mA	0.9	1.3	1.8	0.7	1.1	1.6	2	2
	Dose Rate (mGy/min)	4.30	6.22	8.61	1.33	2.18	3.29	19.43	12.14
Low Dose High Level Fluoro	kVp	79	79	78	71	72	73	110	110
	mA	1.4	2	2.8	1.1	1.6	2.4	2	2
	Dose Rate (mGy/min)	6.69	9.56	13.01	2.09	3.16	4.94	19.43	12.14
Pulsed Normal Fluoro 8pps	kVp	77	78	77	69	70	71	110	110
	mA	1.9	2.8	3.9	1.5	2.2	3.3	4	4
	Dose Rate (mGy/min)	3.43	5.20	7.04	1.05	1.60	2.51	15.55	9.71
Pulsed High Level Fluoro 8pps	kVp	78	78	77	69	70	71	110	110
	mA	3	4.2	5.9	2.3	3.3	5	6	6
	Dose Rate (mGy/min)	5.58	7.81	10.65	1.61	2.40	3.80	23.32	14.57
Pulsed Low Dose 8pps	kVp	77	77	76	69	69	70	110	110
	mA	0.9	1.3	1.7	0.7	1	1.5	2	2
	Dose Rate (mGy/min)	1.62	2.35	2.98	0.49	0.70	1.09	7.77	4.86

Pulsed Low Dose High Level Fluoro 8pps	kVp	77	77	76	68	69	70	110	110
	mA	1.4	2	2.6	1.1	1.5	2.3	2	2
	Dose Rate (mGy/min)	2.53	3.61	4.56	0.73	1.05	1.68	7.77	4.86
Pulse Normal Fluoro 4pps	kVp	78	78	77	69	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	Dose Rate (mGy/min)	1.86	2.60	3.52	0.52	0.80	1.25	7.77	4.86
Pulsed High Level Fluoro 4pps	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6
	Dose Rate (mGy/min)	2.79	3.90	5.58	0.77	1.20	1.71	11.66	7.29
Pulsed Low Dose 4pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	Dose Rate (mGy/min)	0.81	1.05	1.44	0.18	0.30	0.49	3.89	2.43
Pulsed Low Dose High Level Fluoro 4pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	Dose Rate (mGy/min)	1.17	1.62	2.35	0.32	0.52	0.76	3.89	2.43
Pulse Normal Fluoro 2pps	kVp	78	78	77	68	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	Dose Rate (mGy/min)	0.93	1.30	1.76	0.25	0.40	0.63	3.89	2.43
Pulsed High Level Fluoro 2pps	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6
	Dose Rate (mGy/min)	1.39	1.95	2.79	0.38	0.60	0.85	5.83	3.64

Pulsed Low Dose 2pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	Dose Rate (mGy/min)	0.41	0.53	0.72	0.09	0.15	0.24	1.94	1.21
Pulse Low Dose High Level Fluoro 2pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	Dose Rate (mGy/min)	0.59	0.81	1.17	0.16	0.26	0.38	1.94	1.21
Pulse Normal Fluoro 1pps	kVp	78	78	77	68	70	71	110	110
	mA	2	2.8	3.9	1.5	2.2	3.3	4	4
	Dose Rate (mGy/min)	0.46	0.65	0.88	0.13	0.20	0.31	1.94	1.21
Pulsed High Level Fluoro 1pps	kVp	78	78	78	68	70	71	110	110
	mA	3	4.2	6	2.3	3.3	4.5	6	6
	Dose Rate (mGy/min)	0.70	0.98	1.39	0.19	0.30	0.43	2.91	1.82
Pulsed Low Dose 1pps	kVp	77	76	75	66	68	69	110	110
	mA	0.9	1.2	1.7	0.6	0.9	1.4	2	2
	Dose Rate (mGy/min)	0.20	0.26	0.36	0.05	0.08	0.12	0.97	0.61
Pulsed Low Dose High Level Fluoro 1pps	kVp	77	77	77	67	69	70	110	110
	mA	1.3	1.8	2.6	1	1.5	2.1	2	2
	Dose Rate (mGy/min)	0.29	0.41	0.59	0.08	0.13	0.19	0.97	0.61
Digital Spot	kVp							110	110
	mA							16	16
	Dose (mGy)							1.68	1.05
Film	kVp							110	110
	mAs							80	80
	Dose (mGy)							12.96	8.10

1.7. Ingress of Fluids



The X-Ray system is not waterproof. If liquids drip into the equipment, disconnect the power cord and do not operate the system until it can be cleaned and inspected by a qualified service engineer.

Excessive amounts of fluids such as antiseptics, cleaning solutions or bodily fluids may damage internal components if they are allowed inside the equipment. Use drapes, if necessary, to protect equipment when performing procedures and do not apply excessive amounts of fluid when cleaning.

1.8. Cooling Efficiency

The system does not require any special cooling methods, natural cooling is necessary.

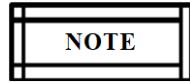
Draping Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product may restrict airflow to components that provide heat sinking and to vents designed to cool the equipment. Drape equipment and cover vents only when exposure to excessive fluids is unavoidable and extended use of the equipment is not required.

1.9. Burns

Extended use of imaging equipment may cause components such as X-Ray tubes to reach temperatures capable of inflicting burns. Use care when positioning equipment to avoid placing hot components in close proximity to patients and personnel. An anesthetized or unconscious patient is incapable of sensing and reacting to a hot component.

1.10. Electromagnetic Compatibility Statement

This product conforms with IEC60601-1-2 Ed3.0 and Ed4.0 EMC standard for medical devices.



This equipment generates, uses, and can radiate radio frequency energy. The equipment may cause or be subject to radio frequency interference with other medical and non-medical devices and radio communications. There may be RISKS of reciprocal interference posed by this equipment during specific investigations and treatments. To provide reasonable protection against such interference, the Brivo OEC 715/785/865 Mobile C-arm X-ray Product system complies with emissions limits for a Group 1, Class A Medical Devices and has applicable immunity level as stated in IEC60601-1-2.

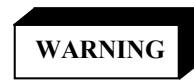
However, there is no guarantee that interference will not occur in a particular installation. The system needs special precautions regarding EMC and needs to be installed and put into service according to the EMC information provided in the Operator Manual. Portable and mobile RF communication equipment can affect system. Keep the power to these RF sources turned off when near this equipment. Recommended separation distances and information regarding compatibility with other equipment are located in the EMC Declaration Table as below.

If this equipment is found to cause interference (which may be determined by switching the equipment on and off), the operator (or qualified service personnel) should attempt to correct the problem by one or more of the following measure(s):

- 1. Reorient or relocate the affected device(s).*
- 2. Increase the separating space between the equipment and the affected device.*
- 3. Power the equipment from a source different from that of the affected device.*
- 4. Consult the point of purchase or service representative for further suggestions.*



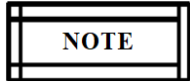
Use of accessories, transducers and cables other than those specified or provided by the manufacturer of this equipment could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation. The manufacturer is not responsible for any interference caused either by the use of cables other than those recommended, or unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the operator's authority to operate the equipment.



Use of this equipment adjacent to or stacked with other equipment should be avoided because it could result in improper operation. If such use is necessary, this equipment and the other equipment should be observed to verify that they are operating normally.



This equipment may be interfered with by other equipment, even if that equipment complies with CISPR EMISSION requirements.



To comply with the regulations applicable to an electromagnetic interface for a Group 1, Class A Medical Device, and to minimize interference risks, the following requirements shall be applied:

- 1. All cables to peripheral devices must be shielded and properly grounded. The use of cables that are not properly shielded and grounded may result in the equipment causing radio frequency interference in violation of the European Union Medical Device directive and FCC regulations.*
- 2. All of the recommended guidance regarding electromagnetic environment shall be followed.*

Guidance and manufacturer's declaration – Electromagnetic Emissions		
The Brivo OEC 715/785/865 Mobile C-arm X-ray Product is intended to use in the specified electromagnetic environment. The purchaser or operator of the Brivo OEC 715/785/865 Mobile C-arm X-ray Product should assure that it is used in an electromagnetic environment as described below:		
Emissions Test	Compliance	Electromagnetic Environment Guidance
RF Emissions (Conducted and radiated) CISPR 11	Group1	The Brivo OEC 715/785/865 Mobile C-arm X-ray Product uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment. The Brivo OEC 715/785/865 Mobile C-arm X-ray Product is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
	Class A	
Harmonic emissions IEC 61000-3-2	Not applicable	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Not applicable	

Guidance and manufacturer's declaration - Electromagnetic Immunity			
The Brivo OEC 715/785/865 Mobile C-arm X-ray Product is intended to use in the electromagnetic environment specified below. The purchaser or operator of the Brivo OEC 715/785/865 Mobile C-arm X-ray Product should assure that it is used in an electromagnetic environment as described below:			
Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air (Ed3.0)	±6 kV contact ±8 kV air (Ed3.0)	Floors are wood, concrete, or ceramic tile, or floors are covered with synthetic material and the relative humidity is at least 30 %.
	±8 kV contact ±15 kV air (Ed4.0)	±8 kV contact ±15 kV air (Ed4.0)	
Electrical fast transient Pulse IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	±2 kV for power supply lines ±1 kV for input/output lines	Mains power quality is that of a typical commercial and/or hospital environment
Surge IEC 61000-4-5	±1 kV Line(s) to Line(s) ±2 kV Line(s) to earth	±1kV Line(s) to Line(s) ±2 kV Line(s) to earth	Mains power quality is that of a typical commercial and/or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11 (250/300 cycle)	0% UT	0% UT	Mains power quality should be that of a typical commercial or hospital environment. If the operator of the Brivo OEC 715/785/865 Mobile C-arm X-ray Product requires continued operation during power mains interruptions, it is recommended that the Brivo OEC 715/785/865 Mobile C-arm X-ray Product be powered from an uninterruptible power supply or a battery.
Power frequency magnetic field (50/60Hz) IEC 61000-4-8 (50Hz or 60Hz)	3 A/m(Ed3.0) 30 A/m(Ed4.0)	3 A/m(Ed3.0) 30 A/m(Ed4.0)	Power frequency magnetic fields are at levels characteristic of a typical location in a typical commercial and/or hospital environment.
NOTE: U_T is the a.c. mains voltage prior to application of the test level.			

Guidance and manufacturer's declaration - Electromagnetic Immunity			
The Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product is intended to use in the specified electromagnetic environment. The purchaser or user of the Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product should assure that it is used in an electromagnetic environment as described below:			
	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment
Conducted RF IEC 61000-4-6	3Vrms , 6Vrms at ISM band 150GHz to 80MHz	3Vrms , 6Vrms at ISM band 150GHz to 80MHz	Portable and mobile RF communications equipment are used no closer to any part of the Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product, including cables, than the recommended separation distance calculated from the equation appropriate for the frequency of the transmitter. Recommended separation distance $d = \left[\frac{3,5}{3}\right]\sqrt{P} \quad 150 \text{ kHz to } 80 \text{ MHz}$ $d = \left[\frac{3,5}{3}\right]\sqrt{P} \quad 80 \text{ MHz to } 800 \text{ MHz}$ $d = \left[\frac{7}{3}\right]\sqrt{P} \quad 800 \text{ MHz to } 2.5 \text{ GHz}$ The formulas are applicable for Ed3.0 only. Note: P is the power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey ^a , are less than the compliance level in each frequency range ^b . Interference may occur in the vicinity of equipment marked with the following symbol: 
Radiated RF IEC 61000-4-3	3V/m 80MHz to 2.5GHz (Ed3.0) 3V/m 80MHz to 2.7GHz, 28V/m (Ed4.0)	3V/m 80MHz to 2.5GHz (Ed3.0) 3V/m 80MHz to 2.7GHz, 28V/m (Ed4.0)	
a: Field strengths from fixed transmitters, such as base stations for cellular telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be estimated accurately. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be performed. If the measured field strength exceeds the RF compliance level above, observe the Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product to verify normal operation in each use location. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the [EQUIPMENT and/or SYSTEM]. b: Over the frequency range 150 kHz to 80 MHz, field strengths are less than 3 V/m. NOTE 1: At 80 MHz and 800 MHz, the higher frequency range applies. NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people. Note3: These are guidelines. Actual conditions may vary.			

The Recommended Separation Distances are listed in the next table, which are applicable for Ed3.0 only.

Recommended Separation Distances for Portable and Mobile RF Communications Equipment and the Brivo OEC 715/785/865 system			
The Brivo OEC 715/785/865 Mobile C-arm X-ray Product is intended to use in the electromagnetic environment where the RF interference is controlled. According to the power rating of the communication equipment, the purchaser or operator of the Brivo OEC 715/785/865 Mobile C-arm X-ray Product can prevent the Electromagnetic interference by the recommended separation distance below:			
Rated Power of Transmitter/W	Separation Distance/m		
	150KHz to 80 MHz	80 MHz to 800 MHz	800 MHz to 2,5 GHz
	$d = [\frac{3,5}{3}]\sqrt{P}$	$d = [\frac{3,5}{3}]\sqrt{P}$	$d = [\frac{7}{3}]\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
For transmitters rated at a power not listed above, the Separation Distance can be estimated using the equation in the corresponding column, where P is the power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
NOTE 1: At 80 MHz and 800 MHz, the higher frequency range applies.			
NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			
NOTE 3: These are guidelines. Actual conditions may vary.			

Guidance and manufacturer's declaration - Electromagnetic Immunity

The Brivo OEC 715/785/865 Mobile C-arm X-ray Product is intended to use in the electromagnetic environment specified below. The purchaser or operator of the Brivo OEC 715/785/865 Mobile C-arm X-ray Product should assure that it is used in an electromagnetic environment as described below:

Test specifications for ENCLOSURE PORT IMMUNITY to RF wireless communications equipment

Test frequency (MHz)	Band ^{a)} (MHz)	Service ^{a)}	Modulation ^{b)}	Maximum power (W)	Distance (m)	IMMUNITY TEST LEVEL (V/m)
385	380 – 390	TETRA 400	Pulse modulation ^{b)} 18 Hz	1,8	0,3	27
450	430 – 470	GMRS 460, FRS 460	FM ^{c)} □ □ 5 kHz deviation 1 kHz sine	2	0,3	28
710	704 – 787	LTE Band 13, 17	Pulse modulation ^{b)} 217 Hz	0,2	0,3	9
745						
780						
810	800 – 960	GSM 800/900, TETRA 800, iDEN 820, CDMA 850, LTE Band 5	Pulse modulation ^{b)} 18 Hz	2	0,3	28
870						
930						
1 720	1 700 – 1 990	GSM 1800; CDMA 1900; GSM 1900; DECT; LTE Band 1, 3, 4, 25; UMTS	Pulse modulation ^{b)} 217 Hz	2	0,3	28
1 845						
1 970						
2 450	2 400 – 2 570	Bluetooth, WLAN, 802.11 b/g/n, RFID 2450, LTE Band 7	Pulse modulation ^{b)} 217 Hz	2	0,3	28
5 240	5 100 – 5 800	WLAN 802.11 a/n	Pulse modulation ^{b)} 217 Hz	0,2	0,3	9
5 500						
5 785						

NOTE If necessary to achieve the IMMUNITY TEST LEVEL, the distance between the transmitting antenna and the ME EQUIPMENT or ME SYSTEM may be reduced to 1 m. The 1 m test distance is permitted by IEC 61000-4-3.

^{a)} For some services, only the uplink frequencies are included.

^{b)} The carrier shall be modulated using a 50 % duty cycle square wave signal.

^{c)} As an alternative to FM modulation, 50 % pulse modulation at 18 Hz may be used because while it does not represent actual modulation, it would be worst case.

It is applicable for Ed4.0 only



Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of this equipment, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.

Minimum separation distances for higher IMMUNITY TEST LEVELS shall be calculated using the following equation:

$$E = \frac{6}{d} \sqrt{P}$$

Where P is the maximum power in W, d is the minimum separation distance in m, and E is the IMMUNITY TEST LEVEL in V/m.

Use Recommendations

This product complies with IEC 60601-1-2 EMC standard for medical devices and with radio frequency emission requirements per CISPR11 Group1 Class A standard limits. The system is intended for use in hospitals.

Adhering to the distance separation recommendation table, between 150 kHz and 2.5 GHz, will reduce disturbances recorded at the image level but may not eliminate all disturbances.

However, when installed and operated as specified herein, the system will maintain its essential performance.

For example, a 1W mobile phone (800 MHz to 2.5 GHz carrier frequency) shall be put 2.3 meters apart from the system (in order to avoid image interference risks).

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMAGNETIC COMPATIBILITY of the system.

1.11. Equipment Malfunction

If either the hospital or equipment circuit breakers trip, an equipment malfunction may be indicated. Do not attempt to operate the equipment until a qualified service engineer has checked it.

If any of the equipment controls fail to respond as indicated in this manual, you should:

1. Remove power to the equipment by placing the power switch in the off position and unplugging the power cord from the AC receptacle.
2. Notify a qualified service engineer.
3. Do not operate the equipment until the service technician advises that it is operating properly.

WARNING

Establish an operation room emergency procedure to ensure patient and user safety in the event of a system failure.

1.12. Mains Isolation

The system has 2 means to isolate its circuits electrically from the supply mains on all poles simultaneously.

1. Switch the breaker to OFF position.
2. Unplug the power cable.

1.13. External Devices

To ensure patient safety, only connect external equipment that has been approved by GE HUALUN Medical Systems Co, Ltd. All equipment such as keyboard, wireless mouse, mobile HDD, etc, attached to the external interface connections must meet the requirements of IEC 60601-1 when operated within the patient environment. When used outside of the patient environment, each externally connected device must comply with the relevant IEC/ISO requirements for that device. In any case, the combination of all externally connected equipment shall not cause the leakage current of any device used within the patient environment to exceed the limits stated in IEC 60601-1.

WARNING

Cover the connector when it is not in use to avoid unintentional touch.

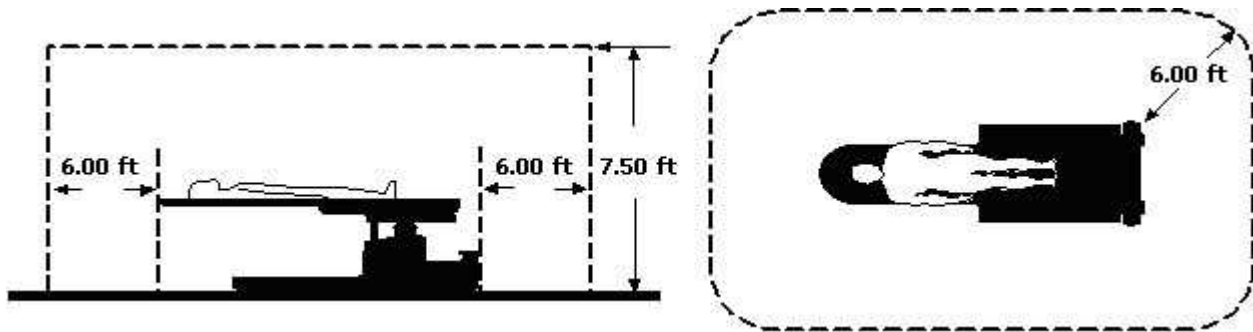
Connect only medical devices that comply with UL/IEC 60601 standards. Contact GE Healthcare for a list of compliant devices. Connecting a non-compliant device could cause injury to the operator, the patient or the equipment.

1.14. Patient Environment

1.14.1. Within the United States

Within the US the Patient Environment is defined by NFPA 99 and UL 60601-1:2003. In areas in which patients are normally cared for, the patient environment is the space with surfaces likely to be contacted by the patient or an attendant who can touch the patient.

This encloses a space within the room 6 ft. beyond the perimeter of the bed (examination table, dental chair, treatment booth, etc.) in its intended location, and extending vertically 7.5 ft. above the floor.

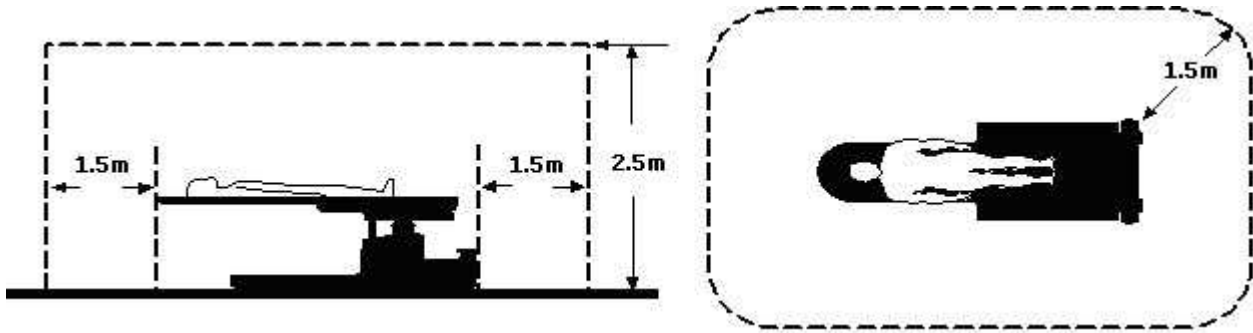


Patient Environment Inside the United States

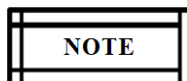
1.14.2. Outside the United States

Outside the US the patient environment is defined by GB9706.15-2008, IEC 60601-1-1:2000, IEC60601-1:2005. In areas in which patients are normally cared for, the patient environment is the space with surfaces likely to be contacted by the patient or an attendant who can touch the patient.

This encloses a space within the room 1.5 m beyond the perimeter of the couch (examination table, dental chair, treatment booth, etc.) in its intended location, and extending vertically 2.5 m above the floor.



Patient Environment Outside the United States



Please consider the dose attenuation caused by the couch (examination table, dental chair, treatment booth, etc.) during the operation.

Chapter2. System Overview

2.1. General Description



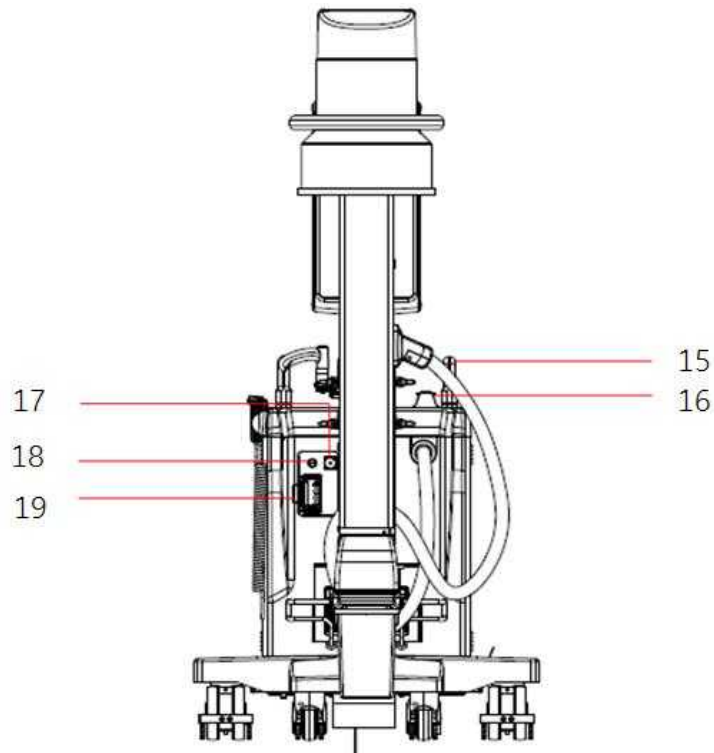
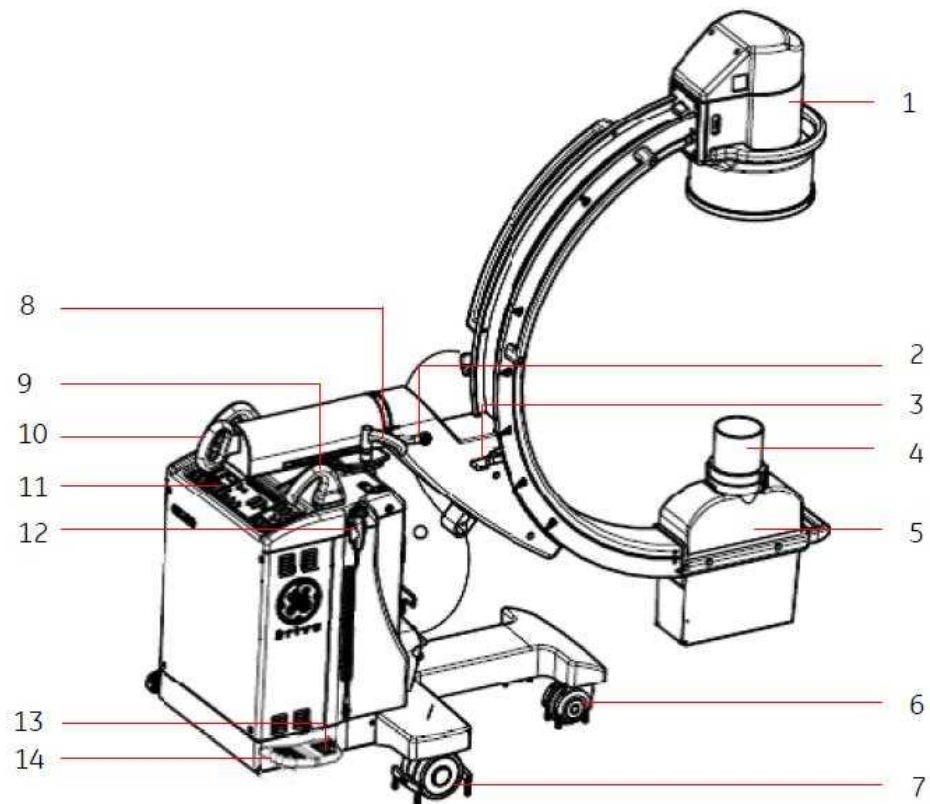
Only properly trained persons can operate this equipment.

Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product includes:

- Mobile C-Arm, console, X-Ray generator, X-Ray image intensifier, display device (monitor frame, monitor, computer and power supply units).
- Hardware Options:
 - Thermal/Film printer
 - Laser aimer on the image intensifier
 - Laser aimer on the tube
 - 30cm skin spacer
 - Film cassette holder
 - Video distributor for external DVI video output
 - Removable filter
 - Removable grid
 - Footswitch
 - DVD Driver
 - Dual 19" Mono LCD Monitor (only for Brivo OEC 785)
 - Wi-Fi Internet Adapter (only for Brivo OEC 865)
- Nominal Length of External Cables:
 - Interconnect cable: 8.0m
 - Bundle cable: 2.5m
 - Power cable: 7.5m
 - Handswitch cable: 0.8m
 - Footswitch cable: 10m/5m for multifunctional two-pedal footswitch with save button
8m/5m for two-pedal footswitch

Please refer to Chapter 12 "Technical Reference" for more details on some hardware options.

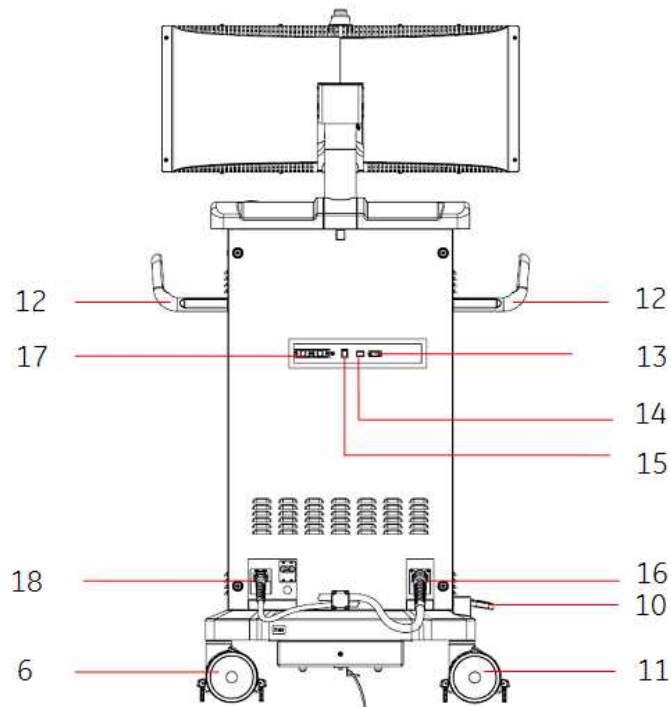
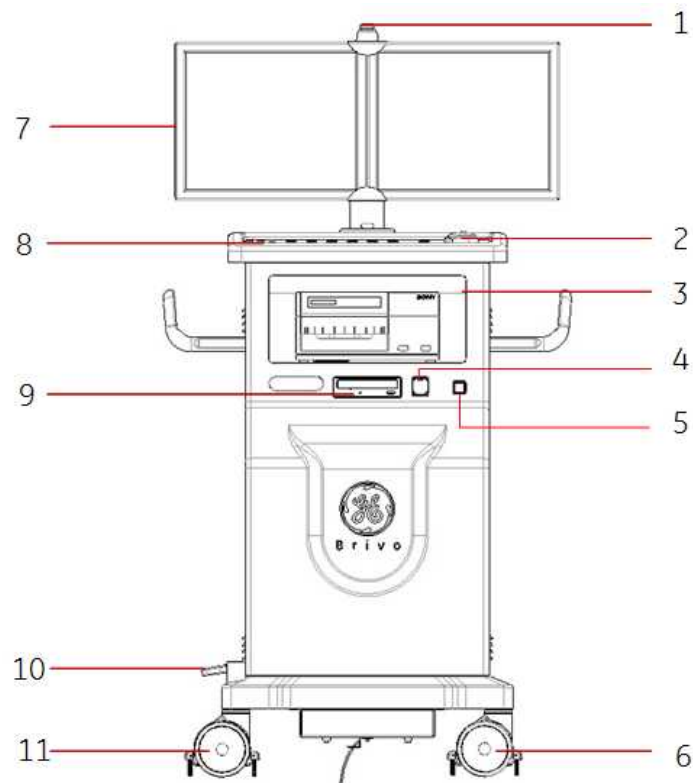
2.2. C-Arm Components



System Overview

1. Image intensifier and CCD camera
2. Lateral rotation movement brake handle
3. Orbital movement brake handle
4. Skin spacer
5. X-Ray tube head (X-Ray tube and collimator)
6. Front wheel
7. Rear wheel
8. Wig-wag movement brake handle
9. Steering handle
10. Horizontal movement brake handle
11. C-Arm control panel
12. Handswitch
13. Brake release pedal
14. Brake pedal
15. Driving handle
16. Emergency switch
17. Footswitch socket
18. Key switch
19. Interconnect cable socket

2.3. Workstation Components



System Overview

1. Exposure indicator
2. Mouse
3. Cabinet for printer
4. USB connector
5. Power on/off
6. Universal wheel (with brake for Brivo OEC 715/785; without brake for Brivo OEC 865)
7. LCD Monitors
8. Workstation membrane keyboard
9. DVD driver (Option)
10. Brake pedal (only for Brivo OEC 865)
11. Central control wheel (for Brivo OEC 865) ; Universal wheel with brake (for Brivo OEC 715/785)
12. Handles
13. Room Control Connector (Room interface)
14. Network interface
15. Wireless network interface switch (Option)
16. Interconnect cable
17. Digital Video Interface (DVI) (Option)
18. Power cable, incoming circuit breaker and power on indicator

Chapter3. Mechanical Positioning

3.1. Overview

This chapter describes:

- Positioning
- Moving

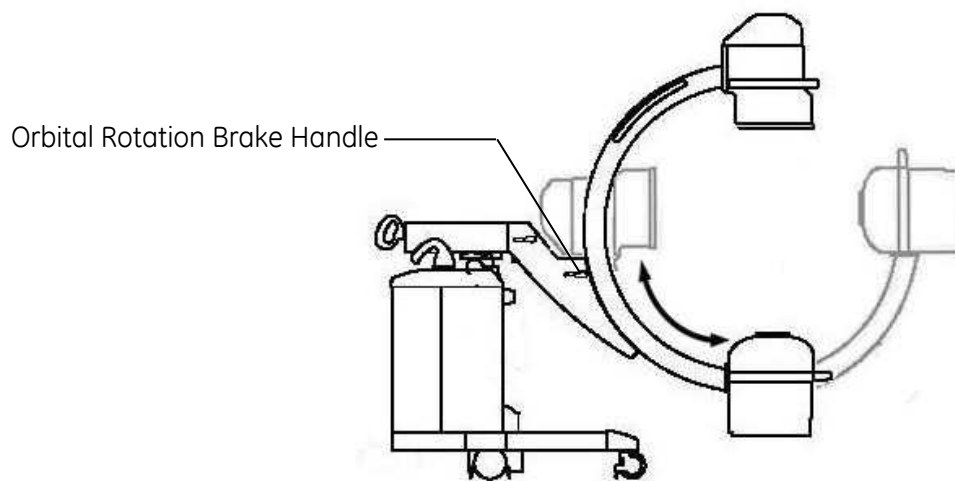
3.2. Positioning

WARNING

Carefully position the C-Arm to prevent patient or other things contact.

3.2.1. C-Arm Orbital Rotation

Brivo OEC 715/785/865 C-Arm X-Ray Image System provides 120° of orbital rotation (90° underscan and 30° overscan). The bearing side of the C-Arm is marked with a scale to aid in positioning.

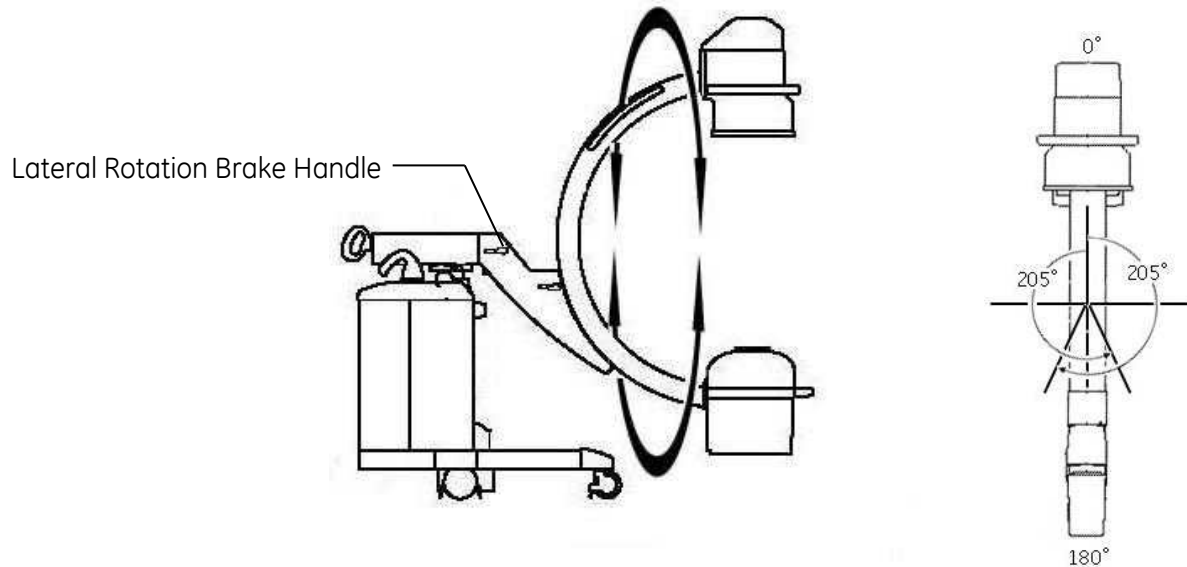


To lock or unlock the orbital rotation brake, turn either of the brake handles located on both sides of the C-Arm support assembly. The locked position is indicated by a “lock” label.

WARNING

Do not place fingers or allow clothing between these assemblies when positioning the C-Arm.

3.2.2. Lateral Rotation



WARNING

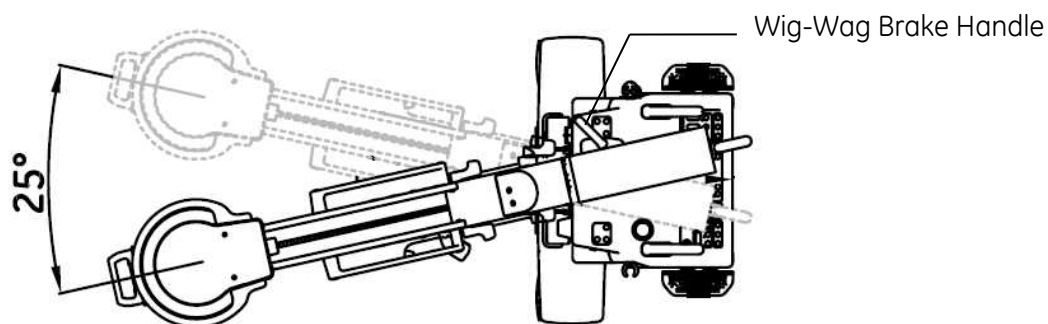
During rotation, there are accessible moving parts, keep hands/fingers away from the moving parts or the location that the moving parts may collide.

The C-Arm can provide 205° of lateral rotation on the C-Arm support axis.

CAUTION

When accessories such as the laser aimer or film cassette are attached to the C-Arm, releasing the orbital or lateral rotation brake could cause the C-Arm to move. Make sure to hold the C-Arm to control its movement whenever a brake is released.

3.2.3. Wig-Wag



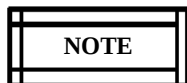
The movement range of the Wig-Wag is 25 (± 12.5).

Release the Wig-Wag brake by placing the brake handle in the position indicated by the “unlock” label to enable the Wig-Wag movement of the C-Arm.

Lock the Wig-Wag brake by placing the brake handle in the position indicated by the “lock” label to lock the Wig-Wag movement of the C-Arm.



Pinch points exist under the Wig-Wag brake handle.



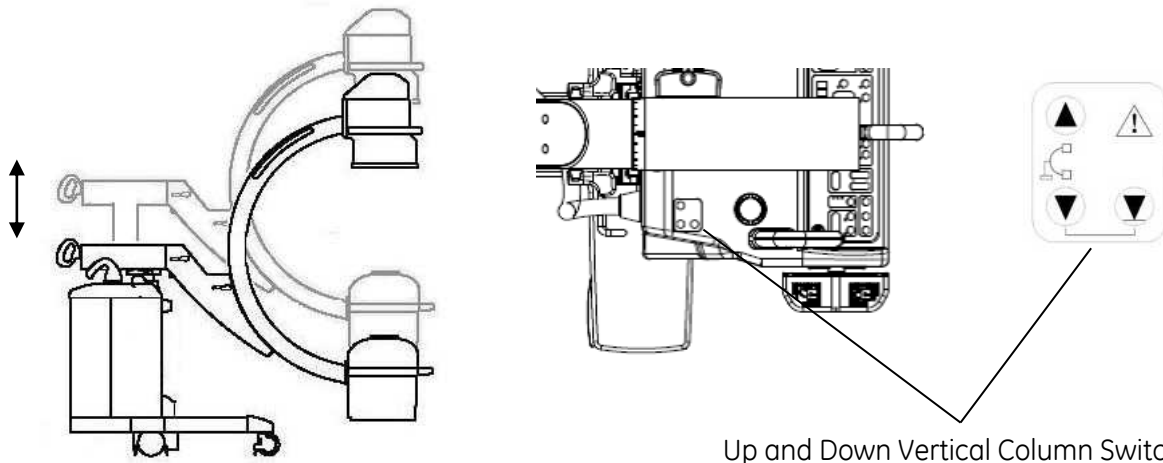
In use of Wig-wag brake, don't lift the brake handle. Once the handle is lifted, it will be in calibration mode, the location of LOCKED will be changed unconsciously.

Calibration of Wig-Wag brake

1. Lift Wig-Wag handle when it's LOCKED.
2. Rotate the handle to any locking position you want.
3. Put down the handle.

Please check if the Wig-Wag brake could effective use after calibration.

3.2.4. Vertical Column Operation



Up and Down Vertical Column Switches

The vertical column motor is actuated by pressing the up and down vertical column switches located on top cover of the mainframe. Use the vertical column to elevate the C-Arm upward and downward.

Press ▲ or ▼ to lift and lower the C-Arm, the upward movement continues until the switch is released or the movement reaches its limits.

When C-Arm can't be driven down by only pressing down switch, please release the switch, make sure there are no obstacles between the C-Arm and the T-Base. Make sure there are no obstacles between

Cross-Arm and console. Press and hold the two switches ▼ ▼ (down and mandatory switches), it will lower C-Arm to the lowest position.

Use the scale located on the vertical column to help position the C-Arm at the height you want.

WARNING

Please make sure C-Arm will not be blocked by any other equipment during the vertical movement. If there're any obstacles, system will sound two beeps and stop movement. Please check and clear the obstacle. Then you can continue to raise or lower the C-Arm. If any failure occurs, please contact GE Healthcare service engineer.

When positioning the vertical column, observe the moving assemblies to ensure the safety of patients and hospital personnel.

WARNING

A possible pinch point exists between C-Arm and the tip of the front cover. Do not place your foot on the tip of the front cover while operating the vertical column or while positioning the C-Arm.

WARNING

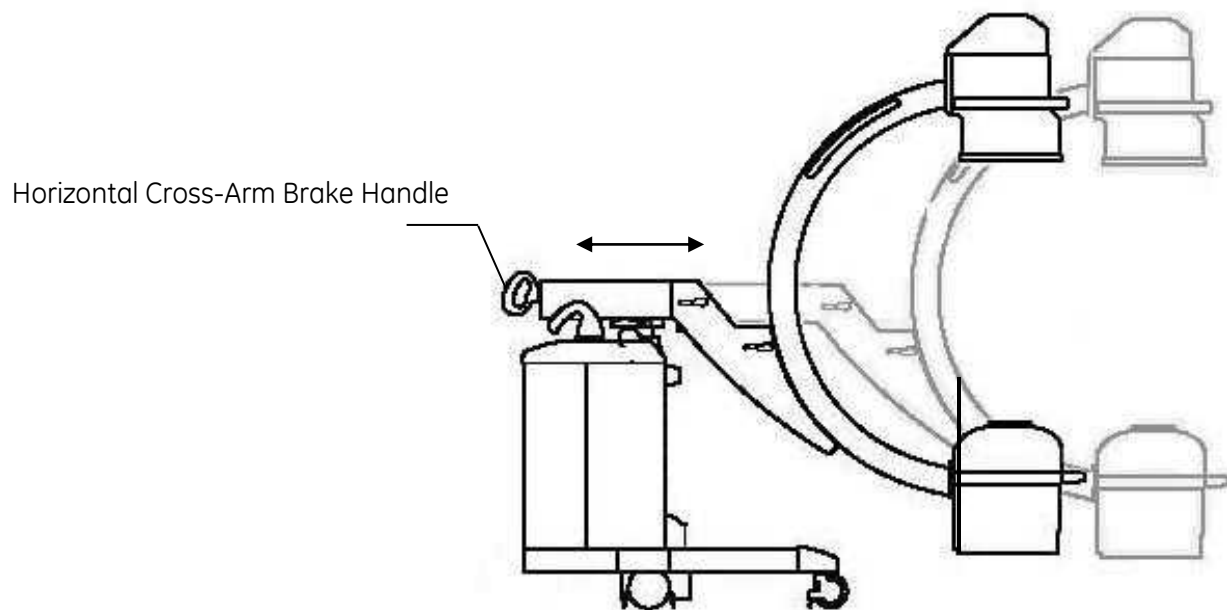
Set all brakes and watch carefully when you move the motorized vertical column. Even a low-speed collision can result in significant personal injury or equipment damage.

3.2.5. Horizontal Cross-Arm

The horizontal Cross-Arm extends a maximum of 20cm.

To position the Cross-Arm:

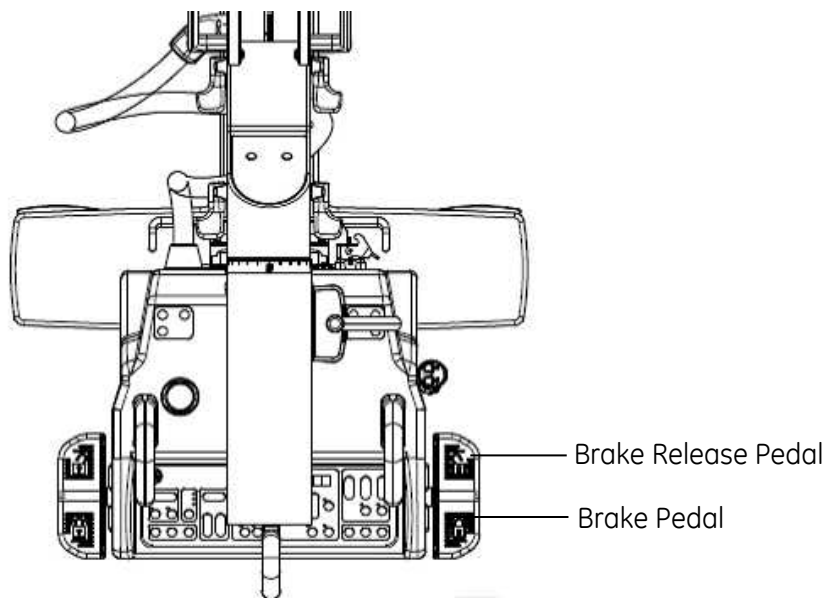
1. Release the Cross-Arm brake by placing the brake handle in the position identified by the **unlock** label.
2. Push or pull the Cross-Arm to the desired position. Use the centimeter scale located on the Cross-Arm as an aid in positioning.
3. Lock the Cross-Arm brake by placing the brake handle in the position identified by the **lock** label.



3.2.6. C-Arm Brakes

To lock the wheel brake and steering handle, press the brake pedal.

Pressing the brake release pedal allows the wheels to rotate freely and the steering handle to steer.

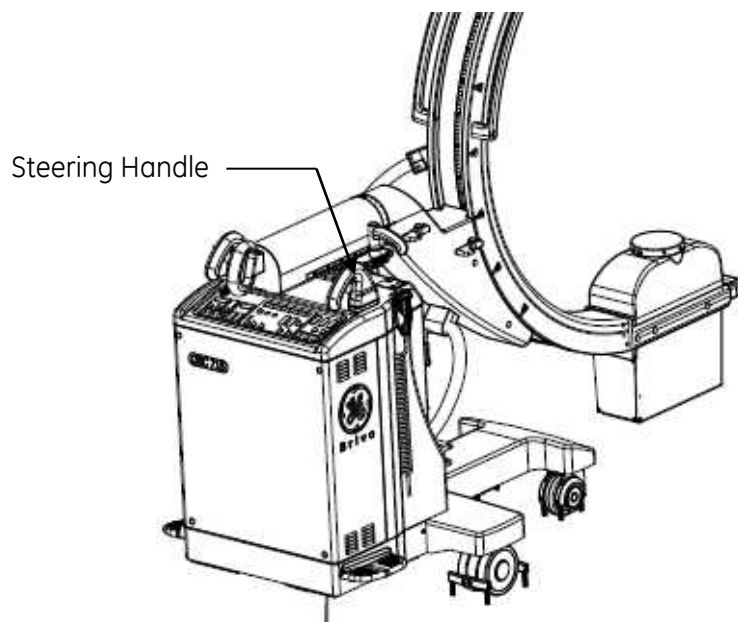


3.2.7. C-Arm Steering Handle

Use the right steering handle to turn the rear wheels $\pm 90^\circ$ to the right or left. Use this feature to position the C-Arm during clinical applications or to negotiate sharp turns during transport. The rear wheels turn at approximately the same angle as the right steering handle.



To avoid losing the control of the Mobile C-Arm, always reduce transport speed before moving the steering handle out of 0 position.

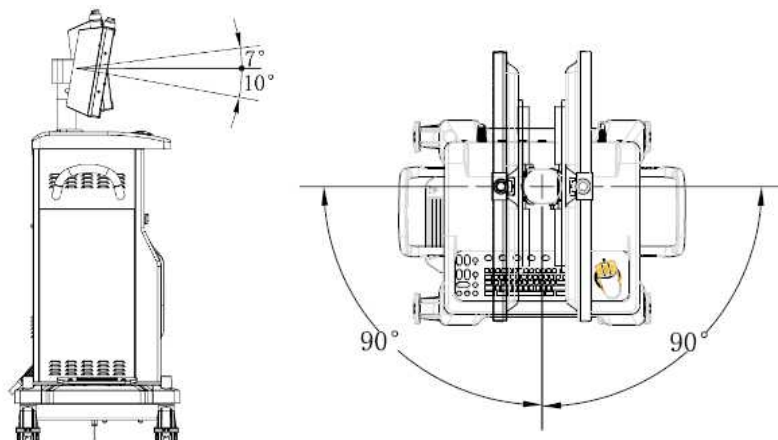


3.2.8. Workstation Monitor Adjustment

The Workstation LCD Monitor can be placed to the proper position for best viewing.

Hold the upper and the lower sides of the monitor to adjust the pitch degree until it moves to the desired position. Tilt arm: $\geq 7^\circ$ up/10° down.

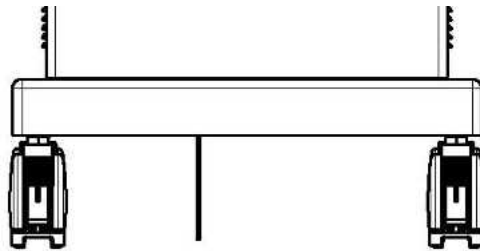
Hold the left and right sides of the monitor to adjust the rotate degree, the Workstation LCD Monitor can rotate $\pm 90^\circ$.



3.2.9. Workstation Brakes

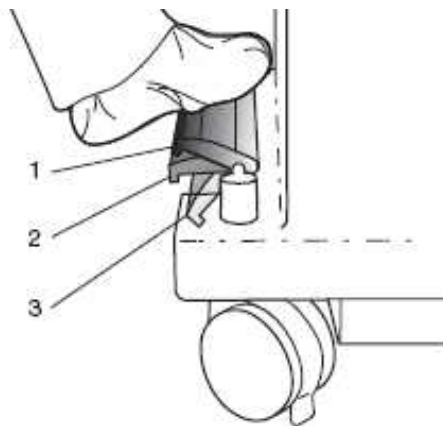
Brivo OEC 715/785 Workstation provides wheel brakes. Pressing the wheel brakes allows you to lock the brakes and prevent the Workstation from moving.

Unlock the brakes by pressing down the wheel brakes again.



Brivo OEC 865 Workstation provides brake pedal located on the left side of the Workstation. It has three positions that control wheel movement. The pedal positions are:

1. The wheels close to the pedal roll in a straight line, and the wheels opposite pivot freely. Place the brake pedal this position to move the Workstation long distances.
2. Allows all the wheels to pivot freely. Place the brake pedal in this position to easily maneuver the Workstation during final positioning.
3. Prevents Workstation from moving.



3.3. Moving the C-Arm and Workstation

3.3.1. Overview

Familiarize yourself with the location and mechanical operation of all controls prior to moving the system.



Neglecting the precautions during moving could cause the system to lose control and may cause injury to the operator and other persons.



Before transporting the system, please make sure the 4 brake handles: the C-Arm orbital rotation brake, the lateral rotation brake, the Wig-Wag brake, and the horizontal Cross-Arm brake are locked. Otherwise, equipment damage or personal injury may result. If any brakes are not firmly locked, don't transport the system and contact GE Healthcare Service Engineer.



Cords and cables connected to the Workstation and C-Arm can become trip and/or snag hazards in the work area. Personnel working around the system should be aware of all cords and cables whenever they move around the system or the system is moved.

Always:

- Check and make sure there are no obstacles on the way.
- Move the system slowly.

3.3.2. Moving the C-Arm

Familiarize yourself with the location and mechanical operation of all controls prior to moving the C-Arm.



Use the handles provided on the C-Arm to position mechanical assemblies.

1. Return all moving assemblies to their most compact positions. Lower the vertical column and retract the Cross-Arm. Remove the cassette holder if fitted. (Do this before switching off the mobile C-Arm).



When the C-Arm is positioned on more than or equal to 10° incline, and the mechanical parts are extended to their greatest length, if the cassette holder is attached, the C-Arm will be unstable. Please be sure to put the C-Arm in its compact position.

2. Lock all movable mechanical assembly brakes: the C-Arm orbital rotation brake, the lateral rotation brake, the Wig-Wag brake, and the horizontal Cross-Arm brake.
3. Press and hold the power on/off key 2 seconds on the Workstation to switch the system off. Unplug the power supply cable and secure the power cable.
4. Disconnect the interconnect cable from the C-Arm and coil and secure the cable around the Workstation's handles.
5. Disconnect the footswitch and secure it on the C-Arm. Store the handswitch in its holder.
6. Place the C-Arm's steering handle in the normal position and unlock the wheel brakes.
7. Guide the C-Arm by pushing with the steering handles.



Do not move the C-Arm over inclines greater than 10°. Do not move the C-Arm up or down stairs or steps.

Do not lock the C-Arm in place on an incline greater than 5°.

Transport the system at normal speed. Get across ramps and thresholds as slow as possible.

8. When you reach your destination, place the C-Arm's brake pedal in the locked position.

3.3.3. Moving the Workstation

To move the mobile Workstation from one place to another, operate as follows:

1. Ensure the system power is off.
2. Unplug the power supply cable and secure the power cable.
3. Disconnect the interconnect cable from the C-Arm and secure the interconnect cable.
4. Ensure that all the cable connected with the Workstation has been removed.
5. Ensure the monitor pitch and rotate degree is back into place.
6. Ensure the Workstation brake pedal is unlocked and guide the Workstation to its destination.
7. When you reach your destination, lock the Workstation brake pedal.

WARNING

Do not move the Workstation over inclines greater than 10 degrees. Do not move the Workstation up or down stairs or steps. Do not lock the Workstation in place on an incline greater than 5 degrees.

WARNING

Do not stretch interconnect cable and power cable when moving the equipment.

Chapter4. Start Up and Operating Control

4.1. Overview

This chapter describes:

- Start Up
- C-Arm Controls
- Workstation Controls
- Image Quality Control

4.2. Start Up

4.2.1. Power On

Press the Power On/Off key on the Workstation to start up Workstation, use Workstation to view images. The C-Arm receives power through an interconnect cable attached to the Workstation, connect C-Arm to produce images.

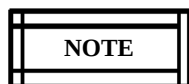


At initial setup or when removing from storage, stabilize for 24 hours at ambient temperature and humidity before applying power. Failure to observe this warning will result in damage to equipment.



Remove any devices from USB connector before booting, otherwise system booting may fail.

1. Plug the Workstation power plug into a properly rated receptacle. Refer to the Chapter 12 “Technical Reference” for information about power requirements.

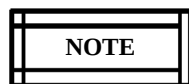


If the system power off automatically due to power lost, user needs to power on the system after power restored to proceed.

Brivo OEC 715/785/865 Mobile C-Arm X-Ray product provides a UPS to protect data loss. If the power lost during operation, UPS battery will supply power to computer and right monitor until system shut down automatically.

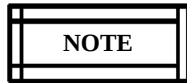


Do not disconnect, repair or replace UPS without permission. Only a GE qualified service engineer can perform repair and replacement. The life cycle of the UPS is about 3-5 years in normal use. It may be different in different conditions.



To use the system for the first time, it is recommended to boot the system for continuous 8 hours to ensure that the UPS is fully charged, otherwise if the system power down accidentally, there is risk that the system will not safely power off during the normal work, when the system power down accidentally, the UPS play a protective role to ensure the system safely power off, and it is recommended to boot the system 8 hours in advance to ensure that the UPS is fully charged the next time an additional charge of the UPS is not necessary when the system does not abnormally power off.

2. Insert the Workstation interconnect cable into the connector located on the top left side of the C-Arm interface panel by pushing the connector in and lock it in place. Users need hear audible sound to determine the socket locked in place.



The Workstation could be powered on without connected with C-Arm by interconnect cable. Operator could review the images by the Workstation.



If the connector does not lock in place, unreliable system operation may result.

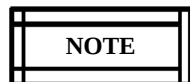


Forbid to use different model or serial number between Brivo OEC 715/785/865 Workstation and C-Arm. System will pop up a warning message of unmatched connection. There is unpredictable risk to continue using the system in this case.

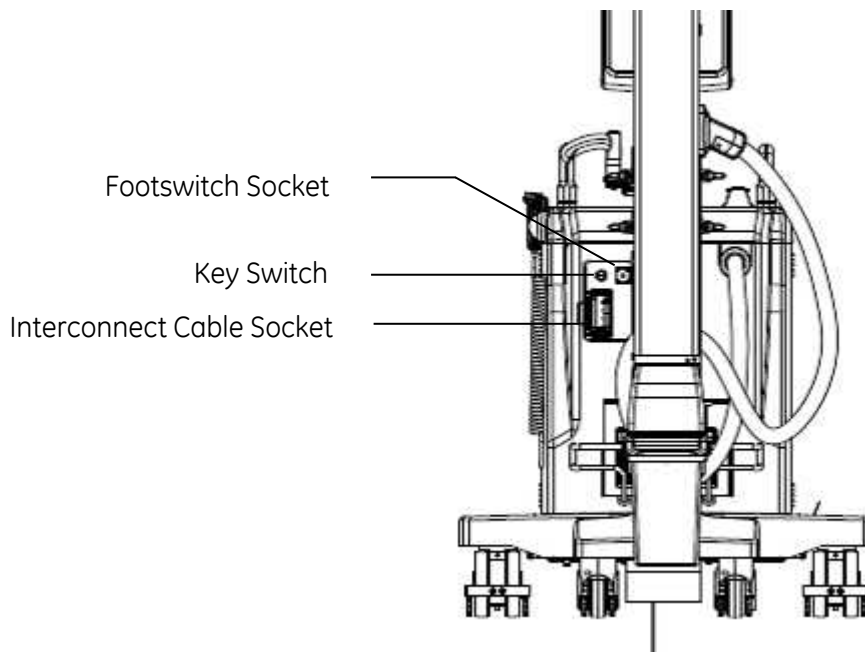
3. Connect the footswitch to the socket on the top left side of the C-Arm interface panel.



Connector covers must be in place when connectors are not in use to avoid inadvertent contact with connectors.

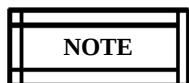


Use of a footswitch cover is recommended during all medical procedures.



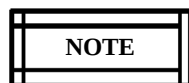
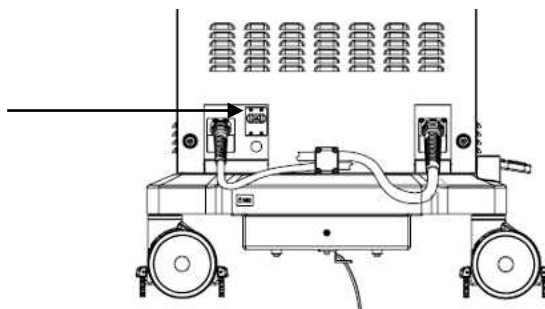
- Turn the key switch located on the C-Arm interface panel to enable X-Rays and motorized mechanical movement.

When the key switch is placed in the on position , the C-Arm is fully operational. When the key switch is placed in X-Ray off position , X-Ray is disabled and the lift column is enabled. When the key switch is placed in the standby position , X-Rays and vertical column movement are disabled, but the C-Arm power remains.



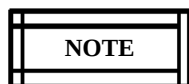
It is recommended to turn the key switch to standby position and safe keep it at all times, except when a procedure is in progress, to prevent the possibility of radiation being emitted through the inadvertent actuation of an X-Ray switch.

- Make sure that the breaker is in ON state before pressing the power on/off button. The power on indicator will light.



Do not position the system to make it difficult to operate the breaker.

- Press the Workstation power on/off button. Both the Workstation and the C-Arm will initial power up sequence. The field size is in its maximum. The auto fluoroscopy mode is selected.
- System startup is complete when the login screen or Exam Management screen is displayed on the right monitor. The boot up time is about 2 minutes.



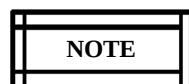
The system won't response power off command by pressing power on/off button until start up sequence complete

If a problem is encountered during power on, error message may be displayed on the right monitor and restart will be required. Refer to Chapter 10 Display Messages for more information about messages. If restart system and failure remains, call GE Healthcare Communication Center for service.



Data loss can occur if the system startup sequence is interrupted. Do not turn off the breaker or unplug the power cable until system startup is complete.

If you are performing a procedure which draping is necessary, drape the C-Arm using the Steri uick draping system.

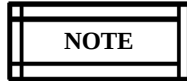


Draping the X-Ray tube housing will inhibit air flow resulting in cooling inefficiencies. This may result in the housing reaching its rated heat capacity sooner. When the housing reaches its maximum rated heat capacity, X-Rays will be discontinued until the housing has cooled.

4.2.2. Standby or Power Off

Standby

1. Turn the C-Arm key switch to the standby position (). The system is still power on.



Placing the equipment in standby disables X-Rays and vertical column operation so these features will not be activated unintentionally.

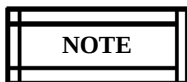
2. Turn the key switch back to on position when you are ready to use the C-Arm.

Power off

1. Press power on/off key on Workstation to power off system.
2. Unplug the Workstation power cord from the AC receptacle.
3. Disconnect the Workstation interconnect cable from the C-Arm.

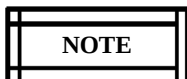


When you turn the system off, wait for at least 1 minute before powering the system back on. This helps prevent from operational problems.



Remove all power from the system before moving the system or if a problem occurs which prevents normal operation. All power should also be removed when periodic maintenance and cleaning is performed.

During shutting down, the power shall not be cut off automatically until the system software exits.



During powering off, the system will ignore power on command by pressing power on/off button until shutdown finished.

4.3. C-Arm controls

This section describes the C-Arm's operating controls. Before you begin imaging, familiarize yourself with the following contents:

- C-Arm Control Panel
- Footswitch and Handswitch
- Emergency Switch



Do not touch tube housing or let it touch anyone during operating due to the potentially high temperature generated by the tube.



This equipment produces scatter radiation. Ensure proper safety practices during operation.



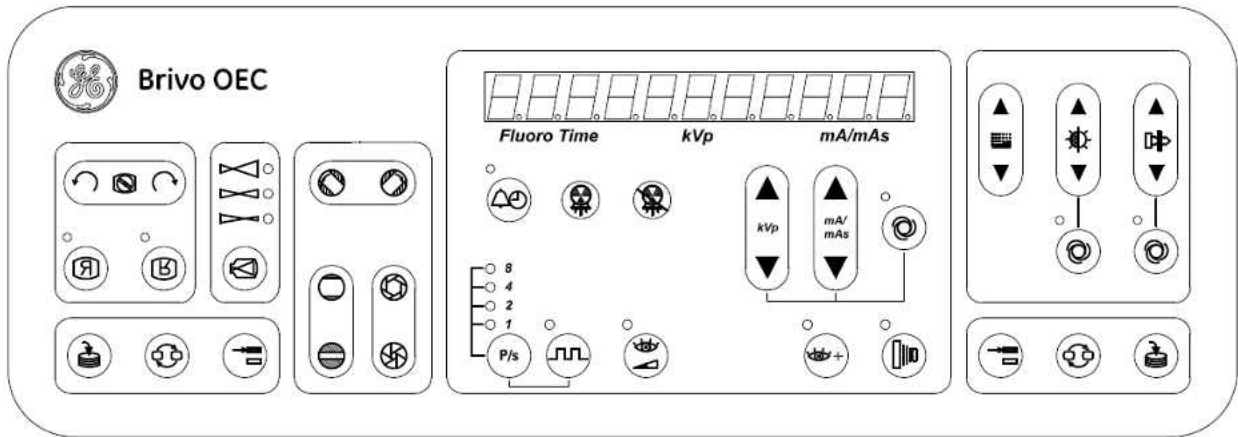
Lack of UPS battery capacity may cause data loss. Then the system is not recommended for use during surgery.

4.3.1. C-Arm Control Panel

The Mobile C-Arm's Control panel allows operator to control the generation, display of images, adjust the parameters of fluoroscopy and radiographic film. The keys and indicators on the control panel are grouped according to their function.

These groups of keys allow you to:

- Adjust Image Orientation
- Select Image Intensifier Field Size
- Control Collimator
- Control Generator
- Select Noise Filter
- Adjust Contrasts/Brightness, Smart metal



Only functions applicable to the mode selected are active. If any function is active, then the corresponding indicator lights are illuminated. The normal auto fluoroscopy mode is the default setting.

Image Orientation

Use these keys to rotate or reverse the image produced on the left WorkStation monitor. Image Rotation and Image Reversal are useful for displaying the image in any orientation that you desire.

Image rotation and image reversal are not available to be used with swapped image and when image annotation or image measurement has been applied.



Image Rotation

Press the left key to rotate the image counter-clockwise or press the right key to rotate the image clockwise.

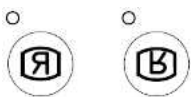


Image Reversal

Press the "Image Reversal" keys to change the orientation of the image displayed on the WorkStation's left monitor. Press the left key to reverse the image from left to right. Press the right key to invert the image from top to bottom.

Image Intensifier Field Size



Field Size

Press the "Field Size" key to select the X-Ray field size. The illuminated LED indicates which field size is selected. Press the "Field Size" key until the field size you want is selected. Refer to the table below:

<u>TEXT</u>	<u>ICON</u>	<u>FIELD SIZE</u>
NORM		9 inches
MAG1		6 inches
MAG2		4.5 inches

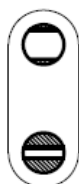
Collimator Control

The X-Ray beam may be collimated by using the iris collimator or the shutter collimator. Press the key until the shutter or iris is in the position you desire.



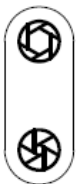
Shutter Collimator Rotation

Press the left key to rotate the shutter collimator counter-clockwise. Press the right key to rotate the shutter collimator clockwise.



Shutter Collimator Open/Close

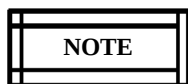
Press the "Shutter Collimator Open/Close" keys to open and close the shutter collimator. Press the bottom portion of the key to close the shutters and press the top portion of the key to open the shutters.



Iris Collimator Open/Close

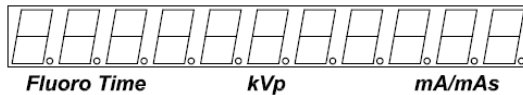
Press the "Iris Collimation Open/Close" keys to open or close the collimator iris. Press the bottom portion of the key to close the iris or the top portion of the key to open the iris.

When the system is transferred to film mode, Iris opens fully. And the parameters set in the fluoroscopy mode are memorized. When return to fluoroscopy mode, it can be restored until a reset is made.



Proper use of the iris can decrease the X-Ray radiation to the patient and improve image quality.

Fluoro time, kVp, mA/mAs Display



Display fluoro time (min), kVp and mA/mAs values.



Continued using without cooling may damage the X-Ray tube. A warning message will display on the Workstation if the tube temperature is too high. Although fluoroscopy is not prevented, you should wait for the tube to cool before making another exposure.

System Indicators



Alarm Reset

The C-Arm counts or accumulates the amount of time that X-Rays are generated when an X-Ray switch is pressed. The amount of accumulated fluoroscopy time is indicated on the C-Arm control panel.

If pulsed imaging is enabled, the amount of time accumulated depends on the length of time an X-Ray switch is pressed and the length of the pulses and the number of pulses per second.

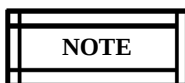
System will sound continuous beeps and illuminate the Alarm Reset LED during any exposure after accumulated fluoroscopy time is 4'30".

To silence the alarm and reset the accumulated fluoroscopy time:

Press Alarm Reset briefly to silence the alarm.

Press and hold Alarm Reset for approximately two seconds to reset the accumulated exposure time to zero and silence the alarm.

In addition, when the continuous fluoroscopy time reaches 9', system will sound continuous beeps. When this time reaches 9'30", X-Rays will be terminated automatically until the timer is reset.



"Automatic Termination at 9'30" in One Shot" could be configured as disable or enable by GE Healthcare service representative.



Repetitious prolonged exposure could increase radiation and probability of harm to the patient and operator.



X-Ray Emission Indicator

In the working mode, when the foot/handswitch is activated to take an exposure, the indicator will light.

System sounds discontinuous beeps warning during any exposure.



X-Ray Off Indicator

When the key switch is placed in X-Ray off position () or standby position (), X-Ray is disabled and this indicator will light.

In the other X-Ray off case, this indicator will light too.

Generator Control



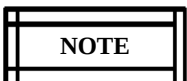
kVp

Press the “kVp” key can adjust kVp manually and override auto technique setting. Press the top portion of the key to increase kVp or press the bottom portion of the key to decrease kVp.



mA/mAs

Press the “mA/mAs” key can adjust mA/mAs manually for fluoroscopy/film respectively and override auto technique setting. Press the top portion of the key to increase mA/mAs or press the bottom portion to decrease mA/mAs.



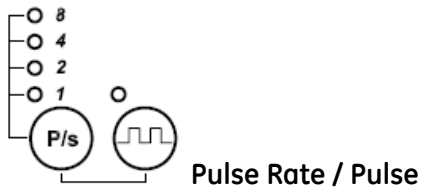
Manual mode can be selected directly by pressing “k p” or “mA/mAs” key during auto mode.



Auto Technique

Pressing the “Auto Technique” key enables the system to produce an optimum image by adjusting the technique parameter (kVp, and mA) automatically. When selected, the LED will illuminate indicating Auto.

Deselecting auto will extinguish the LED and the kVp and mA value can be adjusted manually.



Pulse Rate / Pulse



Press "Pulse" key  to enable pulsed imaging, press "Pulse Rate" key  to select pulse rate. Press the key until the pulse rate you desire is selected. The illuminated LED indicates which pulse rate is selected. Pulsed X-Rays can be used to reduce total radiation dose.



Using a pulsed fluoro mode can significantly reduce the dose received by the patient. However, image quality may be degraded in pulsed fluoro modes.



Low Dose

Press the "Low Dose" key to select the Low Dose mode. The corresponding LED will illuminate to indicate that Low Dose mode is selected.

Select Low Dose mode to reduce the exposure dose rate to the patient.



High Level Fluoro Mode& Digital Spot Mode Enable/Disable

This key is used to enable/disable HLF and Digital Spot mode. It's only effective in Australia and other countries when the laws and regulations require the use of this feature.

When pressing this key in these countries, the corresponding LED will illuminate and the High Level Fluoro Mode, Digital Spot Mode could be enabled by pressing the right foot/handswitch. Press this key again to disable the *High level Fluoro Mode Digital Spot Mode*. Make sure the LED lights before pressing the right foot/handswitch to make exposure.

In the other countries, HLF and digital spot mode are always enabled, invalid key sound will be given when pressing this key.

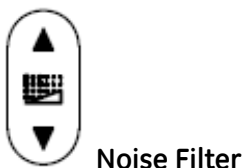


High level Fluoro Mode and Digital Spot Mode are both High Dose Fluoro imaging which can subject the patient and those working around the X-Ray field to a significantly larger dose of radiation than they would receive during normal fluoro operation. To minimize X-Ray exposure hazards, use High level Fluoro imaging with discretion.



Press the “Film” key to enable film operation. The film LED illuminates when film mode is enabled. In this mode, kVp and mAs values can be adjusted manually. Refer to Chapter 7 “Radiographic Film” for details.

Image Processing Keys



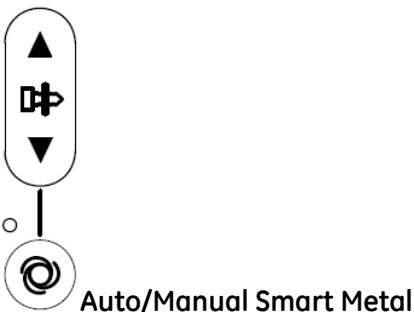
Press the “Noise Filter” key to select relative noise filter. Noise Filter is only available when real time imaging and the image was LIH.

Recursive Filter function provides image noise reduction. Higher levels of recursive filtering produce smoother (or less noisy) image, but cause more lag when imaging moving objects or anatomy.



Press the “Auto/Manual Brightness/Contrast” key to adjust the brightness and contrast level display.

The current contrast or brightness level is shown in a numeric value displayed on the bottom right corner of the image.



Press the “Auto/Manual Smart Metal” key to reduce the appearance of metal in the image. It could provide optimum image quality even when metal is introduced to the field. The current smart metal level is shown in a numeric value displayed on the bottom right corner of the image. When there is no metal in the field, please make the smart metal value as 0 to disable it.

Workstation Operations



Save

Press the “Save” key to save the left monitor image on the system disk.

After you have completed the exposure, press the “Save” key to save the last image displayed on the left monitor to the Workstation hard-drive.



Swap

Press the “Swap” key to exchange images between the left and right monitors. Swap will copy the image to the right monitor in the following cases: 1) only one image on the left monitor and there’s no image on the right monitor; 2) you are taking exposure.



Mode Switch

The “Mode Switch” button is not available for this system. Invalid key sound will be given when pressing this key.

4.3.2. Footswitch and Handswitch

The X-Ray foot/handswitch offer maximum flexibility to control X-Rays generation. Switches may be used to generate X-Rays or save images.

WARNING

Before Exposure, please check and confirm the mode you selected to avoid unexpected X-Ray Exposure.

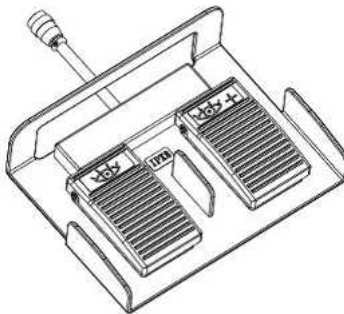
CAUTION

Taking very rapid and short exposures repeatedly over an extended period of time may cause errors to occur. Avoid continuous tapping on the X-Ray switch over long periods of time.

Footswitch

There are 2 kinds of two-pedal footswitch. The footswitch available is determined by which you purchased.

Two-pedal Footswitch



Press the left switch labeled  to produce standard fluoroscopic images.

Press the right switch labeled  to produce high level fluoro or digital spot images.

Multifunctional Two-pedal Footswitch



Press the left switch labeled  to produce standard fluoroscopic images.

Press the right switch labeled  to produce high level fluoro or digital spot images.

Press the save button labeled  to save the image displayed on the left monitor.

Handswitch



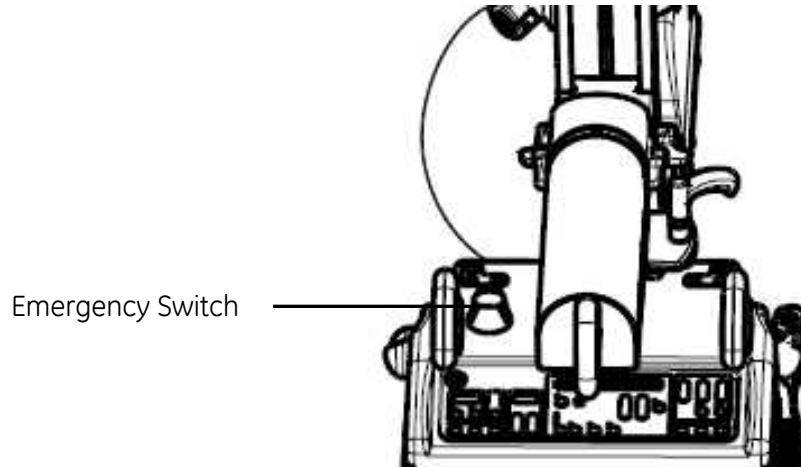
Press the left black switch labeled  to produce standard fluoroscopic images.

Press the right yellow switch labeled  to produce high level fluoro or digital spot images.

Press any switch to generate Radiographic Film.

4.3.3. Emergency Switch

The emergency switch located on the top of C-Arm control panel. When emergency situation occurs, press the switch to stop vertical column mechanical movement and disable X-Rays.



WARNING

If pressing an emergency switch fails to stop motor movement or X-Rays, press the power off switch or disconnect the power plug from the AC receptacle.

If the emergency switch is pressed on purpose or inadvertently, vertical column operation will cease and X-Rays will be disabled. Rotate the Emergency Switch in the direction of the arrow indication to release it and continue to use the system.



Should the system unexpectedly become unresponsive or lockup, restart the system to return it to normal operating condition. If the system fails to respond or continues to lockup, call your service representative.

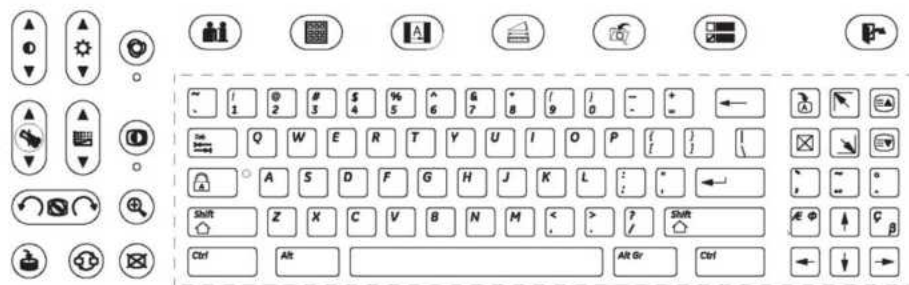
4.4. Workstation Controls

This section describes how to use the Workstation controls for imaging and post-processing. Before using the Workstation, you should become familiar with the controls located on the Workstation membrane keyboard.

The power supply to the monitor will be powered on or off along with system power.

4.4.1. Workstation Membrane Keyboard

The Workstation membrane keyboard controls include function keys, image processing and navigation, text-entry.



Function Keys



Exam Management

Press the “Exam Management” key to display the Exam Management screen. This screen allows you to enter patient information. Refer to Chapter 5.1 “Exam Management” for details.



Image Directory

Press the “Image Directory” key to display the Image Directory screen. This screen allows you to review, print, and archive stored images and dose information. Refer to Chapter 5.3 Image Review, Print and Archive for details.



Annotation

Press the “Annotation” key to display the Annotation screen. This screen allows you to place markers, add comments and crop images. Refer to Chapter 6.1 “Image Annotation” for details.



Measurement

Press the “Measurement” key to display the Measurement screen. This screen allows you to perform distance measurement and angle measurement. Refer to Chapter 6.2 “Image Measurement” for details.



Query & Retrieve

Press the “Query & Retrieve” key to display the Query & Retrieve screen. This screen allows you to query remote server and retrieve images from Query & Retrieve Server. Refer to Chapter 6.3.7 “Query and Retrieve” for details.



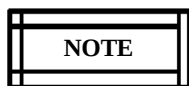
Setup

Press the “Setup” key to display the Setup screen. This screen allows you to configure system setup options. Refer to Chapter 8 “System Setup” for details.



Logout

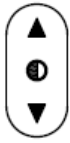
Pressing the “Logout” key allows the user to lock up the system.



The logout function can only be available when you select “Require Users to login” in the Security screen. Refer to Chapter 8.4 “Security”.

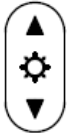
Image Processing Keys

Image processing keys allow you to adjust images displayed on the left monitor.



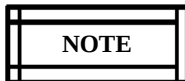
Contrast

Press the “Contrast” key to adjust the contrast level display.

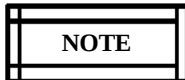


Brightness

Press the “Brightness” key to adjust the brightness level display.



The settings can be applied to static and live images. Values achieved in the last image can be retained. And the values can be saved and recalled with static image.



The contrast/brightness adjust function will only take effect on the image. It can't affect the hardware setting of the monitor.



Auto Brightness/Contrast

Press the “Auto Brightness/Contrast” key to enable the system to automatically select the optimum amount of contrast and brightness, and the Auto LED below this key is lighted.

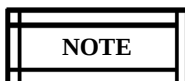


Edge Enhancement

Press the “Edge Enhancement” key to change the current level of edge enhancement. The enhance level is shown on the edge enhancement indicator bar displayed on the left monitor.

Changes to enhancement are indicated by a corresponding rise or fall in the shaded portion of the indicator bar. The indicator bar disappears two seconds after the “EDGE ENHANCE” button is released.

Pressing and holding the key, or repeated pressing, changes the value.



This function can be applied to live and post-processed images. Values achieved in the last image can be retained. And the values can be saved and recalled with static image.



Noise Filter

Press the “Noise Filter” key to select relative noise filter. Noise Filter is only available when real time imaging and the image was LIH.

Recursive Filter function provides image noise reduction. Higher levels of recursive filtering produce smoother (or less noisy) image, but cause more lag when imaging moving objects or anatomy.



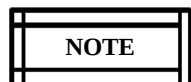
Negate

Press the “Negate” key to activate the negate function, and the Negate LED below this key is lighted.

Negate is applied to the current image on the left monitor and to all subsequent images produced until negate is disabled. The setting of negate can be saved and recalled with the image.

Until disabled, each time Fluoro is pressed on the footswitch or handswitch, the image will be negated.

To disable negate, press the “Negate” key again.



Negate can be applied to static and live images.



Image Rotation

Press the left portion of the key to rotate the image counter-clockwise or press the right portion of the key to rotate the image clockwise.



Zoom & Roam

Press the “Zoom & Roam” key to activate the Zoom & Roam function.

The left monitor image is copied to the right monitor and the left image is displayed by default at one time the magnification.

Select the “2X” button to increase magnification to two times.

Select the “4X” button to increase magnification to four times.

A square box representing the region of interest is displayed on the right monitor. Drag the box to move the region of interest.

Select the “Exit” button to close the Zoom & Roam screen.

**Save**

Press the “Save” key to save the left monitor image on the system disk.

After you have completed the exposure, press the “Save” key to save the last image displayed on the left monitor to the Workstation hard-drive.

**Swap**

Press the “Swap” key to exchange images between the left and right monitors. Swap will copy the image to the right monitor in the following cases: 1) only one image on the left monitor and there’s no image on the right monitor, 2) you are taking exposure.

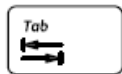
**Blank**

Press the “Blank” key to blank the screen quickly if the screens are displaying information which cannot be seen by someone else. Press any key on the membrane keyboard to re-display the screens’ images.

Navigation and Text-Entry

Workstation navigation keys, text entry and text modification keys allow you to move the cursor and edit text.

Navigation Keys

**Tab**

Move the cursor to the next text-entry field.

**Home****Page up****Page down****End**

Text Entry Keys

The text-entry keys allow you to enter and edit text.



Backspace

Delete one character to the left of the cursor. Also deletes a selected marker or comment.



Caps Lock

The Caps Lock key is used to toggle between all upper case letters and mixed case letters. When Caps Lock is engaged, the Caps Lock LED in the upper right corner of this key is lighted and the membrane keyboard types only upper case letters. When Caps Lock is disengaged, the membrane keyboard types mixed case letters.



Insert



Delete



Arrows

Diacritic Keys



Diacritic keys can be used in any language.

Press this key for marks on the lower left corner of the keys.

Press the Alt Gr+ Diacritic key to access marks in the lower right case position on the keys.

Press shift+ Diacritic key to access marks in the upper left case position on the keys.

Press Alt Gr+ shift+ Diacritic key for marks on the upper right corner of the keys.

4.4.2. Touch Screen

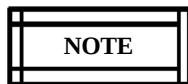
Brivo OEC 865 provides a touch screen on the right monitor for interacting with the Workstation. Using the touch screen and the keyboard for text-entry you can:

- Enter and view patient information
- View and process images
- Configure the X-Ray switches
- Annotate images

To select a touch screen button, or a field for text-entry, firmly touch the desired object with your finger.

4.5. Image Quality Control

To get optimal image quality and X-Ray dose, ensure the use of default setting by turning on Auto Contrast/Brightness key and Auto Technique key on C-Arm control panel. In addition, it is generally advised to position the patient close to the Image Intensifier and to center the region of interest in the imaging field. To get a stable image, please press and hold the foot/handswitch at least 2 seconds during exposure. In case the air covers substantial portion of the image, use the shutter collimator and iris collimator to cover that air portion to minimize the light scatter.



When anatomical thickness changes in region of interest, to get a better image, please stop X-Ray and center the region of interest in the imaging field, select auto mode and then restart the exposure.

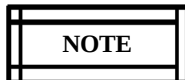
Refer to the table below for common image quality issues from less than optimal settings, and the checklists to resolve those issues.

Item	Issue	Check	Action
1	Image looks too bright	Is "Auto Technique" key turned off	Turn on "Auto Technique" key.
		Is "Auto Brightness/Contrast" key turned off	Turn on "Auto Brightness/Contrast" key.
		Large metal in the field	Turn on "Auto Smart Metal" key and fine adjust auto smart metal level.
2	Image looks too dark	Is "Auto Technique" key turned off	Turn on "Auto Technique" key.

		Is "Auto Brightness/Contrast" key turned off	Turn on "Auto Brightness/Contrast" key.
		When the automatic function is selected in Normal Fluoro mode, does technique reach maximum limitation (110kVp)	Switch to HLF mode or Digital Spot mode.
		Does air cover large portion in the image	Re-position to decrease the portion of air in the image, or use collimator leaf or Iris to cover the portion of air.
		Is the smart metal level too high	Change the smart metal level manually. When there is no metal in the field, adjust the smart metal level to "0".
3	Image looks just grayish (Too small contrast)	Is "Auto Brightness/Contrast" key turned off	Turn on "Auto Brightness/Contrast" key or manually adjust the contrast level.
4	Image looks too black-and-white (Too high contrast)	Is "Auto Brightness/Contrast" key turned off	Turn on "Auto Brightness/Contrast" key or manually adjust the contrast level.
		Does air cover large portion in the image	Re-position to decrease the portion of air in the image, or use collimator leaf or Iris to cover the portion of air.
		Is the smart metal level too high	Change the smart metal level manually. When there is no metal in the field, adjust the smart metal level to "0".
5	Image looks too noisy	Is the image acquired in Pulsed mode	Switch to continuous mode or prolong exposure time.
		Is the image acquired in Low Dose mode	Switch to continuous mode.
		Is the "Noise Filter" set too low	Increase the noise filter setting.
		Is the "Edge Enhancement" level set too high	Decrease edge enhancement level setting.
6	Image looks blurry (details are not clear)	Patient or C-Arm was moving during the acquisition	Decrease the noise filter setting.

		Is the "Edge Enhancement" level set too low	Increase edge enhancement level setting.
		Is the Image Intensifier far away from the patient	Re-position the Image Intensifier closer to patient.
		Is the region of interest not at the center of the image	Re-position the region of interest to the center of the image field.
		Is the "Field Size" set to NORM	Switch to MAG1/MAG2 mode.
7	Small field of view (images look too big)	Is the "Field Size" set to MAG1/MAG2 mode	Switch to NORM field size.
		Is the Image Intensifier far away from the patient	Re-position the Image Intensifier closer to Patient.

Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product also provides the feature of Auto-Adaptive Non-uniformity Correction to improve the uniformity of the image. Because of the curved nature of the image intensifier tube, less X-Ray energy is received at the periphery of the tube than at the center. This causes the image to appear darker around the image edge. Auto-Adaptive Non-uniformity Correction is used to correct it and improve the uniformity of the image.



Correction template can be created automatically in service mode by service engineer, and then the images in live mode can be corrected with the template in real time. This feature is not user selectable.

Chapter5. Usual Operations

5.1. Exam Management

Use Exam Management screen to:

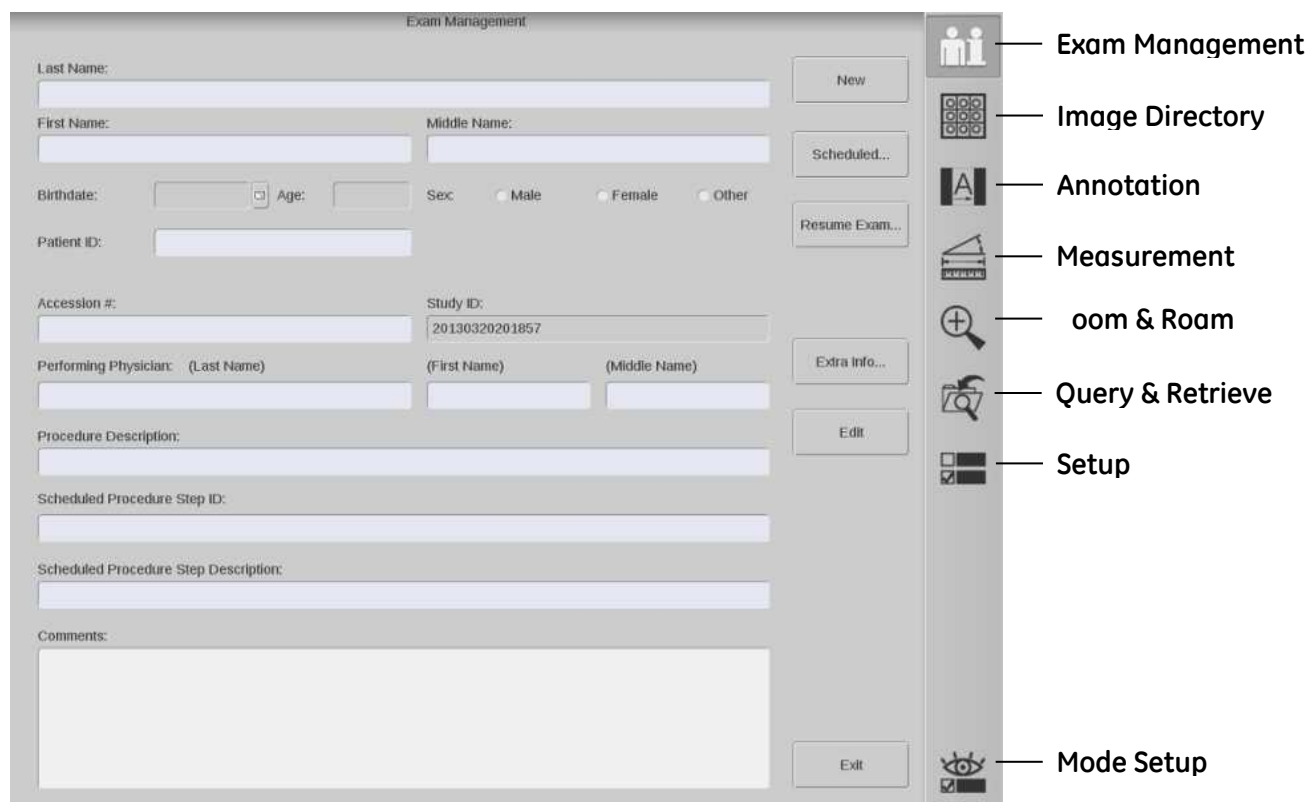
- Create a new exam.
- Display Scheduled Exams screen.
- Display Performed Exams screen.
- Edit patient information.

WARNING

Be cautious while the system is in use. Any sudden, intense impact can damage the system or could result in loss of images and prevent further imaging.

5.1.1. Create a New Exam

The Exam Management screen of a new exam is displayed when a user login. The default Admin login password is "123456".



The screenshot shows the 'Exam Management' application window. The main area contains a form for creating a new exam with the following fields:

- Last Name: [Text Field]
- First Name: [Text Field]
- Middle Name: [Text Field]
- Birthdate: [Text Field] / Age: [Text Field]
- Sex: ☐ Male ☐ Female ☐ Other
- Patient ID: [Text Field]
- Accession #: [Text Field]
- Study ID: [Text Field] (Value: 20130320201857)
- Performing Physician: (Last Name) [Text Field] (First Name) [Text Field] (Middle Name) [Text Field]
- Procedure Description: [Text Field]
- Scheduled Procedure Step ID: [Text Field]
- Scheduled Procedure Step Description: [Text Field]
- Comments: [Text Area]

Buttons on the right side of the form include: New, Scheduled..., Resume Exam..., Extra Info..., Edit, and Exit.

A vertical sidebar on the right contains icons and labels for the following functions:

- Exam Management (Icon: Two people)
- Image Directory (Icon: Grid of squares)
- Annotation (Icon: 'A' in a box)
- Measurement (Icon: Ruler)
- Zoom & Roam (Icon: Magnifying glass)
- Query & Retrieve (Icon: Camera)
- Setup (Icon: Checkmark in a box)
- Mode Setup (Icon: Eye with checkmark)



Press the “Exam Management” key on the Workstation membrane keyboard or select the “Exam Management” button on the system panel to display Exam Management screen in use. Click “New” to create a new exam.

When begin a new exam, enter patient information before a procedure.

- Using the Workstation membrane keyboard, enter patient information. A cursor indicates which field you are in. Use Tab key to move the cursor to the next active field.
- Fill in all the fields to complete patient information. If user starts a new exam without input any patient information, the current patient name will be assigned by system as “Unnamed xxxxx”.
- Click the “Extra Info...” button on the Exam Management screen to display the Additional Information screen. You can add Weight, Height, Referring Physician, Requested Procedure ID, Requested Procedure Description, Protocol Description and Anatomic Regions. Click “OK” to save the additional information and return to Exam Management screen.

Once you have entered patient information, you may immediately begin taking X-Rays for that patient or select “Exit” or switch to other application. The Exam Management screen will close and display the desktop. The input information will be automatically saved.

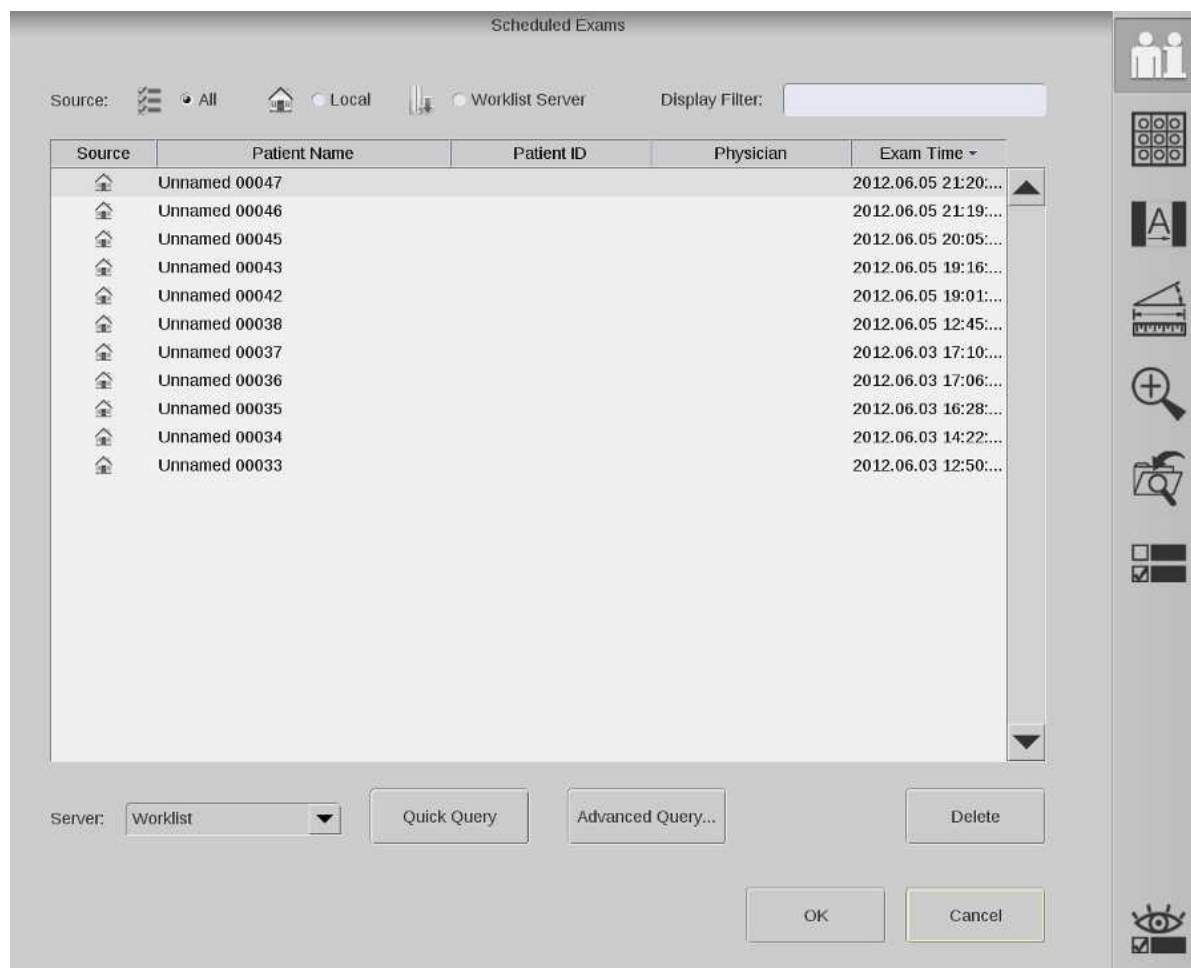
Each time you choose a new exam, resume a performed exam or start a scheduled exam, the C-Arm setting is reset to default parameters. This includes fluoro timer, exposure mode settings, mag mode, image rotation, Image reversal, contrast/brightness, recursive factor, smart metal, negate and edge enhancement. Exposure parameters will be reset to 70kV, 1.4mA. The collimator shutter and iris will adjust to a full open position.

You can set collimator, mag mode, image rotation and flip/flop to retain last value in System Setup screen.

5.1.2. Scheduled Exams Feature

The scheduled exams feature allows you to query the exams from worklist server, view local created and queried scheduled exams.

Select “Scheduled Exams” from the Exam Management screen. Scheduled Exams screen is displayed.



The Scheduled Exams screen displays a list of scheduled exams by Exam Time at descending order. To list by Source, Patient Name, Patient ID, Physician, click the related column header. User can choose to display Patient ID or Accession in System Setup.

Select a server from the server dropdown list. Click "Quick Query" to display a list of scheduled exams queried from the server. Select the "Advanced Query" to set advanced query criteria.

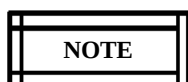
Enter search criteria in the "Display Filter" text field to filter exams listed in the Scheduled Exams.

Detailed information of DICOM Worklist Query, please refer to section 6.3 "DICOM".

Click "Delete" to delete the selected schedules exam. You may select multi exams from the list by using "Ctrl" key.

Select the "OK" button to close the Scheduled Exams screen and display the Exam Management screen for the selected exam. You may begin taking exams for the selected exam.

Select the "Cancel" button to return to Exam Management screen for original patient.

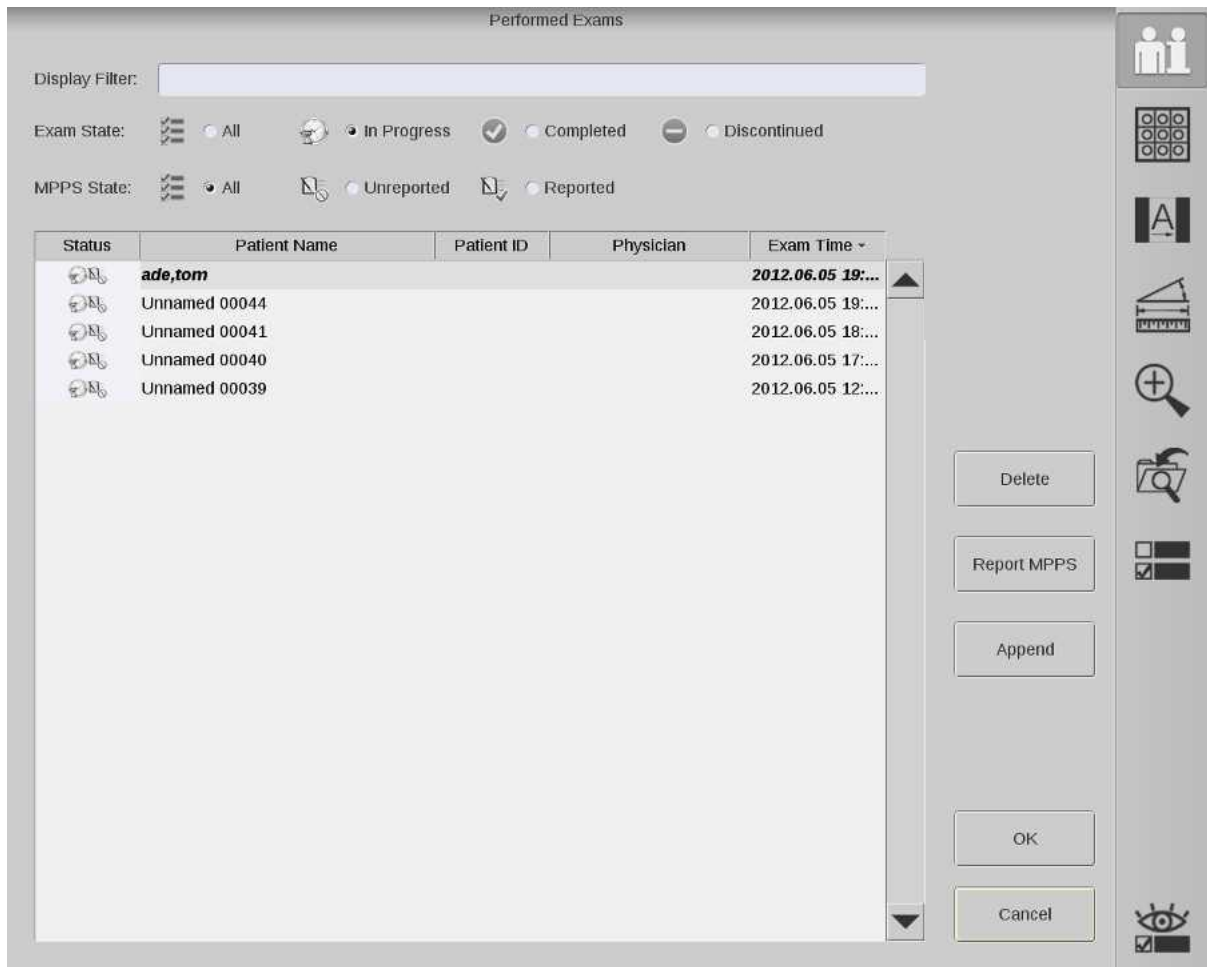


Always ensure the correct patient name is displayed on the Exam management screen.

5.1.3. Resume a Performed Exam

The Performed Exams screen contains a list of previously performed exams.

Select the “Resume Exam” button from the Exam Management screen. A Performed Exams Screen is displayed.



The Performed Exams list displays exams by Exam Time at descending order. To list by Status, Patient Name, Patient ID, Physician, click the related column header. User can choose to display patient ID or Accession in System Setup screen.

Enter retrieve criteria in the “Display Filter”, a list of exams that meet the retrieve criteria are displayed.

Select the option of Exam state: All/ In-Progress/ Completed/ Discontinued, the exams are filtered by the Exam state.

Select the option of MPPS state: All/ Unreported/ Reported, the exams are filtered by the MPPS state.

Selects one of “In-Progress” exams and click “OK” button or double click the selected exam to display the selected exam. The selected exam is set as the current exam.

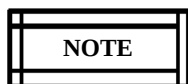
Click “Delete” to delete the selected performed exam.

Click Report MPPS to report current selected exams' MPPS state to configured MPPS server.

Click “Append” to create a new scheduled exam based on the selected exams. Only the completed and discontinued exams could be appended for further X-Ray exposures.

Select the “OK” button to close the Performed Exams screen and display the Exam Management screen for the selected exam.

Select the “Cancel” button to return to Exam Management screen for original patient.



The completed and discontinued exams cannot be selected to resume. An information dialog will pop up then.

5.1.4. Edit Patient Information

To edit patient information on a previously saved record in Scheduled Exams or Performed Exams, recall the exam and follow the instructions for editing information.

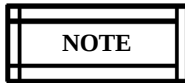
Select the “Edit” button on the Exam Management screen, you can edit the displayed patient's information.

The screenshot displays the 'Exam Management' window. It features a main form area with various input fields and a vertical sidebar on the right containing several icons. The form fields include:

- Last Name:** A text box containing 'Unnamed 00083'.
- First Name:** An empty text box.
- Middle Name:** An empty text box.
- Birthdate:** A date picker.
- Age:** A text box.
- Sex:** Radio buttons for 'Male', 'Female', and 'Other'.
- Patient ID:** An empty text box.
- Accession #:** An empty text box.
- Study ID:** A text box containing '20130320201857'.
- Performing Physician:** Three text boxes labeled '(Last Name)', '(First Name)', and '(Middle Name)'.
- Procedure Description:** A large text area.
- Scheduled Procedure Step ID:** An empty text box.
- Scheduled Procedure Step Description:** A large text area.
- Comments:** A large text area.

On the right side of the form, there are several buttons: 'New', 'Scheduled...', 'Resume Exam...', 'Extra Info...', 'Edit', and 'Exit'. The sidebar on the right contains icons for patient management, exam scheduling, and other functions.

Click “Exit” button to save and return to the desktop.



User cannot edit patient information under following scenarios:

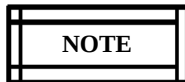
1. *Exams queried from MW server.*
2. *Image(s) of the exam has/have been sent to server.*
3. *MPPS has been reported to server.*
4. *Exam is completed or discontinued.*
5. *Exam is appended from another exam .*

5.2. Imaging Mode

This section describes the basic imaging features. Features related with Radiographic (Film) will be described in chapter 7.



Establish emergency procedures for handling the patient in case of the loss of imaging or other Workstation or C-Arm functions during an exam.

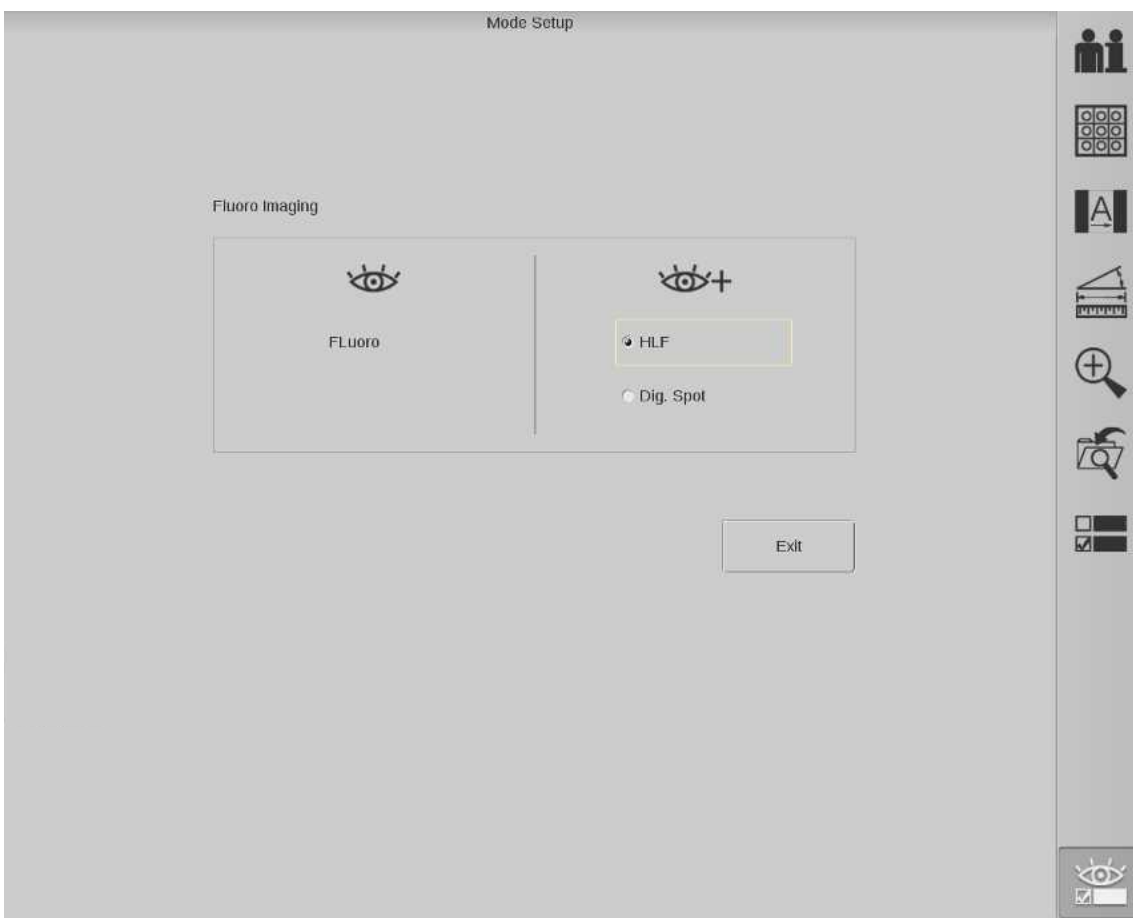


In Digital Spot and Radiographic (Film) mode, the image display delay exists. If you need obtain real-time image, please use the fluoro imaging mode.

Mode Setup Screen



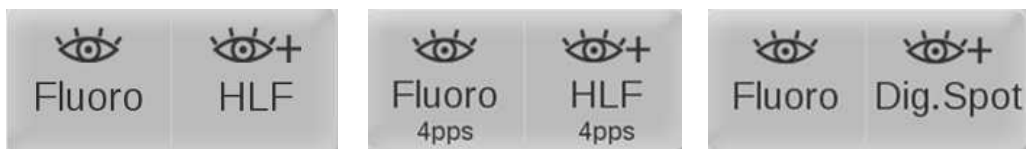
Select "Mode Setup" from the System Panel. Mode Setup screen is displayed on the right monitor. It is provided to enable you to choose between HLF (High Level Fluoro) and Digital Spot mode.



X-Ray Switch Mode Status Bar

During an exam, the configured active mode pair is displayed on the status bar. The status bar is located on the bottom left corner monitor when an image is displayed.

Left half part displays image mode triggered by left switch  on foot/handswitch, right half part displays image mode triggered by right switch  on foot/handswitch.



If the film mode is selected, Film is displayed in the mode status bar.



5.2.1. Normal Fluoro Imaging

Normal fluoro mode displays live fluoroscopic images on the left monitor. After the image has been acquired, image attributes such as edge enhancement or contrast and brightness can be modified.

To produce a normal fluoro image

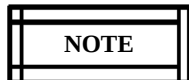
1. Verify Fluoro is displayed on the status bar.
2. Select pulsed X-Ray or low dose if desired. Use the C-Arm Pulse key  to switch between continuous and pulsed X-Rays. Use the C-Arm Low Dose key to switch between continuous and low dose imaging.
3. Press the left X-Ray switch  on the foot/handswitch to take exposure.

5.2.2. High Level Fluoro Imaging

High Level Fluoro (HLF) mode is used for getting high quality images. High Level Fluoro mode decreases noise level and improves image quality by increasing fluoroscopic mA.



High level Fluoro Imaging can subject the patient and those working around the X-Ray field to a significantly larger dose of radiation than they would receive during normal fluoro operation. To minimize X-Ray exposure hazards, use High level Fluoro Mode with discretion.



For each H F shot, please limit the X-Ray time within 20". Otherwise, the system will terminate X-Ray automatically. If you need more H F time, please release the foot/handswitch and re-press it.

To produce a High Level Fluoro image

1. Verify HLF is displayed on the status bar. If Dig. Spot is displayed, use the Mode Setup screen to select HLF.
2. Select pulsed X-Ray generation on C-Arm control panel, if desired. If pulsed operation is chosen, the pulse rate is displayed on the status bar. Use the C-Arm Low Dose key to switch between continuous and low dose imaging.
3. Press the right X-Ray switch  on the foot/handswitch to make High Level Fluoro exposures.

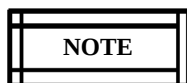
5.2.3. Digital Spot Imaging

Digital Spot mode creates a short-duration, high mA exposure to produce a high quality single image.

After the image has been acquired, image attributes such as edge enhancement or contrast/brightness can be modified. Pulsed operation is not allowed in digital spot mode.

To produce a Digital Spot image

1. Select the Digital Spot from the Mode Setup screen. Verify Dig. Spot is displayed on the status bar.
2. Press the left X-Ray switch () on the foot/handswitch to confirm positioning and obtain appropriate exposure parameters.
3. When the desired image appears on the monitor, release the left X-Ray switch () and then press and hold the right X-Ray switch () on the foot/handswitch to obtain the digital spot image.
4. The system automatically terminates the exposure and saves the Digital Spot image on the hard disk.
5. If the image positioning and exposure parameters have been set, you can press the right X-Ray switch () on the foot/handswitch to obtain the digital spot image.



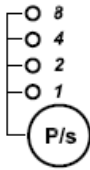
Continue pressing the switch until the system automatically terminates the exposure. If you release the X-Ray switch early, the exposure will prematurely terminate.

5.2.4. Pulsed Fluoro Imaging

This section describes how to switch between continuous and pulsed imaging.

Pulsed mode generates a preset number of X-Ray pulses each second while the X-Ray switch is pressed. Pulsed X-Rays can be used to reduce total radiation.

To produce a pulsed fluoro image

1. Press the "Pulse" key  on the C-Arm control panel to enable pulsed operation. Press "Pulse Rate"  key to select pulse rate. The default pulse rate is 4 PPS. The selected pulse rate displays automatically on the status bar.
2. Press the related X-Ray switch on the foot/handswitch to make Pulsed Fluoro exposures.

3. To reselect continuous operation, press the “Pulse” key  again. The pulse rates will no longer display on the status bar.

5.2.5. Low Dose Imaging

Low dose imaging is to reduce the exposure dose to the patient.

To produce a Low Dose image

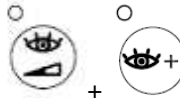
1. Press the “Low Dose” key  on the C-Arm control panel to enable low dose imaging.
2. Press the related X-Ray switch on the foot/handswitch to make low dose fluoro exposures.

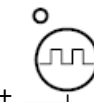
5.2.6. Imaging Mode List

The following list summarizes the modes available on the system. Additionally it shows the X-Ray techniques available with each mode, the status bar and C-Arm control panel display for each mode, and a brief list of examples for each mode. The examples are not exhaustive and other uses may be as valid as the examples shown. The user must be aware of the techniques associated with the various modes and monitor dose rates and doses for each patient.

- Auto: X-ray techniques (kVp/mA) are adjusted automatically by system to optimize the image.
- Manual: X-ray techniques (kVp/mA) are adjusted manually by user to get the desired image.

The user can select Auto or Manual mode listed in the table below as she/he wants.

Mode	X-Ray Techniques	Display	Imaging Task
Normal Fluoro Auto/ Manual	kVp:40 to 110 mA: 0.1 to 4		General Fluoro Imaging
High Level Fluoro Auto/ Manual	kVp:40 to 110 mA: 0.2 to 12		High quality but high-dose Imaging
Low Dose Auto/ Manual	kVp:40 to 110 mA: 0.1 to 2		Continuous fluoro imaging with a lower dose
Low Dose High Level Auto/ Manual	kVp:40 to 110 mA: 0.1 to 6		High quality with a lower dose

Pulsed Normal Fluoro Auto/ Manual	kVp:40 to 110 mA: 0.1 to 4	   	Reduce X-Ray dose to patient
Pulsed High Level Fluoro Auto/ Manual	kVp:40 to 110 mA: 0.2 to 12	   	High quality and low-dose Imaging
Pulsed Low Dose Auto/ Manual	kVp:40 to 110 mA: 0.1 to 2	 	Reduce X-Ray dose to patient
Pulsed Low Dose High Level Auto/ Manual	kVp:40 to 110 mA: 0.1 to 6	  	High quality and low-dose Imaging
Digital Spot Auto	kVp:40 to 110 0.2 to 16 mA 200V, 220V, 230V, 240V; 0.2 to 10 mA 100V, 110V, 120V.		Short duration, high mA and high quality single imaging
Low Dose Digital Spot Auto	kVp:40 to 110 0.1 to 8 mA 200V, 220V, 230V, 240V; 0.1 to 5 mA 100V, 110V, 120V.	 	Short duration, high mA and high quality single imaging with a lower dose
Film Manual	kVp:40 to 110 20 mA 200V, 220V, 230V, 240V; 10 mA 100V, 110V, 120V.		Capture an X-Ray image on film

5.2.7. Modes of Operation Engaging and Disengaging

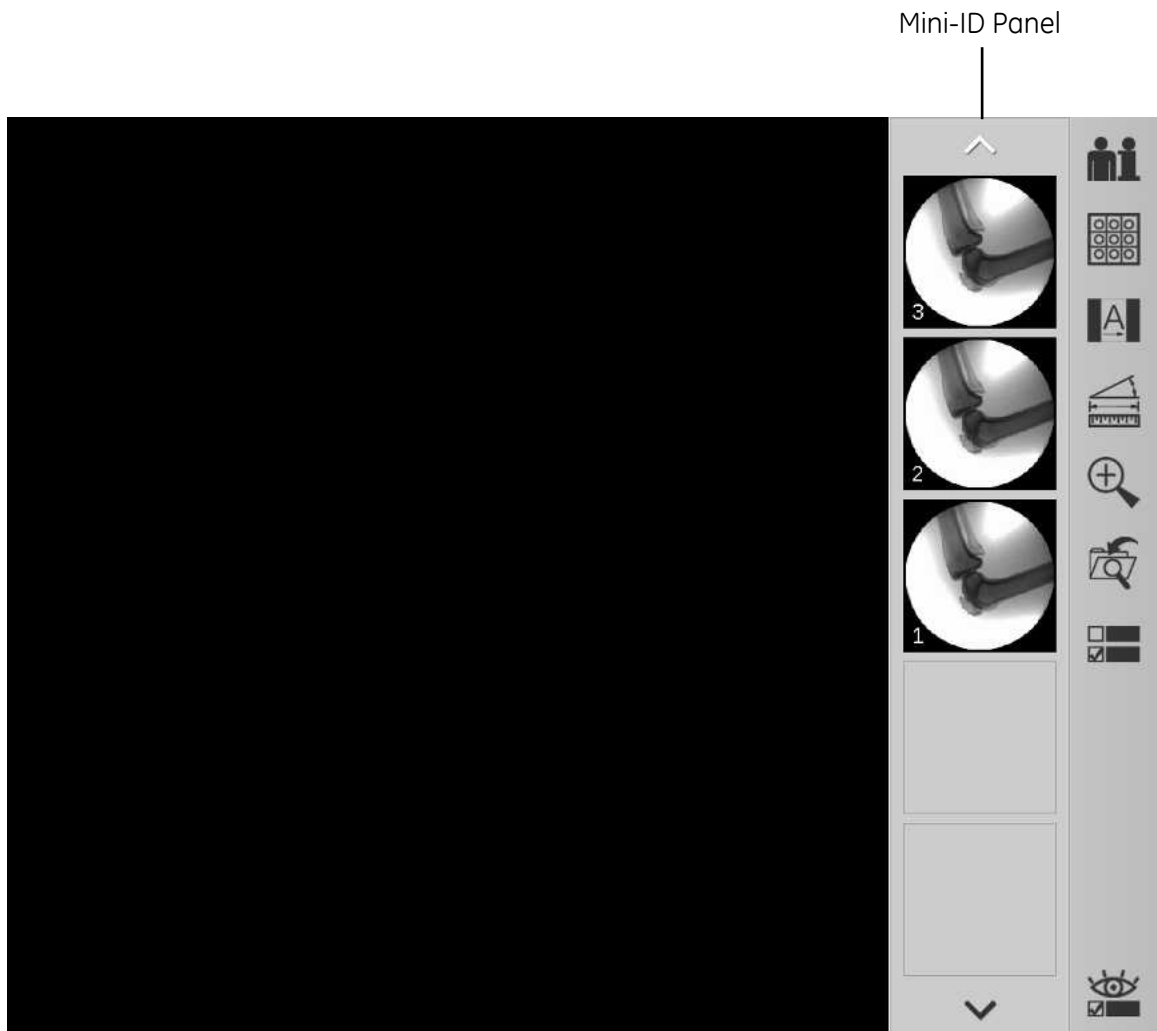
The fluoro modes used on the Workstation can be engaged in a variety of ways including mode keys on the C-Arm control panel, Workstation user interface, and the various footswitches and handswitches used by the systems. Additionally, all modes can be engaged, disengaged, and modified using the Mode Setup Screen displayed on the right monitor of the Workstation. The Mode Setup Screen, in turn, sets up the functions assigned to the various switches. The Mode Setup Screen is accessed by selecting the Mode Setup button on the system panel.

5.3. Image Review, Print and Archive

5.3.1. Review Images

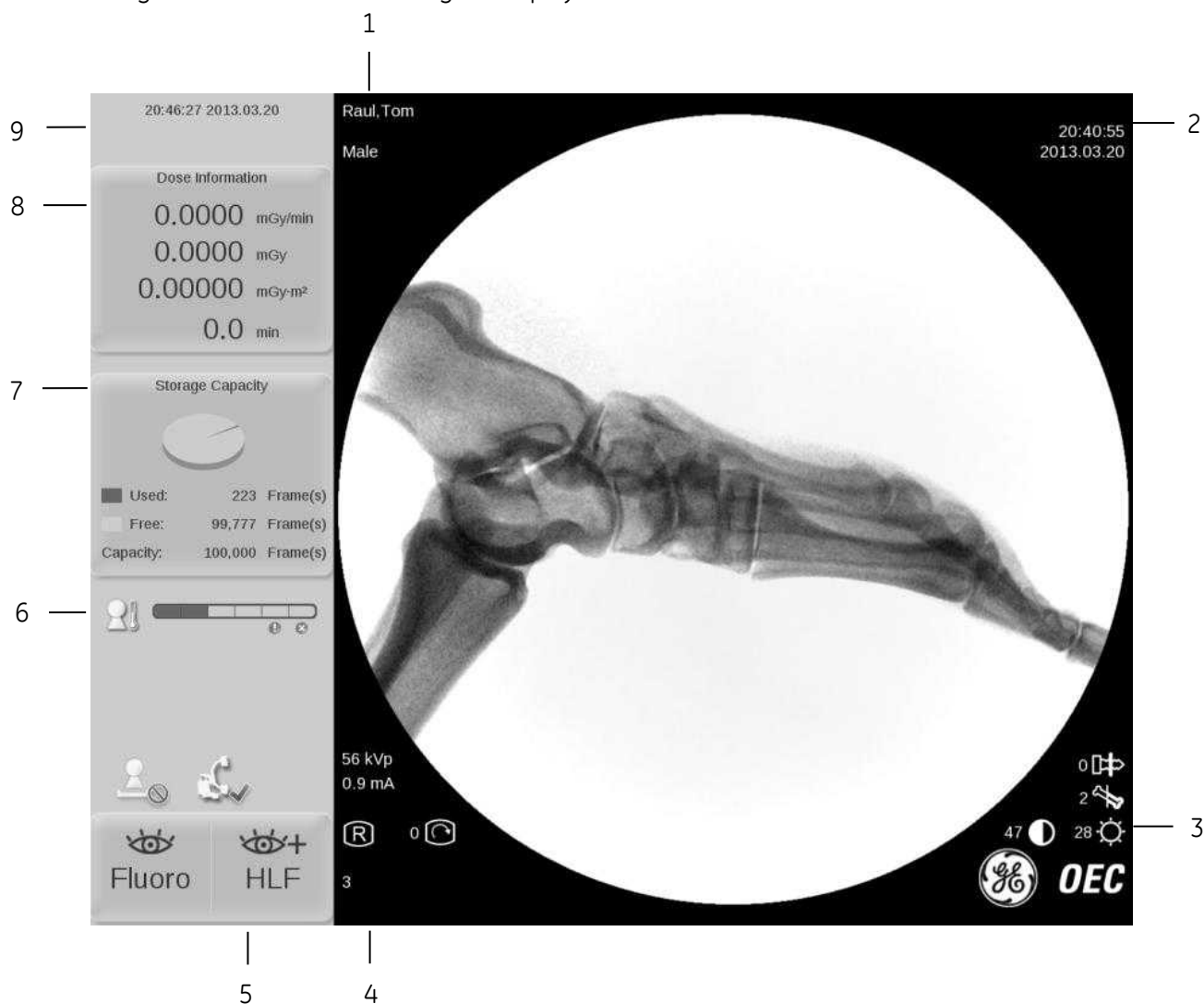
The desktop when exposing is displayed on the right monitor as below. You could review images of current exam by the Mini-ID Panel.

Select the Up/Down on system Mini-ID Panel to scroll previous/next previewed image. Select one previewed image to display the selected image on the left monitor.

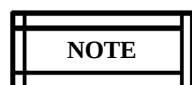


Usual Operations

The following information with the image is displayed on the left monitor:



1. Patient information, physician name
2. Time and hospital when generating the image
3. Image contrast value, brightness value, edge enhance value, smart metal value
4. Technical parameters used during the exposure, flip/flop indicator, rotation indicator and image number
5. X-Ray Switch Mode Status Bar
6. Tube temperature indicator
7. Storage Capacity



When beginning of each exam, please confirm there is enough image storage space. Regularly check the available storage space and archive important records to avoid image loss.

8. Dose information (Dose measured at 30cm above the Image Intensifier, center along the reference axis)

- Dose

Range: 0-80,000 mGy; Accuracy: +-35%

- Dose Rate

Range: 0-120 mGy/min; Accuracy: +-35%

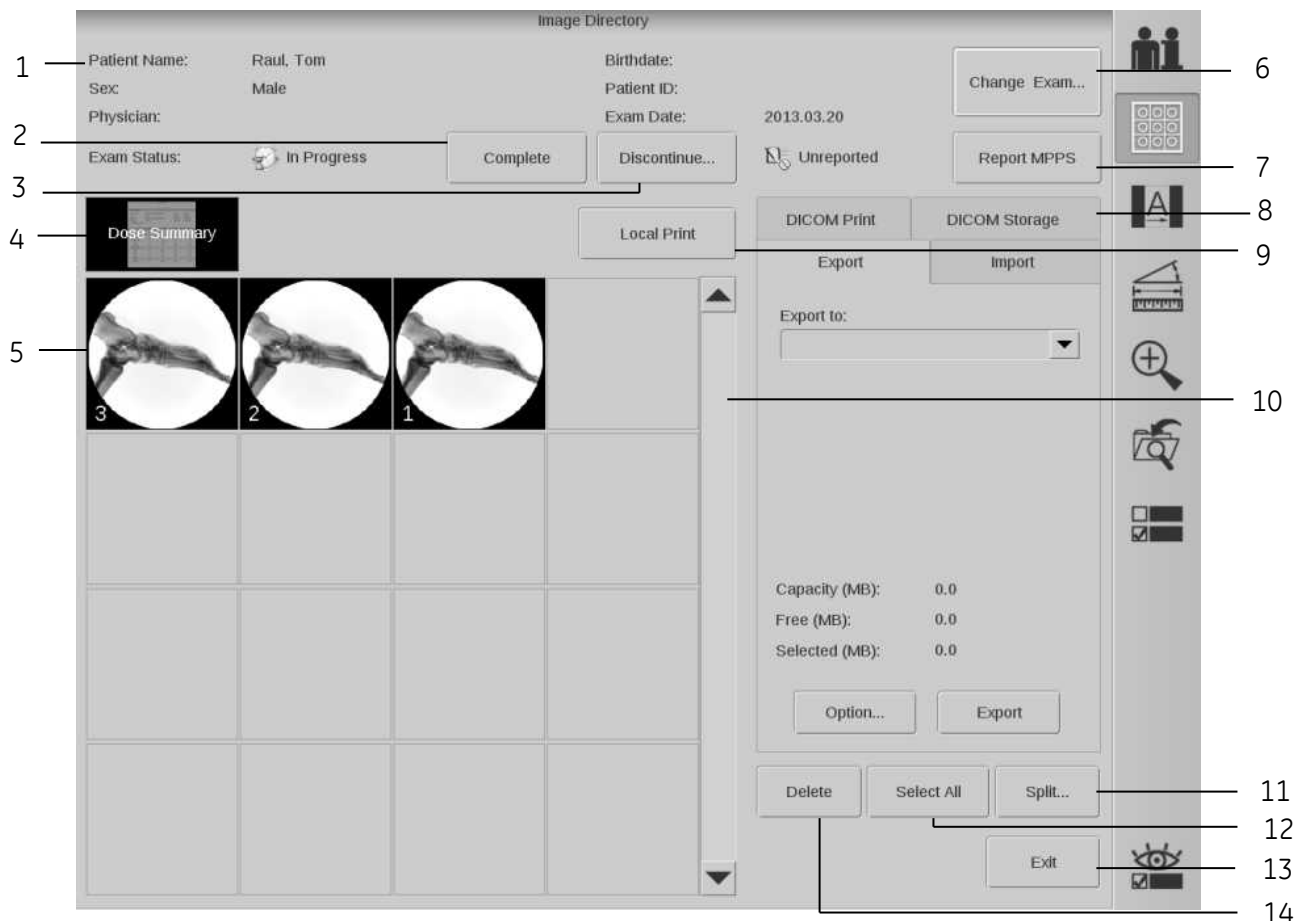
- DAP

Range: 0-20,000 Gy.cm2; Accuracy: +-35%

9. System Date & time, hospital

The Image Directory screen allows you to display saved static images for review and print, archive and review patient information and dose information.

To display the Image Directory screen on right monitor, press the "Image Directory" key  on the Workstation membrane keyboard or select the "Image Directory" button on the system panel.



1. **Patient Information** The patient information is displayed.
2. **Complete** Used to complete the selected exam.
3. **Discontinue...** Used to discontinue the selected exam.

4. **Dose Summary** Display the dose summary information.
5. **Preview Images** Display image thumbnails to select.
6. **Change Exams...:** Used for recalling images that have been previously saved in performed exam.
7. **Report MPPS** Used to report MPPS. This button is only displayed when you choose the MPPS option for some model.
8. **Export, Import, DICOM Print, DICOM Storage:** Used to export exams to media, import exams from media, DICOM print images and archive images to DICOM Storage Server.
9. **Local Print** Click to print locally.
10. **Scroll Bar** Used to scroll through all previewed images.
11. **Split...:** Used to move the selected images which are not archived to DICOM server to a new exam and remove them from the original exam.
12. **Select All** Used to select all images.
13. **Exit** Exits the Image Directory.
14. **Delete** Used to delete the selected images.

5.3.2. Review Dose Information

The Dose Summary screen contains X-Ray exposure data of the current exam.

Select the “Dose Summary” button from the Image Directory screen to display the Dose Summary on the left monitor.



- Dose Range: 0-80,000 mGy; Accuracy: +-35%
- DAPRange: 0-20,000 Gy.cm2; Accuracy: +-35%

Country/ Region	MODE (Dose measured at 30 cm above the Image Intensifier)					
	Fluoro kVp 40 to 110 mA 0.1 to 4 	Pulsed Fluoro kVp 40 to 110 mA 0.1 to 4 	High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Pulsed High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Digital Spot kVp 40 to 110 mA 0.2 to 16 	Film kVp 40 to 110 mA 20 mAs 1 to 80 
Algeria	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Argentina	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Australia	50 mGy/min	50mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Austria	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Bangladesh	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Belgium	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Bolivia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Bosnia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Brazil	50 mGy/min	50mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Brunei	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Bulgaria	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Cambodia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Canada	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Chile	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
China	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy

Country/ Region	MODE					
	(Dose measured at 30 cm above the Image Intensifier)					
	Fluoro kVp 40 to 110 mA 0.1 to 4 	Pulsed Fluoro kVp 40 to 110 mA 0.1 to 4 	High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Pulsed High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Digital Spot kVp 40 to 110 mA 0.2 to 16 	Film kVp 40 to 110 mA 20 mAs 1 to 80 
Colombia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Costa Rica	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Cyprus	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Czech Republic	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Denmark	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Dominican Republic	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Egypt	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Estonia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Finland	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
France	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Georgia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Germany	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Greece	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Guatemala	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Honduras	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Hongkong	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy

Country/ Region	MODE (Dose measured at 30 cm above the Image Intensifier)					
	Fluoro kVp 40 to 110 mA 0.1 to 4 	Pulsed Fluoro kVp 40 to 110 mA 0.1 to 4 	High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Pulsed High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Digital Spot kVp 40 to 110 mA 0.2 to 16 	Film kVp 40 to 110 mA 20 mAs 1 to 80 
/Macau						
Hungary	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Iceland	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
India	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Indonesia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Iraq	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Ireland	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Israel	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Italy	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Jamaica	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Japan	50mGy/min	50mGy/min	50mGy/min	50mGy/min	2mGy	20mGy
Jordan	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Kuwait	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Laos	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Latvia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Lebanon	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy

Country/ Region	MODE					
	(Dose measured at 30 cm above the Image Intensifier)					
	Fluoro kVp 40 to 110 mA 0.1 to 4 	Pulsed Fluoro kVp 40 to 110 mA 0.1 to 4 	High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Pulsed High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Digital Spot kVp 40 to 110 mA 0.2 to 16 	Film kVp 40 to 110 mA 20 mAs 1 to 80 
Libya	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Lithuania	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Liechtenstein	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Luxembourg	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Macedonia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Malaysia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Malta	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Mexico	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Montenegro	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Morocco	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
New Zealand	50 mGy/min	50 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Netherlands	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Norway	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Oman	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Pakistan	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Panama	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy

Country/ Region	MODE					
	(Dose measured at 30 cm above the Image Intensifier)					
	Fluoro kVp 40 to 110 mA 0.1 to 4 	Pulsed Fluoro kVp 40 to 110 mA 0.1 to 4 	High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Pulsed High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Digital Spot kVp 40 to 110 mA 0.2 to 16 	Film kVp 40 to 110 mA 20 mAs 1 to 80 
Paraguay	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Peru	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Philippines	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Poland	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Portugal	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Puerto Rico	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Romania	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Russia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Saudi Arabia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Singapore	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Slovenia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
South Africa	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
South Korea	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Spain	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Sweden	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Switzerland	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy

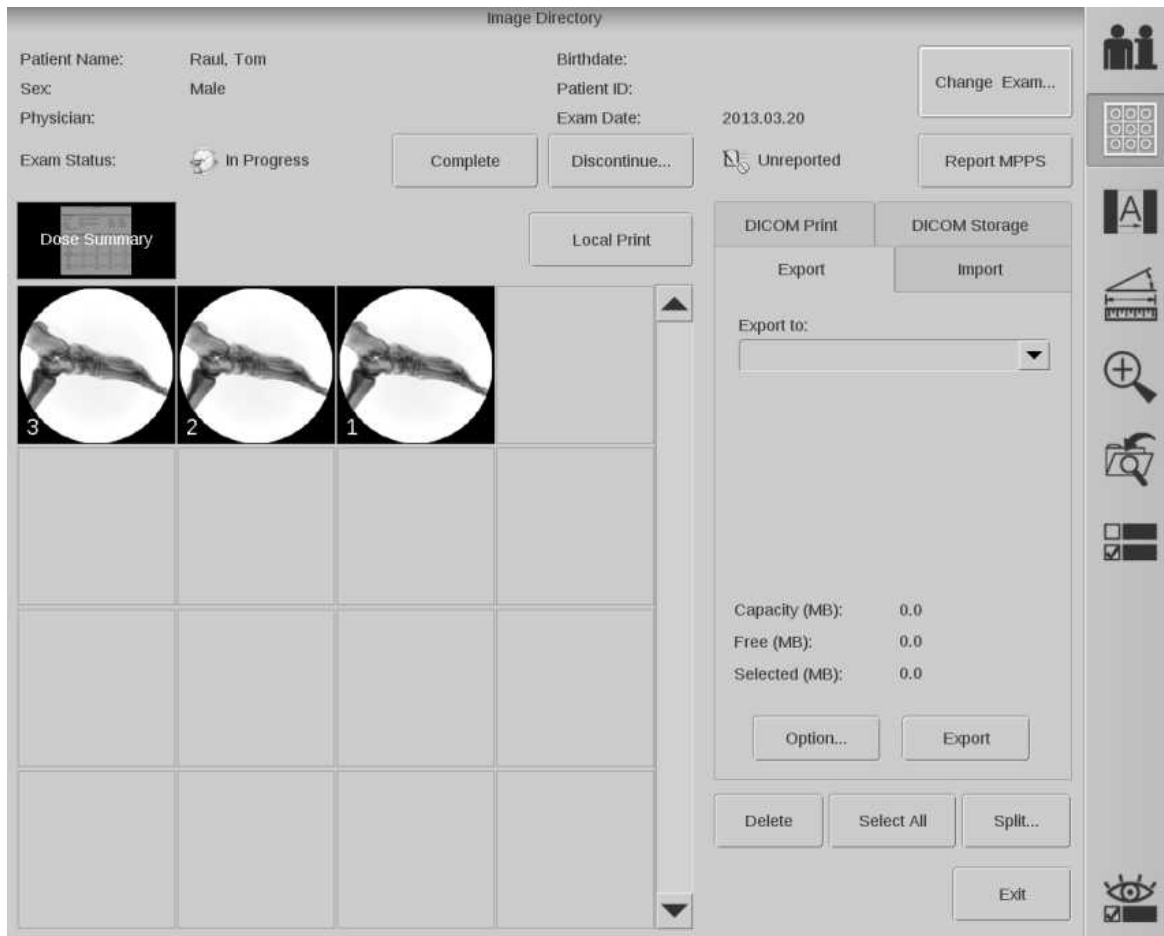
Country/ Region	MODE					
	(Dose measured at 30 cm above the Image Intensifier)					
	Fluoro kVp 40 to 110 mA 0.1 to 4 	Pulsed Fluoro kVp 40 to 110 mA 0.1 to 4 	High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Pulsed High Level Fluoro kVp 40 to 110 mA 0.2 to 12 	Digital Spot kVp 40 to 110 mA 0.2 to 16 	Film kVp 40 to 110 mA 20 mAs 1 to 80 
Taiwan (Republic of China)	87 mGy/min	87 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Thailand	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Trinidad & Tobago	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Tunisia	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Turkey	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
United Arab Emirates	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
USA	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
United Kingdom	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Uruguay	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Venezuela	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy
Vietnam	88 mGy/min	88 mGy/min	100 mGy/min	100 mGy/min	2mGy	20mGy

The highest value for the Low Dose imaging doesn't exceed 50% of the highest value for the Normal Fluoro. The highest value for the Low Level Pulsed Fluoro doesn't exceed 50% of the highest value for the Pulsed Normal Fluoro.

5.3.3. Print, Export and Import

Print to local printer

1. Press the "Image Directory" key  on the Workstation membrane keyboard or select the "Image Directory" button on the system panel to display the Image Directory screen.



2. Select the image you want to print. Press and hold on "Ctrl" key on the membrane keyboard to select multiple images if you want.
3. Press the "Local Print" button.

Export to archive devices

The Workstation supports a variety of archive options. You can archive images, patient information and dose summary.

Available storage options include:

- CD/DVD: DVD+R, DVD-R, DVD-RW, CD-R, CD-RW
- USB: Some removable disks with USB connector can't be used. Brivo OEC 715/785/865 only supports USB disks in Fat16, Fat32, Ext2, Ext3 and Ext4 format.

This section describes how to export from the system to CD/DVD as an example.

Detailed instruction for DICOM is provided in Chapter 6.3.

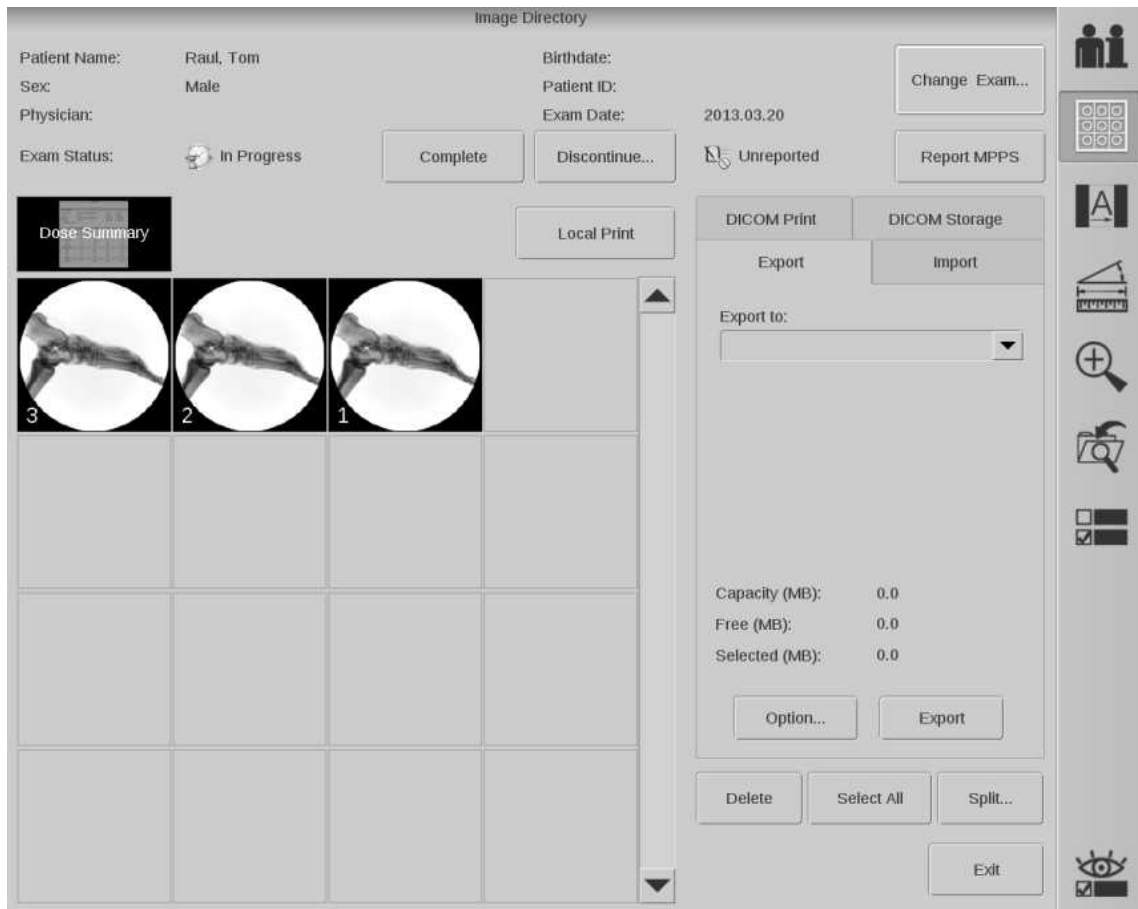


When exporting to CD/DVD, wait for the drive to load the disk before beginning the copy. The disk is loaded after the disk is inserted into the drive and the green LED on the front of the drive turns off.



Do not turn off the system while the system is writing information to a disk in the CD/DVD drive, when the green activity light on the drive is on.

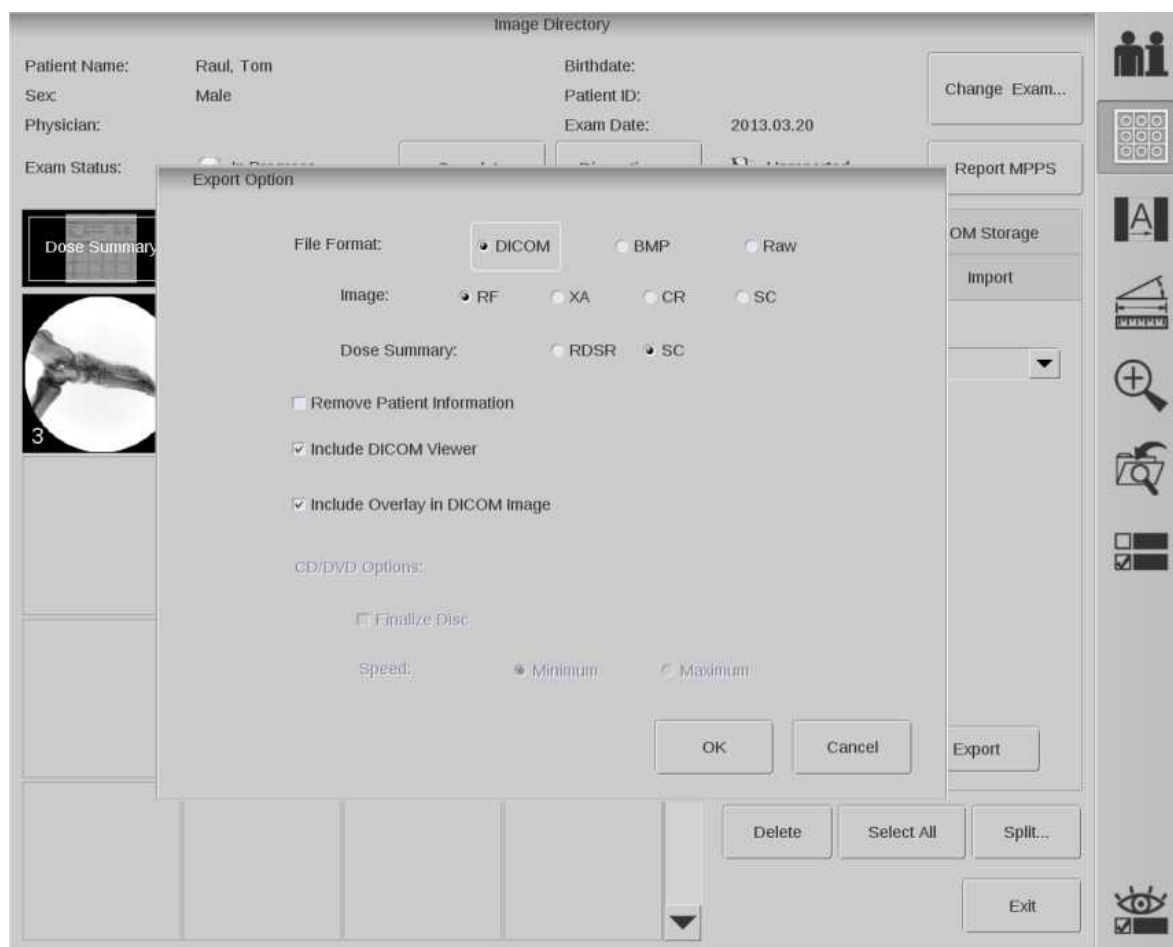
1. Select the "Export" tab on the Image Directory screen.



The number of images you can archive on the device media depends on the selected file format and storage capacity of the media.

When archiving images, patient information can be archived with the images.

2. Select DVD in "Export To" dropdown list.
3. Click "Option" button to display Export Options screen.



The Export Options screen allows you to set options for storing images.

Image Option

File Format

DICOM: Standard DICOM format image

BMP: BMP format image

Raw: System original format image, used to import exams among Brivo OEC 715, Brivo OEC 785 and Brivo OEC 865 systems

Image The Workstation supports the following modalities:

RF: X-Ray Radio Fluoroscopic Image Storage. Default option. 1000 x 1000 pixel, static.

XA: X-Ray Angiographic Image Storage. 1000 x 1000 pixel, static.

CR: Computed radiography image storage. 1000 x 1000 pixel, static.

SC: Secondary Capture Image Storage, screen capture type image with Annotation and markers.
1000 x 1000 pixel, static.

Dose Summary

RDSR: X-Ray Radiation Dose Structured Report

SC: Secondary Capture image

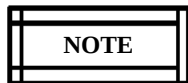
Remove Patient Information Select "Remove Patient Information" to remove all patient identification information of selected images to copy. This function has no impact on images.

Include DICOM Viewer "Include DICOM Viewer" will record "Media Viewer" application to Media. This program can browse images on the disk. "Media Viewer" will run automatically once you insert disk to CD/DVD driver in other computer that installed Windows operating system.

Include Overlay in DICOM Image Select "Include Overlay with DICOM Image" to store the overlay information associated with an image as data separate from the image, such as annotation and measurement.



The DICOM viewer should not be used for diagnostic purposes. Its use is limited to reviewing images.



The viewer can only be used in computer with Java virtual machine version 1.5 or above. Media viewer will install it automatically if there's no Java virtual machine.

CD/DVD Option

Finalize Disc Select this option will close current disk, no data could be written into this disk any more.

Speed Two write speeds can be used in CD/DVD device.

Minimum Make driver to record in the minimum speed for the type and quality of medium in driver. For lower-quality medium or to ensure the compatibility of selected information, use the minimum option.

Maximum Allow driver to achieve the maximum write speed for the type and quality of medium in driver. CD-R medium with high quality allows highest maximum speed. DVD RW medium will bring lowest maximum speed generally.

4. Select correct settings, then click "OK" button.
5. Select the preview images to archive. Select Dose Summary to archive. Press and hold on "Ctrl" key on the membrane keyboard to select multiple images from the previews in the directory. As you select images, the capacity, free space of the CD/DVD and size of selected images are updated in the Image Directory screen.
6. If the size of all the images in the directory is smaller than the disk's free size, choose "Select All" to select all of the images in the directory.
7. Select "Export" button on the Image Directory screen. To cancel the export process, click the "Cancel" button on the message screen.



Cancelling during the copy process may damage the disk.

- After the copy is finished, the CD/DVD will eject automatically.

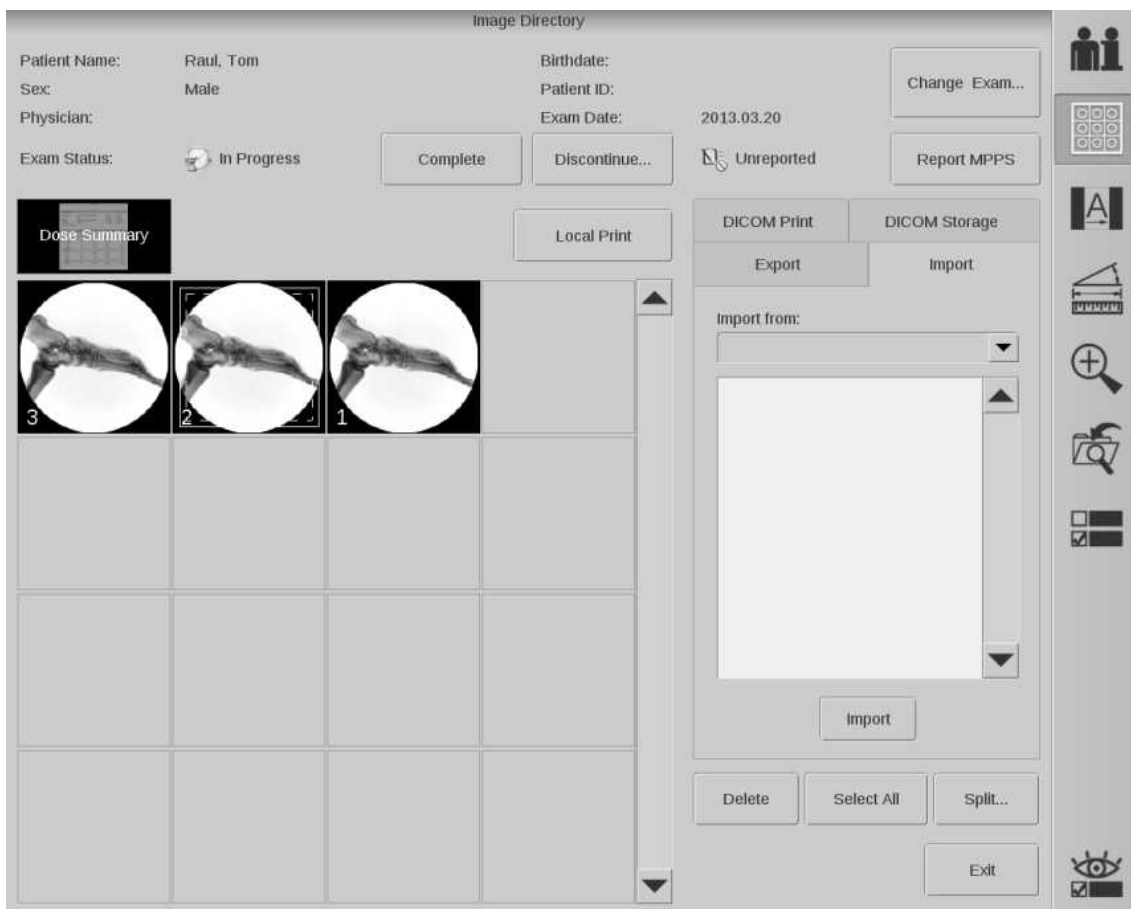
DICOM Viewer

The system provides a DICOM Viewer to view the images exported to CD/DVD/USB media. DICOM Viewer can be exported together with the images. DICOM Viewer is able to run on Windows OS including Windows XP and Windows 7.

If exported image is in DICOM format, read the media on the PC, double click CDVIEWER.EXE. Select the exam you wish to view.

Import from archive devices

- Select the "Import" tab on the Image Directory screen.
- Select the images of the patient you want to import to the system.
- Click "Import" button.

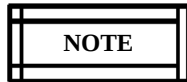


5.3.4. Retrieve a Performed Exam

Select "Change Exam" from the Image Directory screen. The Performed Exams screen is displayed on the right monitor.

Select an exam and then click "OK" button, this exam is displayed.

Select an image displayed in the Image Directory. The image is displayed on the left monitor.



When you access the performed exams list from image directory and select an exam, the current exam does not change. If you take an exposure and save an image while a performed exam is displayed on the Image Directory screen, the image is saved with the current exam, not with the performed exam displayed on the Image Directory screen.

5.4. Pediatric Use

5.4.1. General Instructions for Small or Pediatric Patients

Reduction of radiation dose must be balanced with safe, accurate and effective completion of the procedure. Not all steps listed below may be possible depending on patient size, technical challenge and critical nature and type of procedure. The goal through planning is to minimize the dose to the patient while providing important and necessary medical care while maximizing patient safety.

1. Remove grids when performing examinations on small patients.
2. Think about the position and aperture of the collimators before beginning the procedure, and during the procedure as conditions and field of view change.
3. Collimate tightly. Exclude eyes, thyroid, breast, gonads when possible.
4. Adjust acquisition parameters to achieve lowest dose necessary to accomplish procedure.
5. Keep II tower as close to patient as possible.
6. Use pulse rather than continuous fluoroscopy and with as low a pulse as possible.
7. Position and collimate with fluoroscopy off, with brief fluoroscopy to check position.
8. Pause when you pulse. Tap on fluoroscopy button/pedal and review anatomy on last image hold rather than with live fluoroscopy; minimize fluoroscopy time.
9. Minimize use of electronic magnification; use digital zoom whenever possible.
10. Use last image grab to record information whenever possible instead of exposures.
11. Ensure that the equipment is checked regularly by a physicist or designee with appropriate skills.

The information and list above has been taken from the Pause and Pulse Checklist found on the Alliance for Radiation Safety in Pediatrics Imaging's Image Gently website. For more information about pediatric patient safety in fluoroscopy and other x-ray emitting technologies you can consult:

1. Image Gently at <http://www.pedrad.org/associations/5364/ig/>
2. FDA at <http://www.fda.gov/RadiationEmittingProducts/RadiationEmittingProductsandProcedures/MedicalImaging/ucm29889>
3. International Atomic Energy Association's Radiation Protection of Patients site at <http://rpop.iaea.org>

5.4.2. General Dose Planning

There are several dose reduction measures available in the system. They are: Low Dose Mode, Pulsed Mode using reduced frame rates, removal of the contrast grid, and additional filtration.

- As a general guideline, Low Dose Mode will typically reduce the dose by 50%. This will typically result in reduced image quality.
- Pulse Fluoroscopy can result in dramatic dose reduction where image update frequency is not critical. Images may be noisy /grainy, especially in dense anatomy. The slowest pulse rates provide the greatest dose reduction. Clinical need should determine the appropriate pulse rate.
- Contrast grid removal is not recommended for anatomical thickness greater than 20 cm, although this is where the greatest dose reduction would occur, but this would also result in the greatest loss of image quality.
- Added filtration is recommended for all pediatric applications. Dose reduction will be greatest for light anatomy and will result in some image contrast reduction.
- It may be possible to use these dose reduction features in combination, resulting in even greater dose reduction. However, there will be incremental image quality reduction for each dose reduction feature.

It is not advised to remove the grid for anatomy greater than approximately 15 cm in thickness, since image contrast can be severely impacted. Narrowing the beam through collimation will help minimize the effects of scattered radiation on image quality. This also reduces the exposure area on the patient.

5.5. Emergency Mode

When user forgets the login password, emergency mode can be used to produce images in emergency case.

Press “Emergency Login” to login emergency mode screen. User can access Exam Management, Zoom & Roam and Mode Switch only. Other applications are disabled. In Exam Management screen, user cannot access Resume Exam and Schedule Exam feature.



The image shows the GE Healthcare login interface. At the top left, the text "GE Healthcare" is displayed. At the top right is the GE logo. Below the text, there is a login form. The "User:" label is followed by a dropdown menu showing "Admin". To the right of the dropdown is a "Login" button. Below the "User:" field is the "Password:" label followed by a password input field. Below the password field are two buttons: "Change Password" on the left and "Emergency Login" on the right. The background of the screen is a light gray with a subtle, abstract pattern.

Chapter6. Special Applications

6.1. Image Annotation

This section explains how to create annotation and how to use image annotation to place markers, add comments, crop images. Image annotation is available based on which system you have purchased.

Press the "Annotation"  key from the Workstation membrane keyboard or select the "Annotation" button on the system panel, the Annotation screen is displayed with the image of major view on the right monitor.

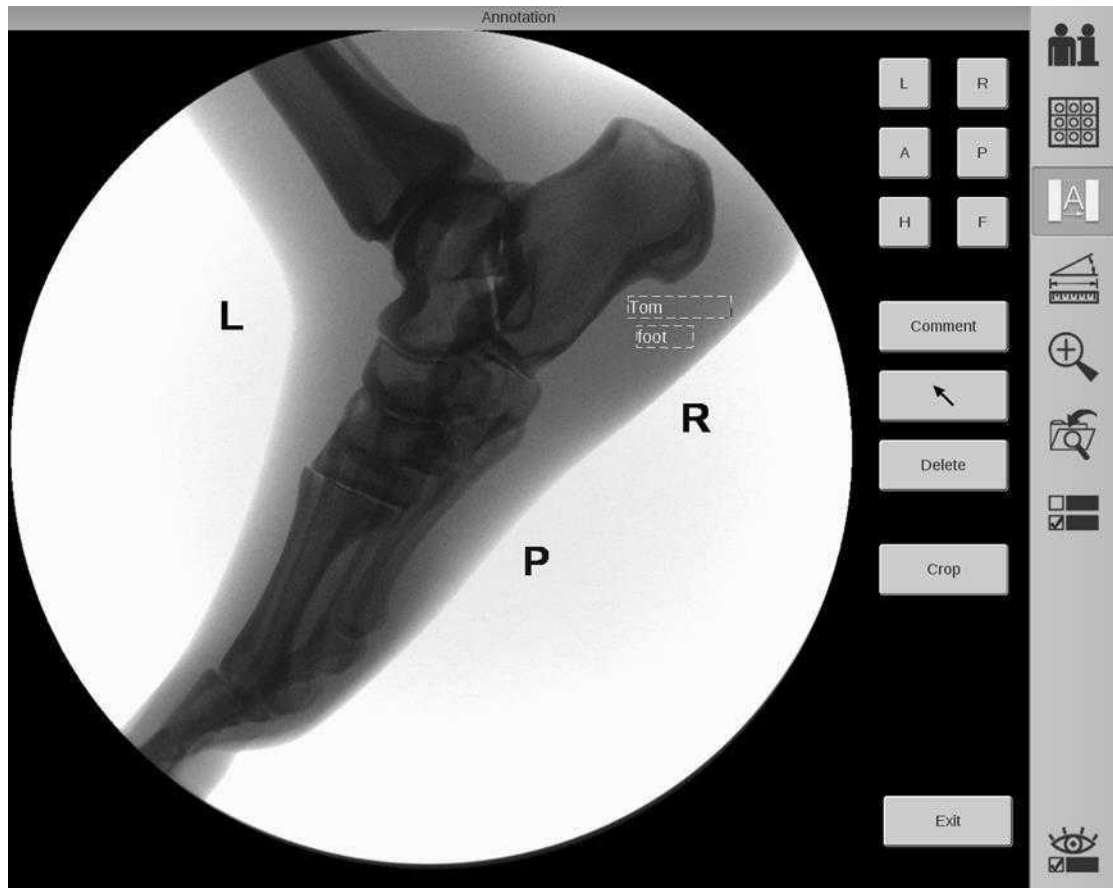


Select "Exit" button to close the Annotation.

6.1.1. Place Markers on an Image

The markers object bar includes a directional arrow and L (Left), R (Right), A (Anterior), P (Posterior) H (Head) and F (Foot) markers.

1. Select a button in the object bar to select a marker. The marker is displayed on both the left and right monitors.



2. To position a marker, select the marker displayed on the right monitor image, and drag it into position.
3. To add additional markers, repeat steps one and two. You can add up to five arrows and up to two for each type patient orientation maker per image.

To delete a marker: select it on the right monitor and press "Delete" button on the screen.

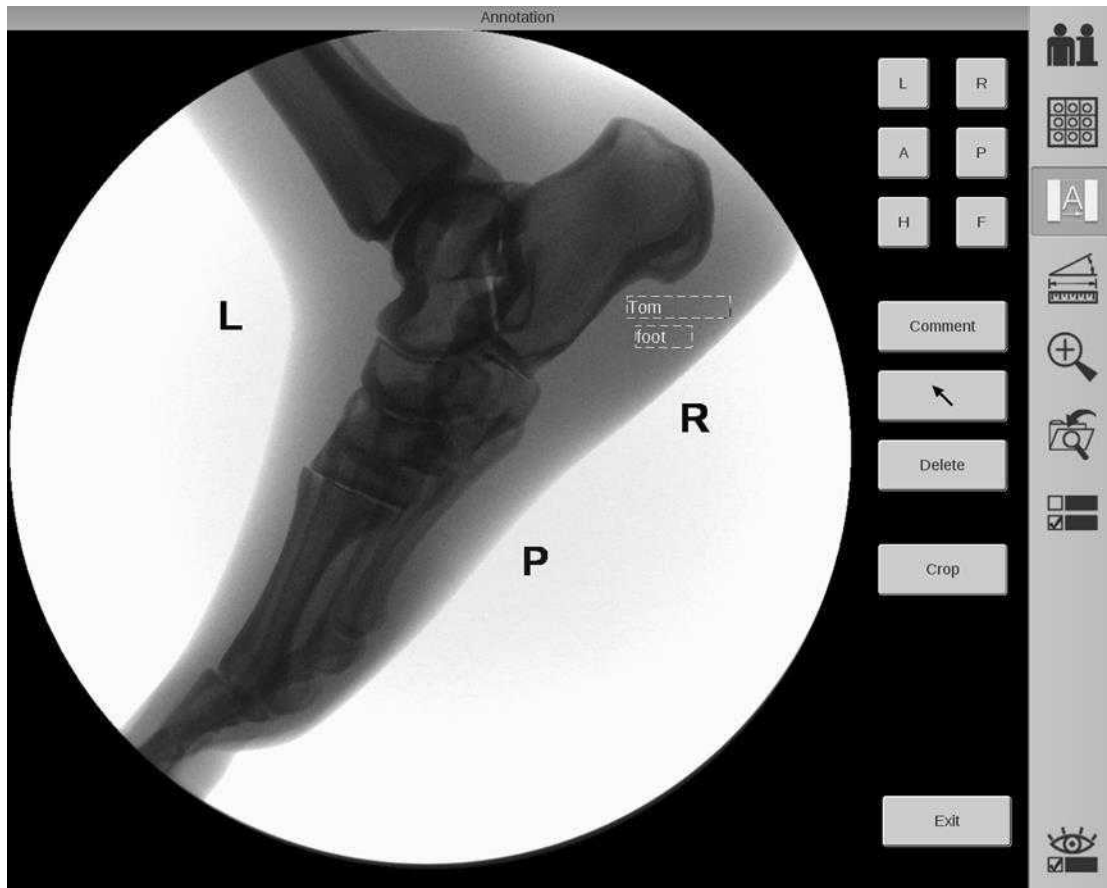
To save a copy of the image with the markers: press the "Save" key



4. Select the "Exit" button to close the Annotation screen. The annotated image remains displayed on the left monitor until you press an X-Ray switch or recall a saved image.

6.1.2. Add Comments to an Image

1. Select the "Comment" button on the Image Annotation screen. The comment box is displayed on both the left and right monitors.



2. Enter text in the comment box. You can enter up to 80 characters.
3. To position the comment box on the image, select the border line of the comment box on the right monitor and drag it.
4. To add additional comments, select the "Comment" button and enter another comment. You can enter up to five comments per image.

To delete a comment: select it and press "Delete" button on the screen.

To save a copy of the image with the comment: press the "Save" key  on the Workstation membrane keyboard.

5. Select the "Exit" button to close the Annotation screen. The annotated image remains displayed on the left monitor until you press an X-Ray switch or recall a saved image.

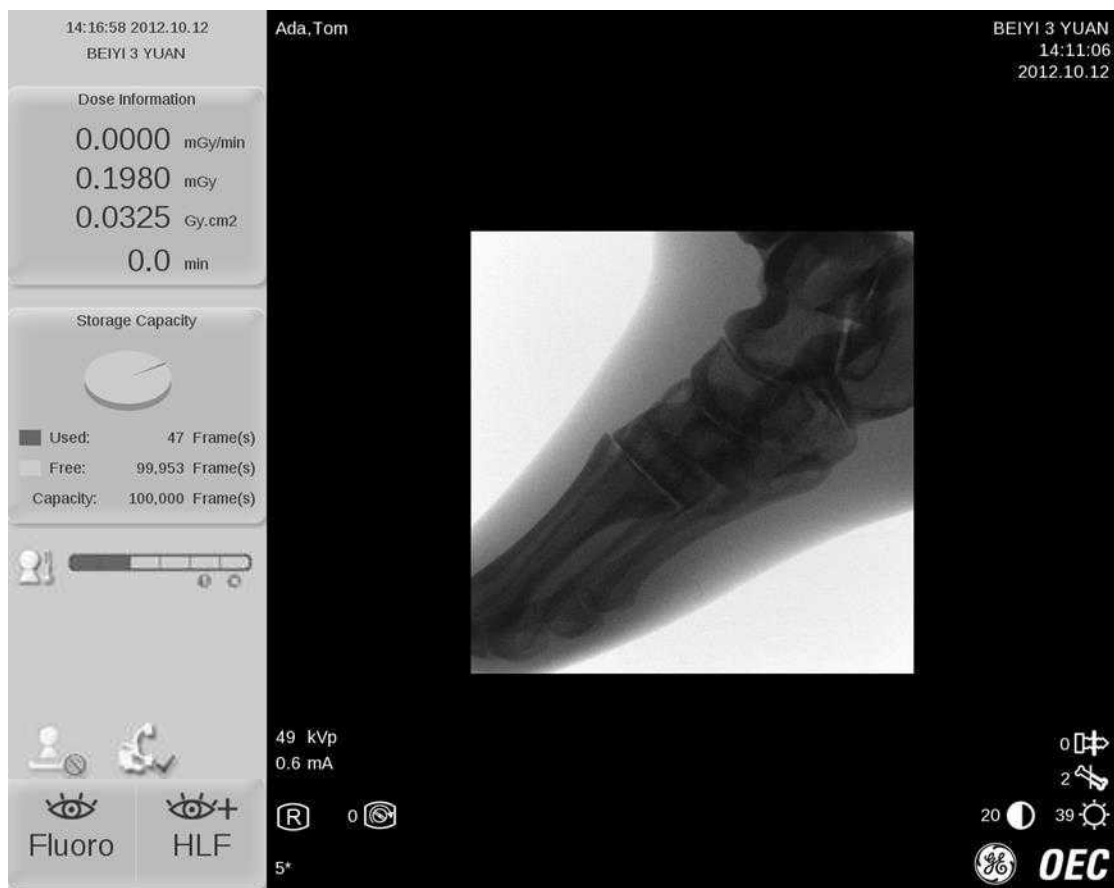
6.1.3. Crop an Image

Crop shutters allow you to shade out portions of the image on the left monitor.

1. Select the "Crop" button from the Annotation screen. The crop shutter is displayed on the right monitor.



2. Select the resize anchors and move them to crop the image. As you move the anchors, the cropped area of the left monitor image is hidden from view.



3. Reselect the "Crop" button, the crop is deleted on both right monitor and left monitor. The part of image hidden is back to display.

To save a copy of the cropped image: press the "Save"



key on the Workstation membrane keyboard.

4. Select the "Exit" button to close the Annotation screen. The left monitor image remains cropped until you press the X-Ray switch or display a saved image.

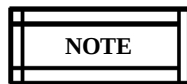
6.2. Image Measurement

This section explains how to measure an image. Image Measurement is available based on which system you have purchased.

Measurement includes the following options.

- Calibration
- Distance
- Angle

Measurements can be applied to a saved image, a recalled image, and the current image displayed.



Because of patient movement or measurement subject is not located in the region of interest or inadequate calibration, measurement result may be inaccurate. Measurement result is only for reference.

Select the “Measurement” key  on the Workstation or select the “Measurement” button on the system panel, the left monitor image is copied to the right monitor and displayed in the Measurement screen.



Click on “Exit” button on Measurement screen, Measurement is closed. The Measurement screen also exits

when X-Ray is on.

6.2.1. Calibration

To perform a calibration between two points, a measuring device should be placed in the plane of interest. The distance on the image should be marked during calibration and the length value entered in the value box. Once the value is entered in the box, the reference value for the fixed length is displayed in the upper right hand corner of the Measurement screen.



When calibration is changed during a procedure, the values in length for distance adjust to the new calibration values.

When zoom is enabled on an image, the reference calibration values are scaled to match the zoom factor.

To Perform Calibration

1. Select the "Calibrate" button on the Measure screen to display the Calibration screen.
2. Move the Calibration Line to set the distance.
3. Enter the measured length or catheter size in the calibration value box. Select the calibration unit.

6.2.2. Distance Measurement

A Calibration must be performed before distance measurements can be placed.

The distance calculation is displayed once both end points have been placed. This value is displayed

adjacent to the end point.

When calibration is changed during the procedure, the distance measurements are recalculated and display the new values.

To perform distance measurements

1. Perform Calibration.
2. Select the "Distance" button. A default measurement line is displayed. This line is based on the calibration performed.



3. Select the distance measurements unit, it includes mm, inch and French.
4. Select the line and drag the anchor to the appropriate position. The distance measurement will display adjacent to the anchor.
5. You can enter up to five distances per image.

To delete a "Distance": select it on the right monitor and press "Delete" button on the screen.

To save a copy of the image with distance: press the "Save" membrane keyboard.

key  on the Workstation

6.2.3. Angle Measurement

To perform angle measurements

1. Select the "Angle" button. An angle is displayed on both the left and right monitors.
2. Select an angle, adjust length and angle degree by anchor dragging, the angle degree value is displayed at bottom of the angle.



3. You can place up to five angles per image.
To delete an "Angle": select it on the right monitor and press "Delete" button on the screen.

To save a copy of the image with angle: press the "Save" membrane keyboard.

key  on the Workstation

6.3. DICOM

6.3.1. Overview

The American College of Radiology (ACR) and the National Electrical Manufacturers Association (NEMA) developed the Digital Imaging and Communications in Medicine (DICOM) standard.

This section contains information about the DICOM application available on the Workstation that enhances the functionality of the Workstation. The DICOM application allows the Workstation to connect to and use existing DICOM networks.

You can download the DICOM conformance statement file for Brivo OEC 715, Brivo OEC 785, Brivo OEC 865 mobile C-Arm X-Ray product from the website: www.ge.com/dicom.

This section describes how to:

- Connect the Workstation to the network
- Configure the Workstation and DICOM devices
- Query and display scheduled exams' information
- Query remote server and retrieve the image
- Print patient information and images
- Archive patient information, images and dose report
- Report MPPS

During network transfers, both images and patient information are sent. The patient information may be the result of a DICOM Worklist Query, or be manually entered on the Workstation in the Exam Management screen by the technician.

Some DICOM features are not available on all systems.

6.3.2. Network Connection

To use the DICOM interface you must connect your Workstation to the network.

Plug the network cable into the network connector or open Wireless Network Interface on the Workstation.

6.3.3. DICOM Setup

The Workstation must be configured to access the DICOM network. Configuration identifies the server to the network and defines the DICOM network nodes to the Workstation. The DICOM features include:

- Worklist
- DICOM Print
- Storage and Commitment
- Query & Retrieve
- MPPS

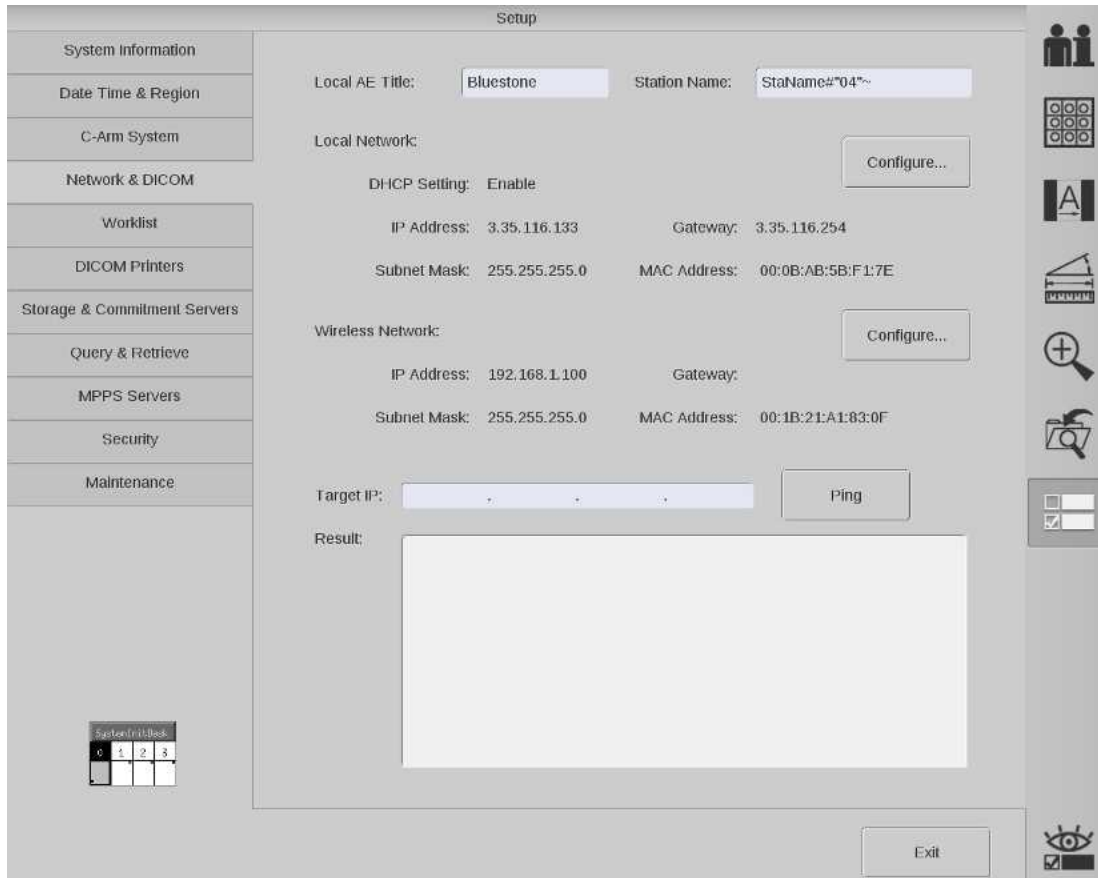
Press the “Setup”  key on the Workstation membrane keyboard.

Use this screen to access the “Worklist”, “DICOM Printers”, “Storage and Commitment Server”, “Query & Retrieve”, “MPPS Server” and Network setup screen.

Network Setup

Click “Network & DICOM” button. The Network & DICOM setup screen shows on the right monitor.

Input the “Local AE Title” and “Station Name”. The default AE Title is “Brivo OEC”. You can change it according to the DICOM service’s requirement.



The screenshot shows the "Setup" window with a sidebar on the left and a main configuration area. The sidebar contains buttons for System Information, Date Time & Region, C-Arm System, Network & DICOM (selected), Worklist, DICOM Printers, Storage & Commitment Servers, Query & Retrieve, MPPS Servers, Security, and Maintenance. The main area is titled "Setup" and contains fields for Local AE Title (Bluestone), Station Name (StaName#*04*~), Local Network settings (DHCP Setting: Enable, IP Address: 3.35.116.133, Subnet Mask: 255.255.255.0, Gateway: 3.35.116.254, MAC Address: 00:0B:AB:5B:F1:7E), and Wireless Network settings (IP Address: 192.168.1.100, Subnet Mask: 255.255.255.0, Gateway: , MAC Address: 00:1B:21:A1:83:0F). There are "Configure..." buttons for both network types. At the bottom, there is a "Target IP:" field, a "Ping" button, and a "Result:" label above a large empty box. An "Exit" button is at the bottom right. A small "System/Block" keypad is visible in the bottom left corner of the window.

Local Network Setup

Click "Configure..." of local network to setup the local network.

Input the target IP Address, Gateway and subnet mask on the screen.

Fill in the information on this screen. You can obtain the correct information from your network administrator. You don't need to input IP, Gateway and subnet mask information when using DHCP.

Click "OK" to return to Network & DICOM screen.

Input a target IP Address and click "Ping" button to verify the network connection. The results will show in the text box.

Wireless Network Setup

Click "Configure..." of wireless network to setup the wireless network information.

Input a target IP Address and click "Ping" button to verify the network connection. The ping results will show in the Result box.

Setup Worklist

The DICOM Worklist server, such as Radiology Information System (RIS), provides scheduled exams' information.

To setup the Worklist:

1. Select "Worklist" button from the setup screen. The Worklist screen is displayed.

The screenshot shows the 'Setup' window with the 'Worklist' tab selected. The sidebar on the left contains the following categories: System Information, Date Time & Region, C-Arm System, Network & DICOM, Worklist, DICOM Printers, Storage & Commitment Servers, Query & Retrieve, MPPS Servers, Security, and Maintenance. The main area is titled 'Setup' and contains the following elements:

- Exam Date:** Radio buttons for Today (selected), Next 2 Days, Next 3 Days, and Next 7 Days. A 'Date Range' section with 'From' and 'To' date pickers (both set to 03/20/2013).
- Modality:** Radio buttons for All (selected), RF, XA, CR, and SC.
- Table:** A table with three columns: Default, Alias, and AE Title. It contains one entry: 'worklist1' under Default and 'JdcmRisServer1' under AE Title.
- Buttons:** 'Add...', 'Edit...', and 'Delete' buttons are located to the right of the table.
- Exit:** An 'Exit' button is located at the bottom right of the window.

On the right side of the window, there is a vertical toolbar with icons for user management, a grid, a large 'A' (likely for annotations), a ruler, a magnifying glass, a camera, and a checkmark icon.

2. Setup Scheduled Exam Date and Modality.
3. Click "Add" button to display the Worklist Server Config screen. Obtain the following information about the worklist server from your network administrator and enter it on this screen:

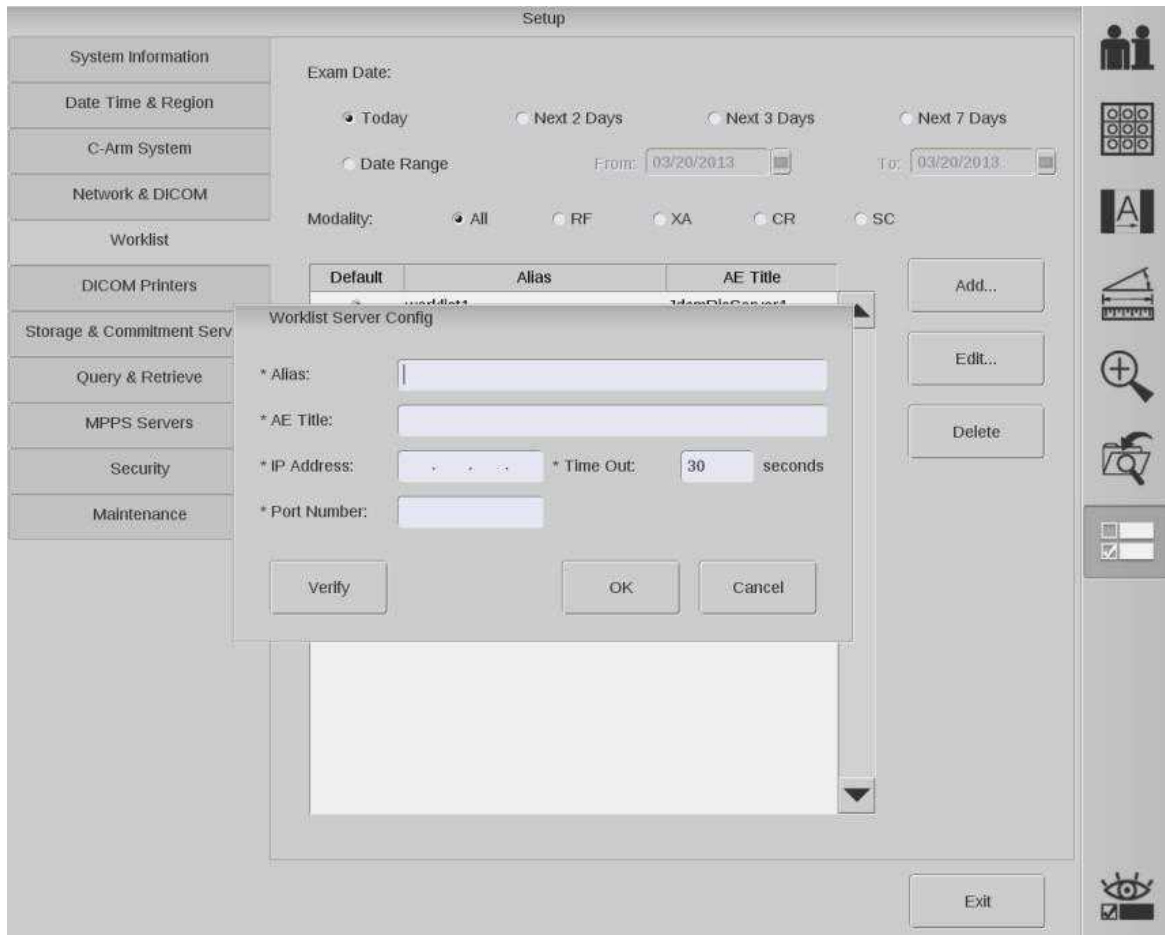
Alias Unique name, internal to WorkStation, used to identify the server in the list of available destination devices.

AE Title DICOM Application Entity Title.

IP Address Unique IP address assigned to the server you are configuring.

Time Out The time in seconds that the Workstation waits for a response from the server before timing out.

Port Number Port number assigned to the query server.

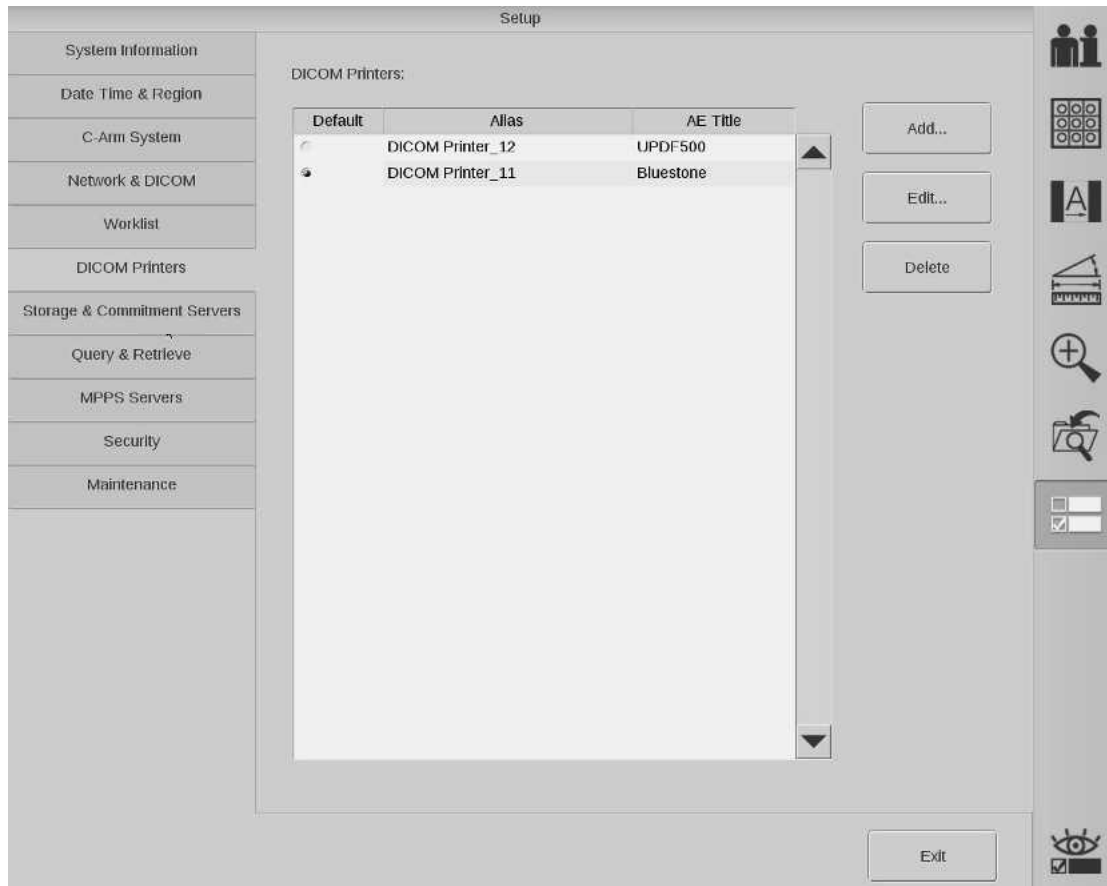


4. Select the "Verify" button to verify that the server is connected. A verified successful message appears if the server has connected. Select "OK" to close the screen.
5. Repeat steps three to four for each DICOM Worklist server on the network. You can edit or delete server by clicking the relevant button.
6. When you have finished configuring all servers, select the "Exit" button to close the Worklist Screen.

Setup DICOM Printer

DICOM printer servers are network nodes that provide print functions in compliance with the DICOM standard.

1. Select "DICOM Printers" from the Setup screen.



2. Select "Add" button.

Obtain the following information from the Network Administrator for your facility and enter it in the text-entry fields on this screen:

Alias Unique name, internal to the OEC WorkStation, used to identify the server in the list of available destination devices.

AE Title DICOM Application Entity Title.

IP Address Unique IP address assigned to the server you are configuring.

Time Out The time in seconds that the Workstation waits for a response from the server before timing out.

Port Number Port number assigned to the print device.

Maximum Maximum print density.

Minimum Minimum print density.

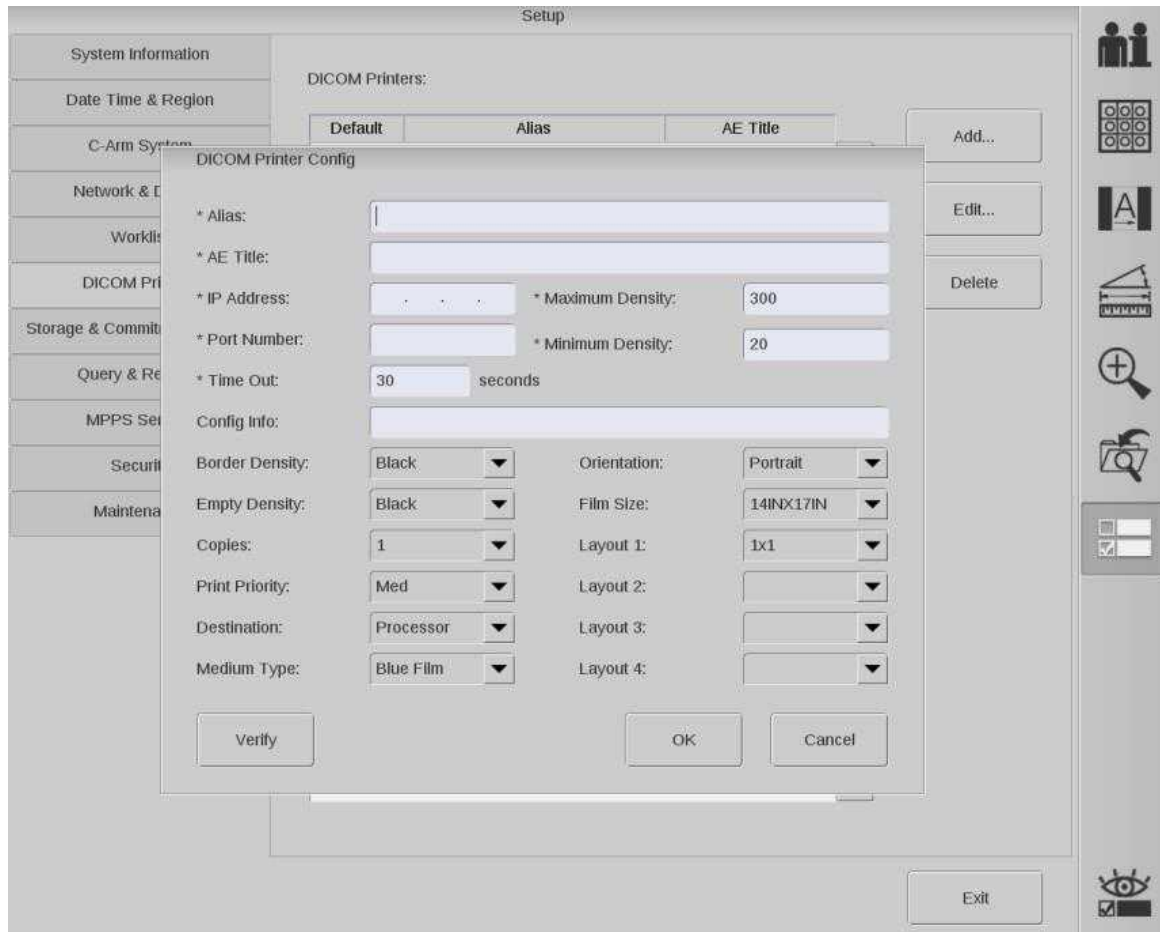
Config Info. This value specifies the DICOM printer parameters. It is contained in the DICOM Printer's Conformance Statement file. The DICOM tag is (2010, 0150) Configuration Information.

3. Select the "Border Density..." button and choose the image border density. The available options are:

- BLACK
- WHITE

4. Select the "Empty Density..." button and choose the image empty density. The available options are:
 - BLACK
 - WHITE
5. Select the "Copies..." button and choose the number of copies that will be printed each time a print command is sent to this printer. The available options are 1 to 10.
6. Select the "Print Priority..." button and choose a print priority. The available options are:
 - HIGH
 - MEDIUM
 - LOW
7. Select the "Destination..." button and chose a print destination. The available options are:
 - MAGAZINE
 - PROCESSOR
8. Select the "Medium Type...." button and choose a medium type. The available options are:
 - PAPER
 - CLEAR FILM
 - BLUE FILM
 - MAMMO CLEAR FILM
 - MAMMO BLUE FILM
9. Select "Orientation" button and choose an orientation. The available options are:
 - PORTRAIT
 - LANDSCAPE
10. Select the "Film Size..." button and choose a film size. The available options are:
 - 8IN×10IN
 - 10IN×12IN
 - 10IN×14IN
 - 11IN×14IN
 - 14IN×14IN
 - 14IN×17IN
 - 24CM×30CM
 - 24CM×24CM
11. Select a film layout for the X-Ray images. You can select up to four layouts for each DICOM printer device you are configuring.

The film matrix determines the printed layout for the X-Ray images in columns and rows. The matrix choices are limited to those available from the selected printer. The total number of images printed on one film sheet is the product of the two numbers.

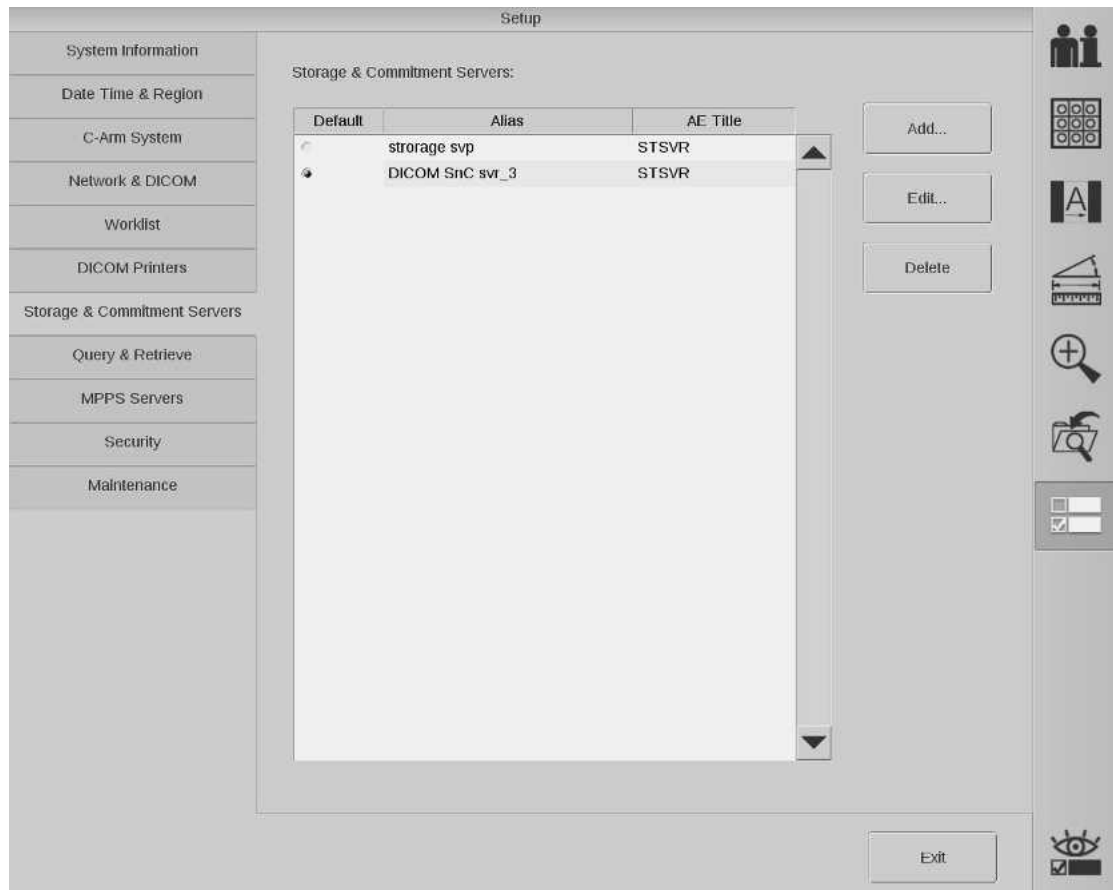


12. Select the "Verify" button to verify that the printer server is connected. A verified successful message appears if the server has connected. Select "OK" to close the screen.
13. You can edit or delete servers in the list by clicking relevant button.
14. Select the "Exit" to close the DICOM Printer screen.

Setup Storage and Commitment Server

A DICOM storage server, such as PACS system, is a networked system used to archive images.

1. Select "Storage & Commitment Servers" from the Setup screen.



2. Select the “Add” button to add storage server you want to configure.

3. Obtain the following storage information from your network administrator and enter it:

Alias Unique name, internal to WorkStation, used to identify the server in the list of available destination devices.

AE Title DICOM Application Entity Title.

IP Address Unique IP address assigned to the server you are configuring.

Port Number Port number assigned to the storage device.

Time Out The time in seconds that the Workstation waits for a response from the server before timing out.

Overlay Choose Never Send or Always Send if Available.

Image The Workstation supports the following modalities:

RF X-Ray Radio Fluoroscopic Image Storage. Default option. 1000 x 1000 pixel, static.

XA X-Ray Angiographic Image Storage. 1000 x 1000 pixel, static.

CR Computed radiography image storage. 1000 x 1000 pixel, static.

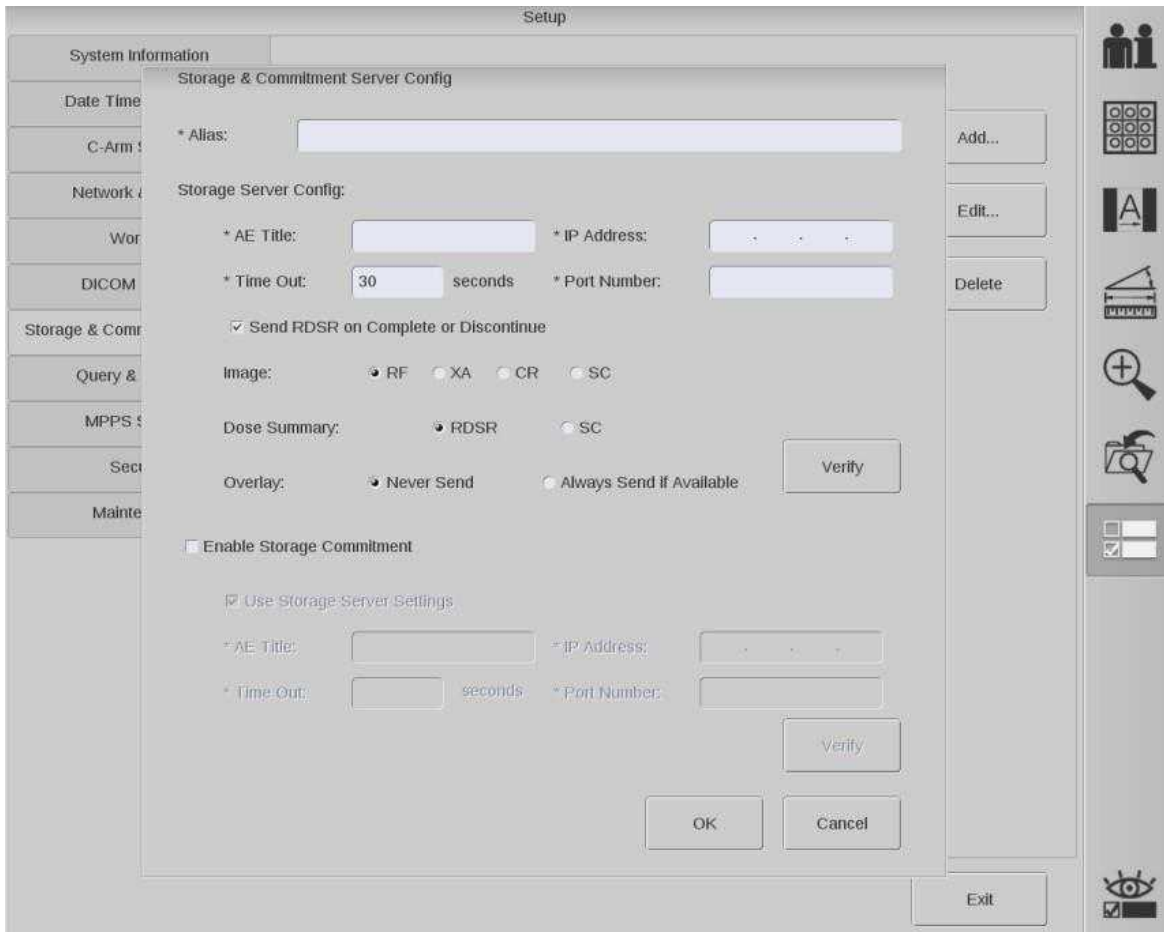
SC Secondary Capture Image Storage, screen capture type image with Annotation and markers.
1000 x 1000 pixel, static.

Dose Summary

RDSR X-Ray Radiation Dose Structured Reporting

SC Secondary Capture image

Send RDSR on Complete Check it when you want to send RDSR on completing an exam.



4. Select the "Verify" button to verify that the server is connected. A verified successful message appears if the server has connected.

5. Check "Enable Storage Commitment" to setup commitment server.

6. Obtain the following commitment information from your network administrator and enter it:

AE Title DICOM Application Entity Title.

IP Address Unique IP address assigned to the server you are configuring.

Port Number Port number assigned to the commitment device.

Time Out The time in seconds that the Workstation waits for a response from the server before timing out.

7. Check "Use Storage Server Settings" if you want to use storage server setting. The commitment information will be displayed automatically.

8. Select the "Verify" button to verify that the commitment server is connected. A verified successful message appears if the server has connected.

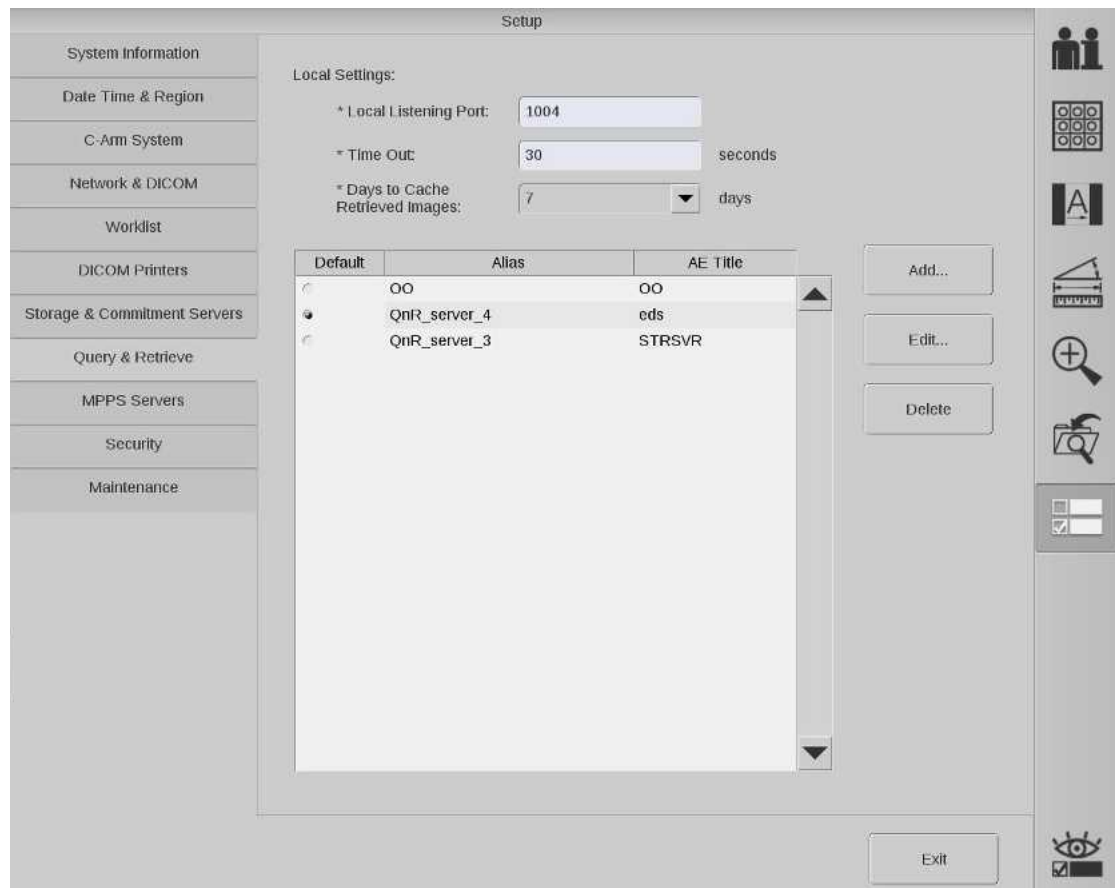
9. Select "OK" button to close the screen.
10. Repeat steps one through nine for each Storage and Commitment server on the network. You can edit or delete server by clicking the relevant button. When you have finished configuring all servers, select the "Exit" button to close the Storage and Commitment Screen.

Setup Query & Retrieve

Query & Retrieve Server, are for retrieving the DICOM image information from remote server.

To configure the Query & Retrieve server:

1. Select "Query & Retrieve" button from the setup screen.



2. Setup Local Setting.

Local Listening Port

Time Out The time in seconds that the Workstation waits for data sent back by DICOM Retrieve Server.

Days to Cache Retrieved Images

3. Click "Add" button to display the Query & Retrieve Config screen. Obtain the following information about the Query & Retrieve server from your network administrator and enter it on this screen:

Alias Unique name, internal to WorkStation, used to identify the server in the list of available destination devices.

AE Title DICOM Application Entity Title.

IP Address Unique IP address assigned to the server you are configuring.

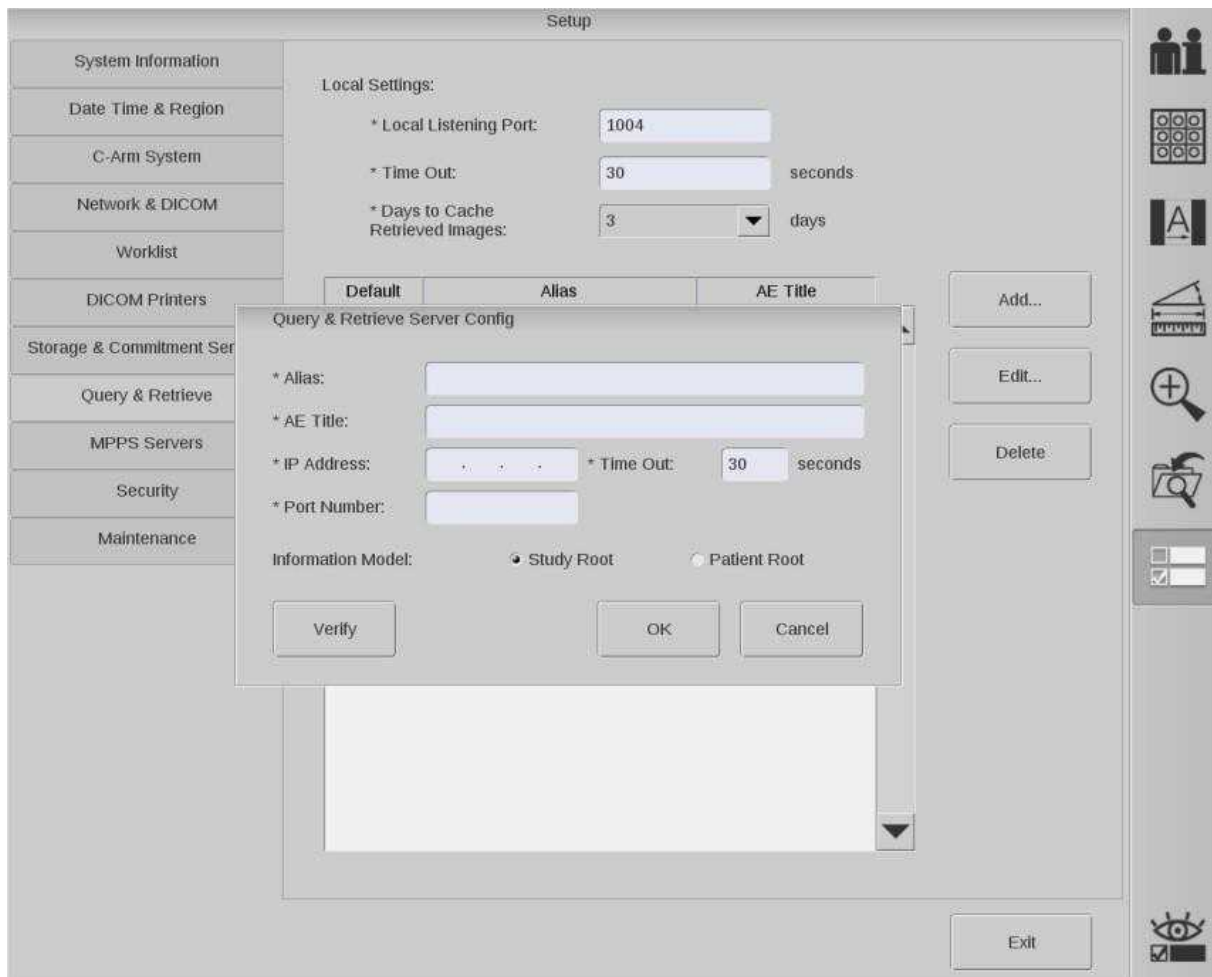
Port Number Port number assigned to the query server.

Time out The time in seconds that the Workstation waits for a response from the server before timing out.

Information Model There are two items of Information Model:

Study Root - Starts queries at the study level;

Patient Root - Starts queries at the patient level.

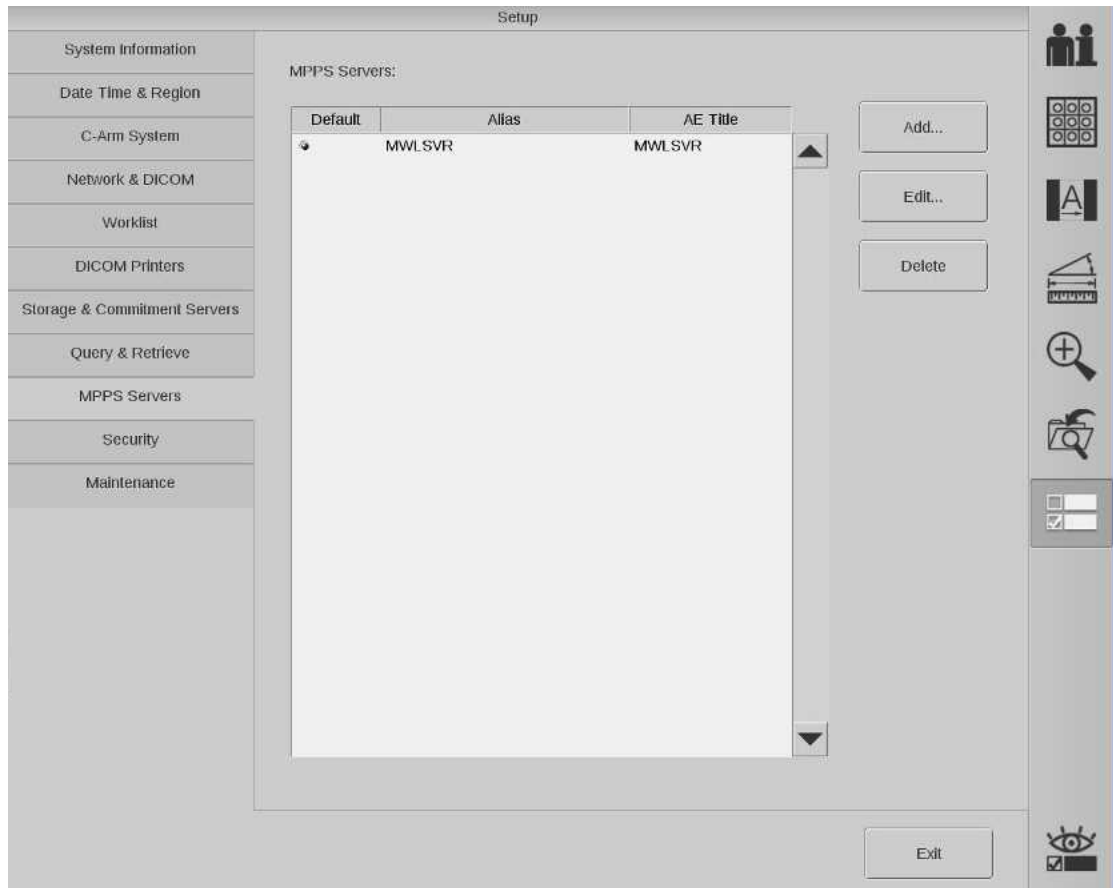


4. Select the "Verify" button to verify that the server is connected. A verified successful message appears if the server has connected. Select "OK" to close the screen.
5. Repeat steps one through four for each Query & Retrieve server on the network. You can edit or delete server by clicking the relevant button.
6. When you have finished configuring all servers, select the "Exit" button to close the Query & Retrieve Screen.

Setup DICOM MPPS Server

To configure the MPPS server:

1. Select "MPPS Servers" button from the Workstation setup screen.



- Click "Add" button to display the MPPS Server Config screen. Obtain the following information about the MPPS server from your network administrator and enter it on this screen:

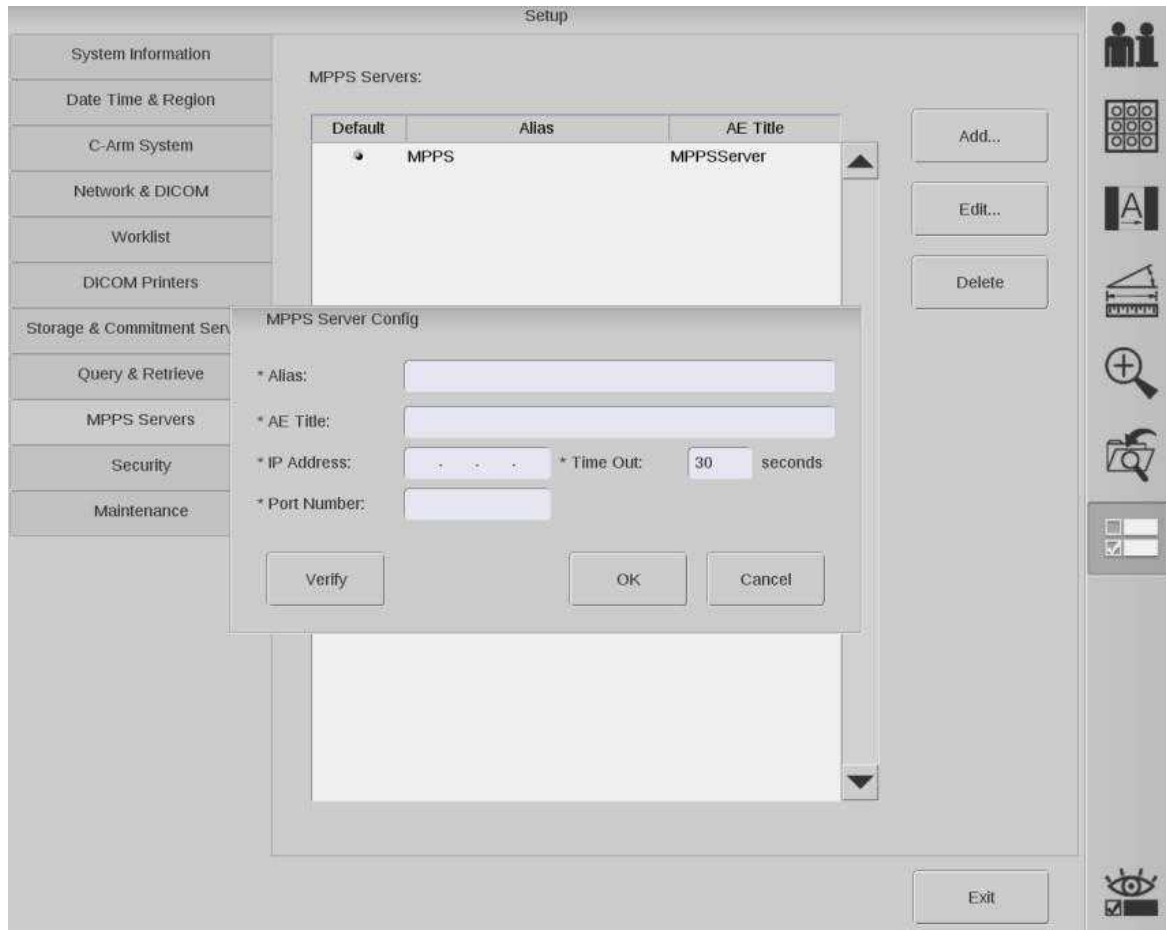
Alias Unique name, internal to WorkStation, used to identify the server in the list of available destination devices.

AE Title DICOM Application Entity Title.

IP Address Unique IP address assigned to the server you are configuring.

Port Number Port number assigned to the query server.

Time Out The time in seconds that the Workstation waits for a response from the server before timing out.



3. Select the "Verify" button to verify that the server is connected. A verified successful message appears if the server has connected. Select "OK" to close the screen.
4. Repeat steps one through three for each MPPS server on the network. You can edit or delete server by clicking the relevant button.
5. When you have finished configuring all servers, select the "Exit" button to close the MPPS Server Screen.

6.3.4. Worklist

DICOM Worklist Query feature allows you to query the scheduled exams' information from DICOM server. You can use the query result to start the exam directly.

1. Make sure the Worklist server is setup successfully.
2. Select "Scheduled..." from the Exam Management screen.

The screenshot displays the 'Exam Management' interface. It features a form with various input fields for patient and exam data. On the right side, there is a vertical toolbar with several icons and buttons. The 'Scheduled...' button is highlighted, indicating the next step in the process.

Exam Management

Last Name:

First Name: Middle Name:

Birthdate: Age: Sex: ☐ Male ☐ Female ☐ Other

Patient ID:

Accession #: Study ID:

Performing Physician: (Last Name) (First Name) (Middle Name)

Procedure Description:

Scheduled Procedure Step ID:

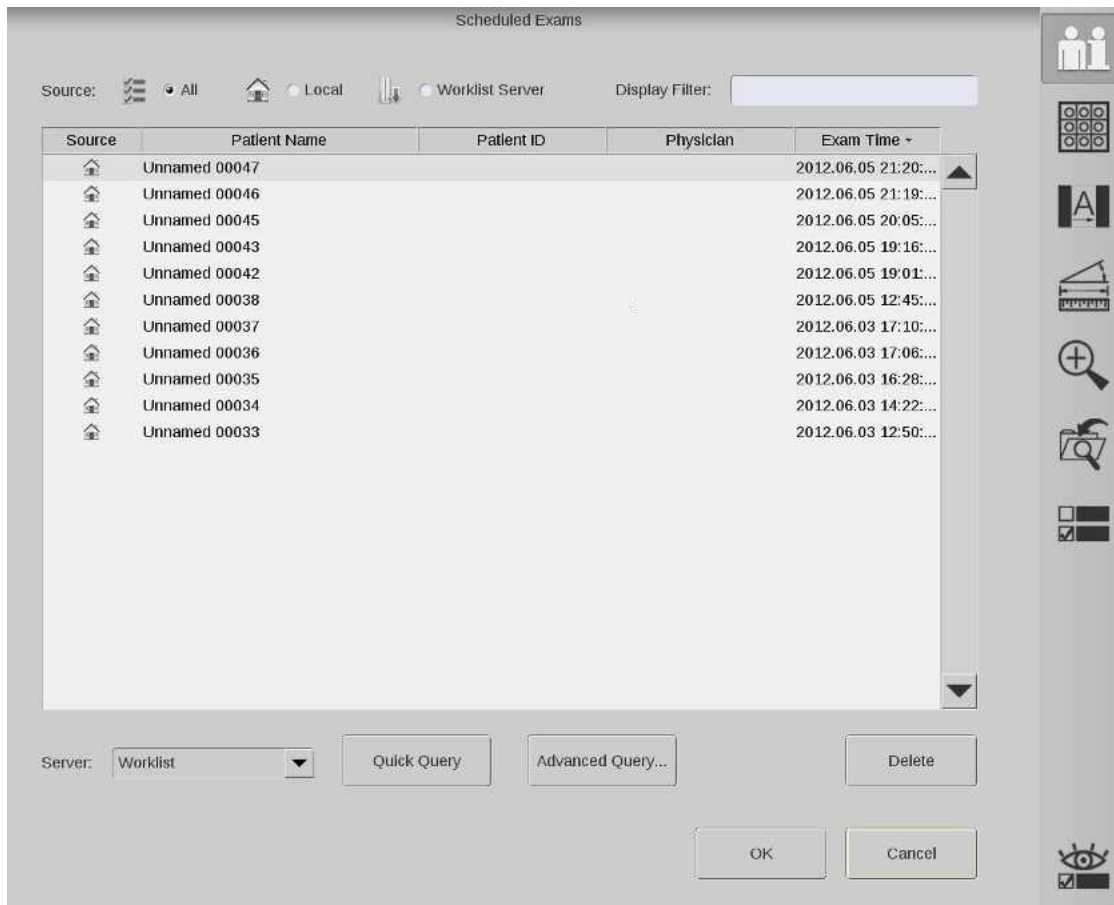
Scheduled Procedure Step Description:

Comments:

Buttons: New, Scheduled..., Resume Exam..., Extra Info..., Edit, Exit

Toolbar Icons: User icon, Grid icon, A icon, Ruler icon, Magnifying glass icon, Camera icon, Checkmark icon, Eye icon

Scheduled Exams screen is displayed:



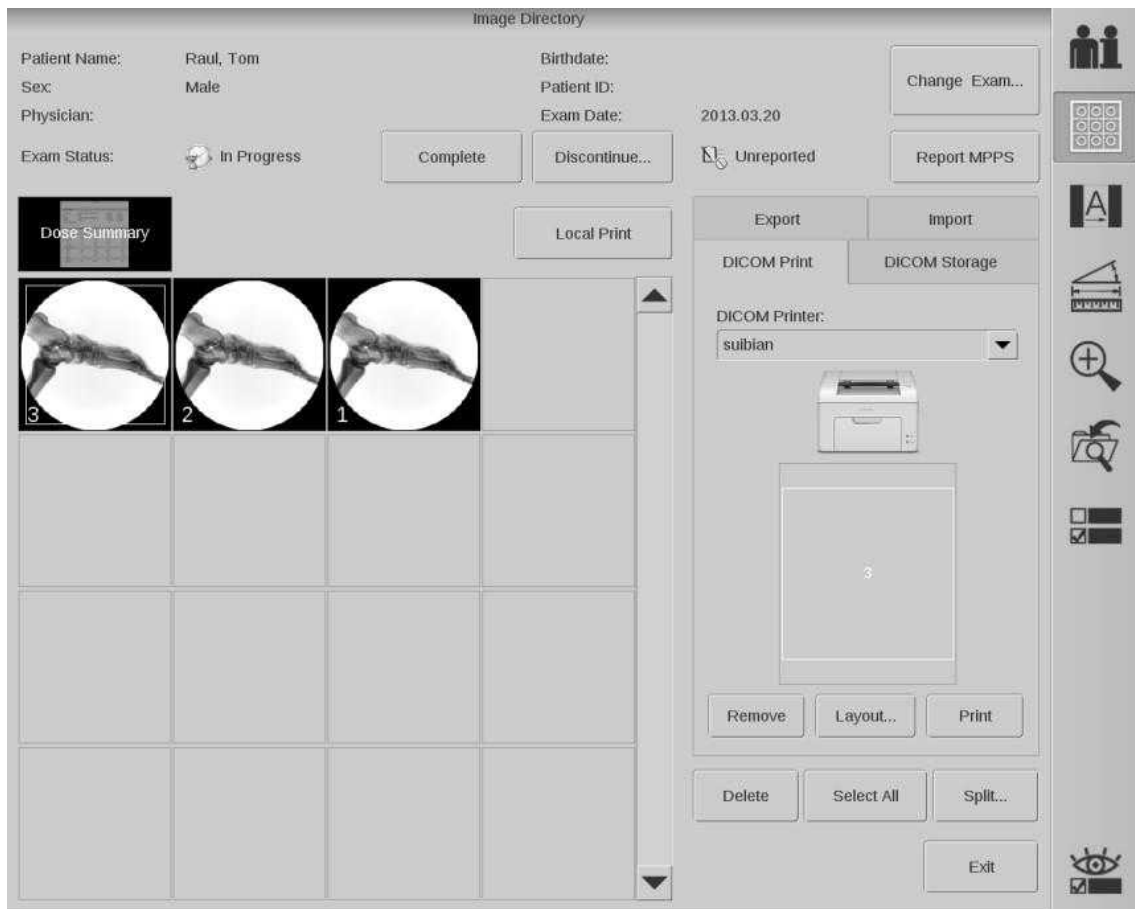
The Scheduled Exams screen displays a list of scheduled exams.

3. Choose a worklist server at the bottom from the server dropdown list.
4. Click "Quick Query" button to perform a DICOM query. When the search is complete, the results are displayed in the Scheduled Exams screen.
5. Click "Advance Query" to setup advance query criteria.
6. Further search can be performed by entering the relevant information in Display Filter to display desired results matched in scheduled exam list.
7. Click "Delete" button, the selected exam will be deleted. Press and hold "Ctrl" key on the Workstation membrane keyboard to select multiple exams from the list.

6.3.5. DICOM Print

Use these steps to print on a DICOM printer.

1. Make sure the DICOM Print server is setup successfully.
2. Press "Image Directory"  key on the Workstation membrane keyboard. Select DICOM print tab.

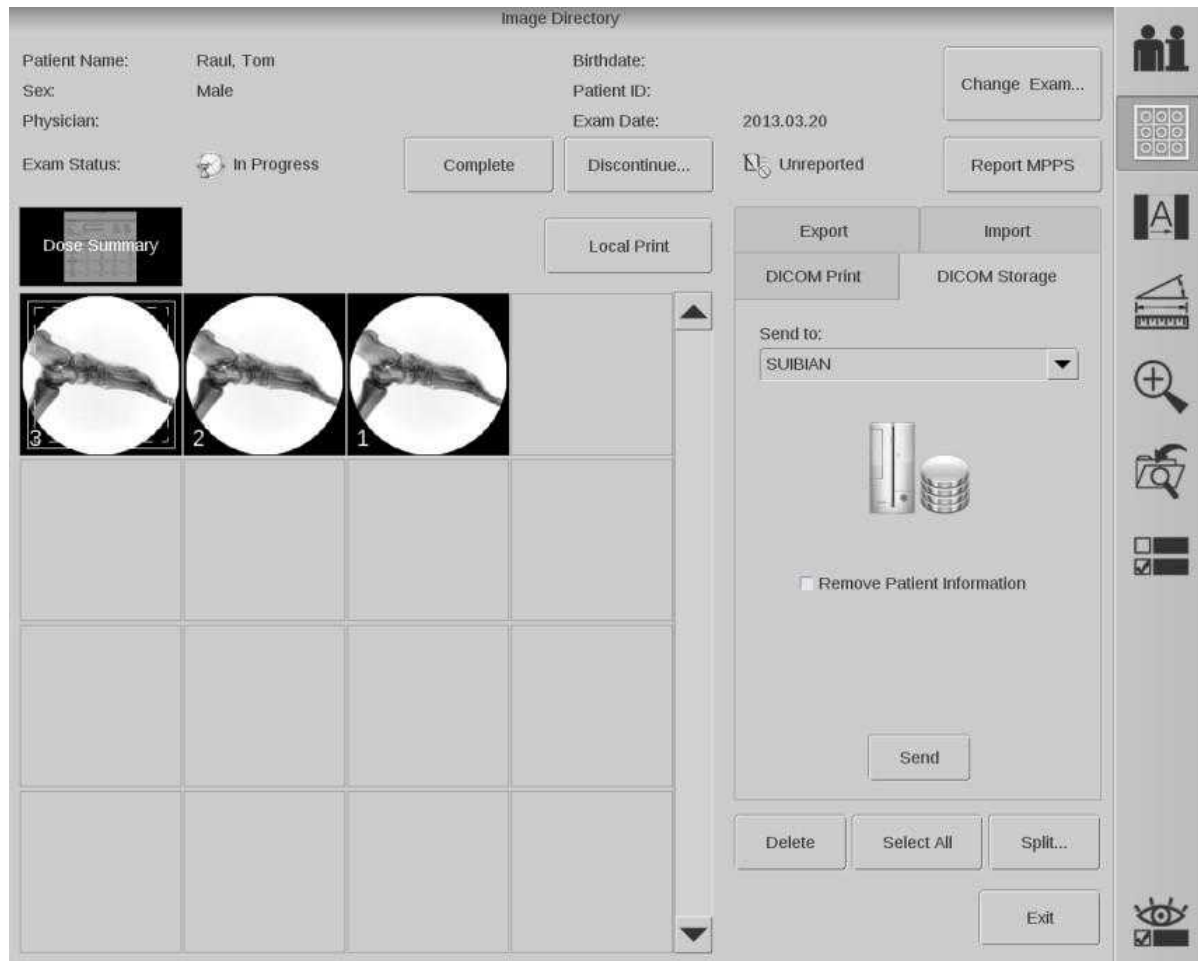


3. Choose a layout format.
4. To add an image to the queue box, select the preview image and then click the queue box. You can continue to select images and load them into the queue until all available positions are filled.
If you select a preview image and then click a box already containing an image, you will replace the existing image with the preview image you selected most recently.
5. Select the "Print" button.
6. As images are sent to the printer, the print in progress screen is displayed. To cancel the print process, click the "Cancel" button on the screen.

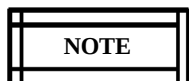
6.3.6. Storage and Commitment

The Workstation allows to storage images and storage commitment on a DICOM storage server.

1. Make sure the Storage and Commitment server is setup successfully.
2. Select the “DICOM Storage” tab from Image Directory and then select a DICOM storage server from the “Send to” dropdown list.



3. Select the image you want to archive. Press and hold “Ctrl” key on the Workstation membrane keyboard to select multiple images from the previews in the directory.
4. Select “Send” button. The Progress screen is displayed.

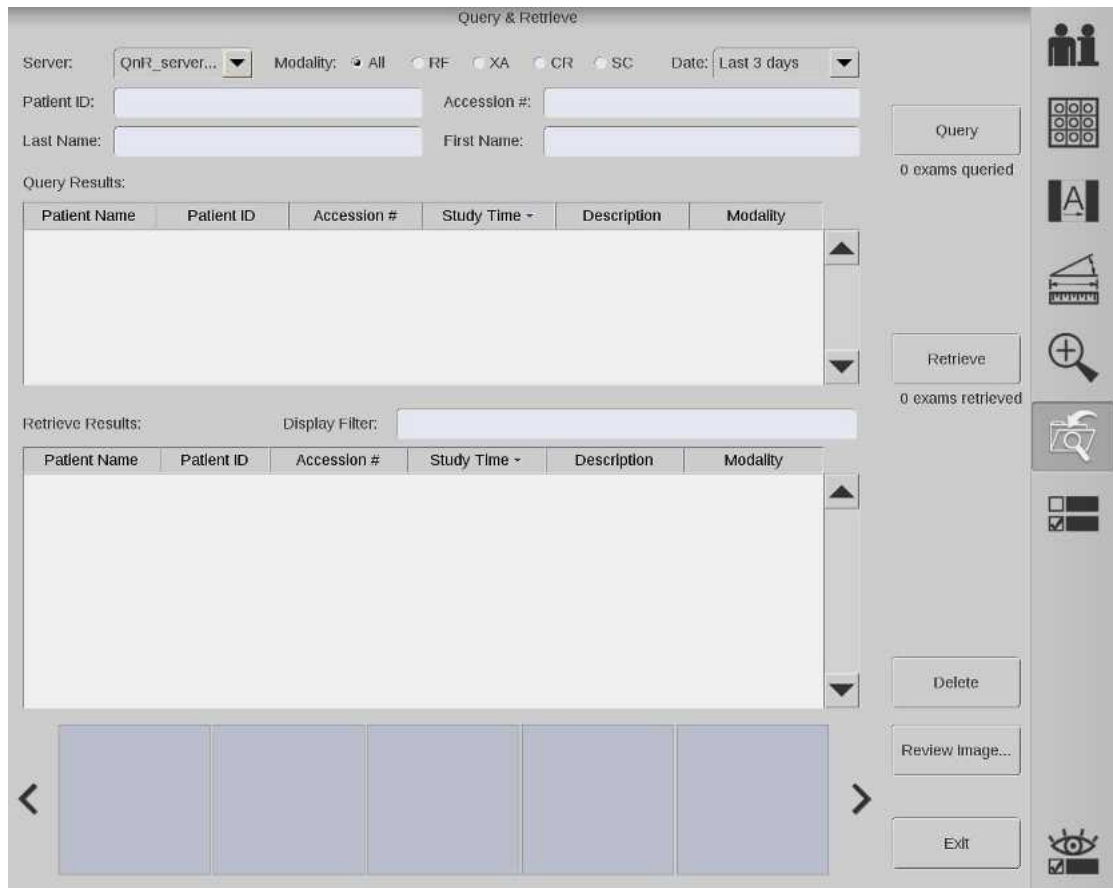


Check regularly for the available storage capacity and secure/archive important records.

6.3.7. Query and Retrieve

DICOM Query and Retrieve allows you to query the remote server to get DICOM exam information, retrieve images of desired exams from the Query and Retrieve Server, filter retrieved exams, review retrieved images.

1. Make sure the Query & Retrieve server is setup successfully.
2. Click the "Query & Retrieve"  button on the Workstation membrane keyboard or select the "Query & Retrieve" button on the System Panel to display the Query & Retrieve screen.



Query & Retrieve

Server: QnR_server... Modality: ☒ All ☐ RF ☐ XA ☐ CR ☐ SC Date: Last 3 days

Patient ID: Accession #: Last Name: First Name:

Query Results:

Patient Name	Patient ID	Accession #	Study Time	Description	Modality

Retrieve Results:

Display Filter:

Patient Name	Patient ID	Accession #	Study Time	Description	Modality

Query: 0 exams queried

Retrieve: 0 exams retrieved

Delete

Review Image...

Exit

3. Select a Query & Retrieve server from the server dropdown list.
4. Setup the query criteria in Modality, Date, Last Name, First Name, Patient ID and Accession .
5. Select the "Query" button to display the results in the table when the search is complete.
6. Select one or more exams in the query result table by using "Ctrl" key. Click the "Retrieve" button to retrieve selected exams from server and display the retrieved exams in the retrieve result.

Query & Retrieve

Server: Modality: ☒ All ☐ RF ☐ XA ☐ CR ☐ SC Date:

Patient ID: Accession #:

Last Name: First Name:

Query

7 exams queried

Query Results:

Patient Name	Patient ID	Accession #	Study Time	Description	Modality
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:...	fgfdfsfajkdsakl...	XA

Retrieve

7 exams retrieved

Retrieve Results:

Display Filter:

Patient Name	Patient ID	Accession #	Study Time	Description	Modality
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:3...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:3...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:3...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:3...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:3...	fgfdfsfajkdsakl...	XA
dsssfjkdjkdjkd...	jjdkkdkk	ffdsfjsjfkdkjffd	2009.07.16 23:3...	fgfdfsfajkdsakl...	XA
Unnamed 017	PUnn07241...		2008.07.24 18:0...		XA

Delete

Review Image...

Exit

7



7. Select a retrieved exam and select Previous/Next button to display the previous/ next image thumbnail.
8. Select "Review Image" button to display Image Viewer, the selected image will be displayed. Operator could use mouse to adjust the image contrast and brightness.
9. Select a retrieved exam and click "Delete" button if you wish to delete the exam.

6.3.8. MPPS

A DICOM Modality Performed Procedure Step (MPPS) is an information object that describes the activities, conditions and results of an exam (imaging procedure) performed on a modality.

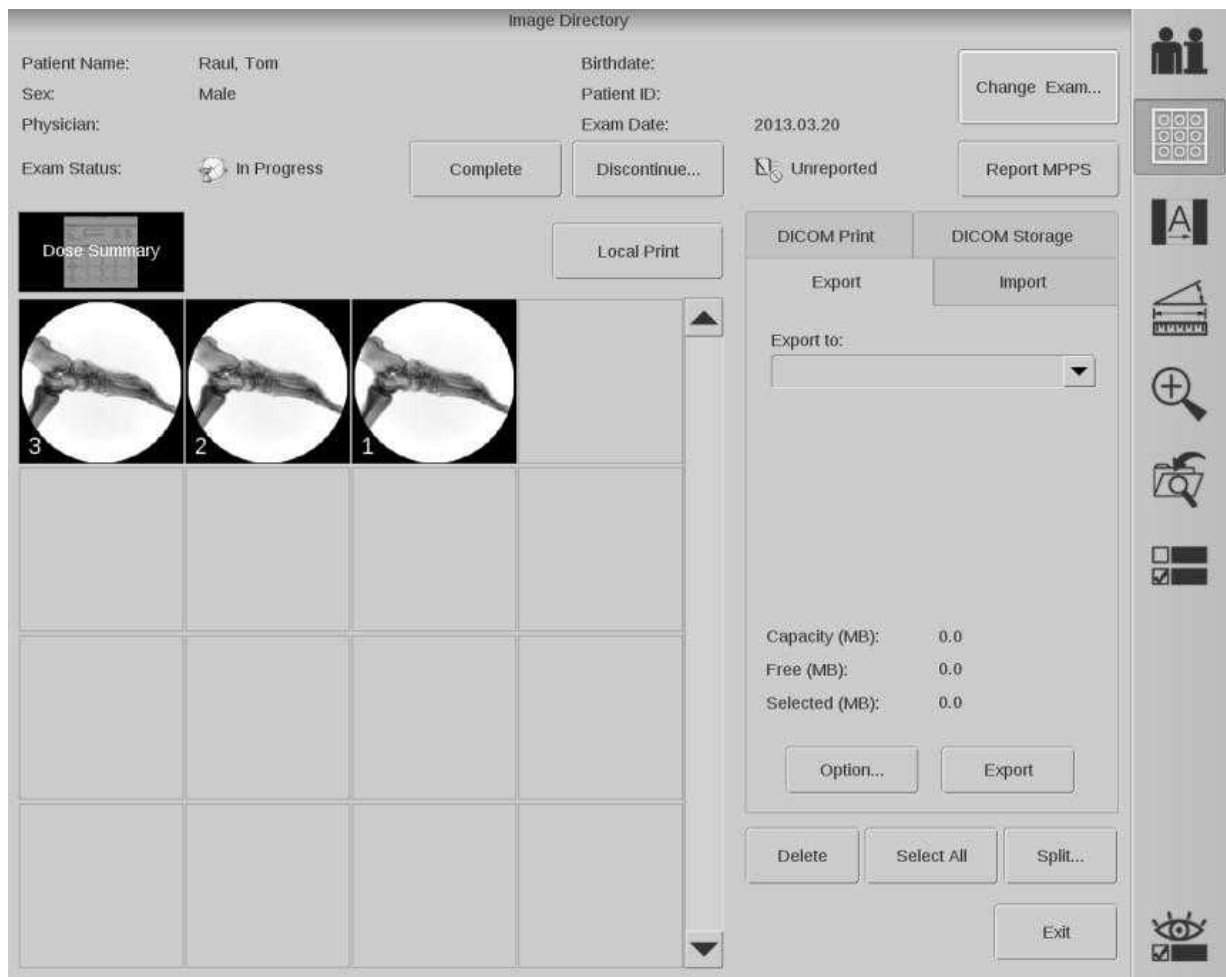
The MPPS package provides capabilities of reporting this information back to the connected RIS (Radiology Information System).

With the DICOM MPPS package the system supports DICOM MPPS as SCU with the following capabilities:

- Provide RIS with exam information
- Report status of the current exam back to RIS

MPPS is only available for some configuration. Check if you have this option before use.

1. Make sure the MPPS server is setup successfully.
2. Click "Complete" or "Discontinue" to complete or discontinue the exams on Image Directory screen.

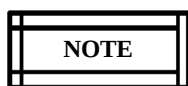


3. Click "Report MPPS" to report the current exam state to MPPS server.

Chapter7. Radiographic Film

7.1. Overview

Use Film mode to produce radiographic films. Film mode is not available in some countries as local regulations requirement. The film cassette holder described in this chapter is available as an option and should be used if you use film mode.

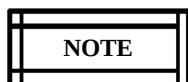


For film imaging, a cassette holder should be placed on the face of the image intensifier.

7.2. Setup and Make a Film Exposure

Perform the following steps to make a film exposure:

1. Press the “Film”  key on the C-Arm control panel.



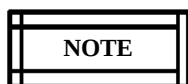
Once film mode has been selected, the field size and collimator settings are locked. If further adjustments are required, you must first reenter into Fluoro mode.

2. Place the cassette holder over the face of the image intensifier with the handle opened out.
3. Rotate the cassette holder handle to securely attach the cassette holder to the image intensifier.



Verify that the cassette holder is securely attached to the image intensifier. Unsecured cassette holders may fall, injuring patients or personnel.

4. Insert a film cassette into the cassette holder and center it.

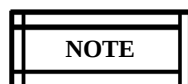


The cassette holder uses friction to hold the cassette. Refer to the “Technical Reference” chapter for cassette sizes.



Verify that the cassette is held securely within the cassette holder. Unsecured film cassettes may fall, injuring patients or personnel.

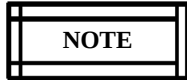
5. Adjust the radiographic parameters (kVp and mAs) to the desired values.
6. Press and hold any X-Ray switch.



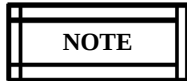
During the radiography, press the X-Ray switch and hold on the switch until the exposure indicator light is off. The duration of the exposure is signaled by a continuous beep. If the X-Ray switch is released before the exposure indicator off, system will give four quick beeps, which means the exposure may not be successful.

7. Release the X-Ray switch at the end of the exposure.
8. Remove the film cassette by pulling the cassette out of the cassette holder.

9. To remove the cassette holder, open the cassette holder handle and press firmly against the side of the cassette holder.



The cassette holder is designed to snugly attach to the image intensifier lip when the handle is engaged. It may be necessary to strike the cassette holder with the palm of your hand to dislodge the cassette holder from the image intensifier.



If the film mode is misused by user for real-time imaging, the image will never display on the monitor.

Chapter8. System Setup

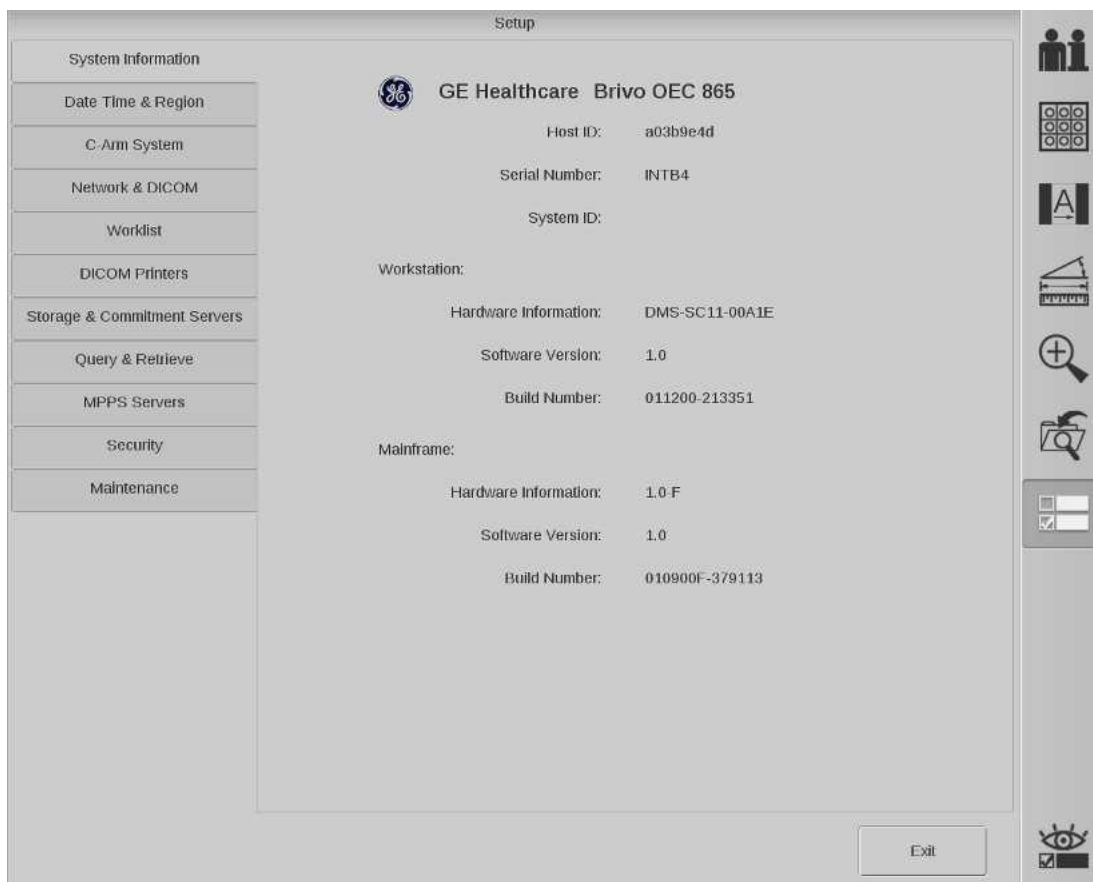
This chapter explains how to use the Setup function to tailor the setup of Workstation operation to your specific requirements. Features related with DICOM setup such as Network & DICOM, Worklist, DICOM Printers, Storage & Commitment Servers, Query & Retrieve and MPPS Servers have been already described in Section 6.3.

Press the “Setup”  key on the Workstation membrane keyboard or select the “Setup” button on the system panel to access setup features.

8.1. System Information

The System Information screen displays information about Host ID, Serial Number, System ID, Workstation information and Mainframe information.

Select “System Information” tab to display System Information Setup screen similar to the following. Take the actual show version of the software and hardware as standard.



Select “Exit” button to return to the desktop.

8.2. Date Time & Region

Select “Date Time & Region” tab to:

- Enter Date, Time, Time Zone and Language
- Select Date, Time and Number Format
- Select Weight and Height Unit

The screenshot shows the 'Setup' window with the 'Date Time & Region' tab selected. The left sidebar contains the following menu items: System Information, Date Time & Region, C-Arm System, Network & DICOM, Worklist, DICOM Printers, Storage & Commitment Servers, Query & Retrieve, MPPS Servers, Security, and Maintenance. The main area displays the following settings:

- Date: 2012.06.05
- Time: 21:57:47
- Time Zone: GMT-00:00
- Language: English
- Date Format: ☐ MM/DD/YYYY ☐ DD.MM.YYYY ☒ YYYY.MM.DD
- Time Format: ☒ 24 Hour ☐ 12 Hour
- Number Format: ☒ 1,000.00 ☐ 1.000,00
- Weight Unit: ☐ kg ☒ lb
- Height Unit: ☐ cm ☒ inch

An 'Exit' button is located at the bottom right of the window. A vertical toolbar on the right side of the window contains icons for user management, grid, AEC, collimator, ruler, zoom, and other functions.

Select “Exit” button to save the setup and return to the desktop.

8.3. C-Arm System

Select “C-Arm System” tab to:

- Enter Hospital Name
- Select Retain Last value at Exam Change or Boot up: Collimator, Mag, Rotation, Flip or Flop
- Enable/ disable Screen Saver
- Setup Screen Saver Wait Time
- Setup Exam List Display: Patient ID or Accession
- Enable/ disable Auto Save at End of Exposure
- Enable/ disable Auto Swap
- Enable/disable Room Interface
- Setup Voltage for Door Opening: High or Low
- Select Exposure Sound
- Setup Dose Warning Threshold

Select “Exit” button to save the setup and return to the desktop.

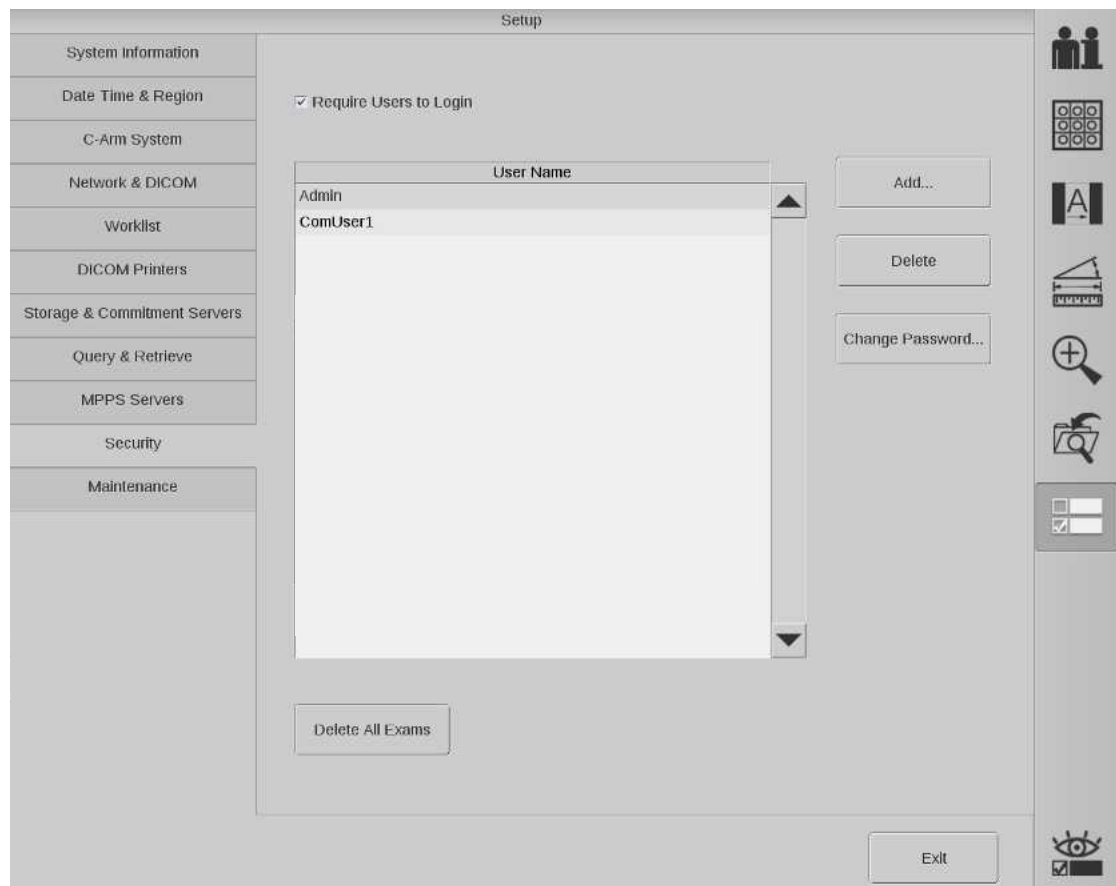
8.4. Security

The “Security” screen is used to setup Workstation for HIPAA compliance.

This feature is only open to the administrator. Click “Security” and enter the administrator password to display security setup screen. If the administrator password has not been set, the default administrator password is “123456”.

Use the “Security” setup screen to:

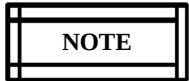
- Require user to login to the Workstation using a password.
- Add /Remove user and password.
- Change the passwords.
- Delete all exams.



Selection of “Require Users to Login”, a login screen will be displayed when the Workstation boot up. The user must enter a password before proceeding to use the Workstation.

Use the “Add” button to setup a user and its password. The account will be displayed in the account list.

Select an account and click “Remove” button, the account will be removed.



The administrator account could not be removed.

Use the “Change Password” button to change the password.

1. Select the user you want to change its password.
2. Select the “Change Password...” button.
3. Enter the new password in the New Password field.
4. Retype the new password in the Confirm Password field.
5. Select “OK” to return to the “Security” Setup screen.

Click “Delete All Exams”. Then a confirmation screen will display. Select “Yes” to delete all exams.

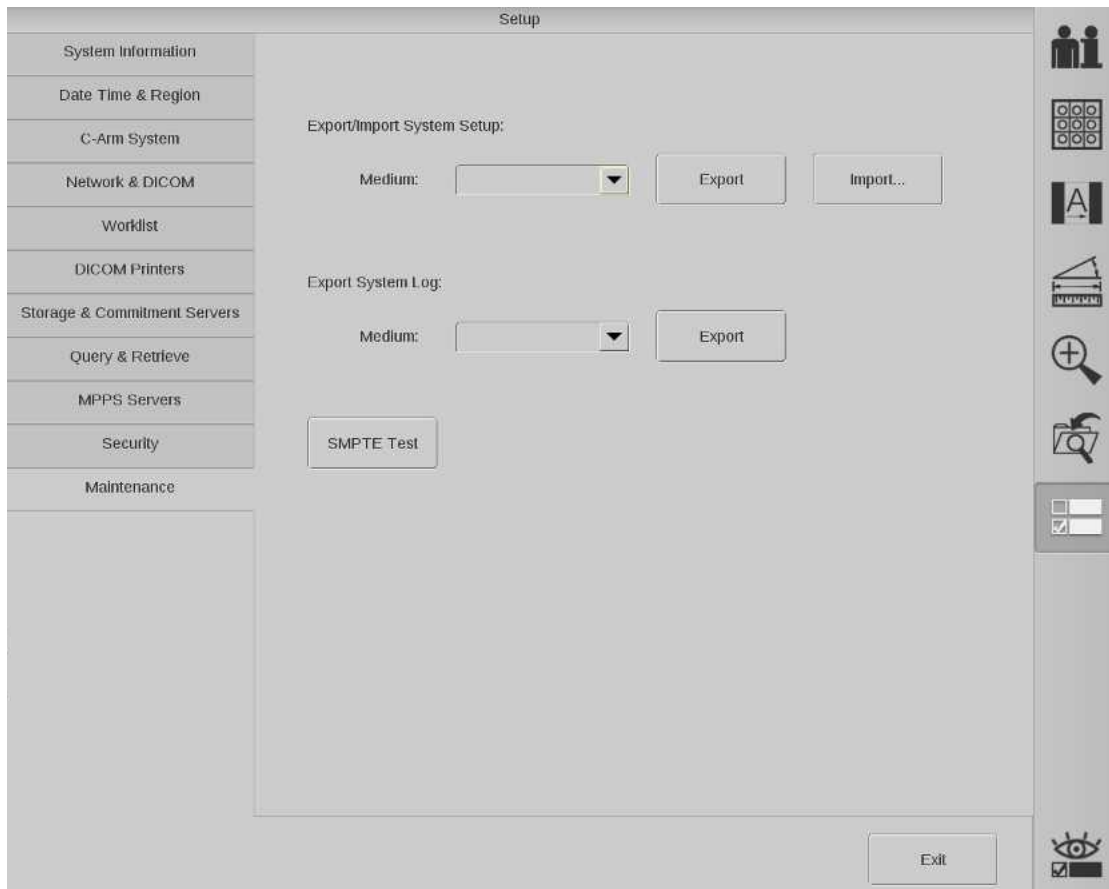


The “Delete All Exams” function deletes all patient information and images from the Workstation.

Select “Exit” button to save the setup and return to the desktop.

8.5. Maintenance

The “Maintenance” setup screen is used for system maintenance.



Export/ Import System Setup

Insert a medium, select the medium from “Medium” dropdown list. Clicking “Export” button allows export of Setup file/Connectivity configuration file. After the process is completed, a message is shown to indicate file exported successfully.

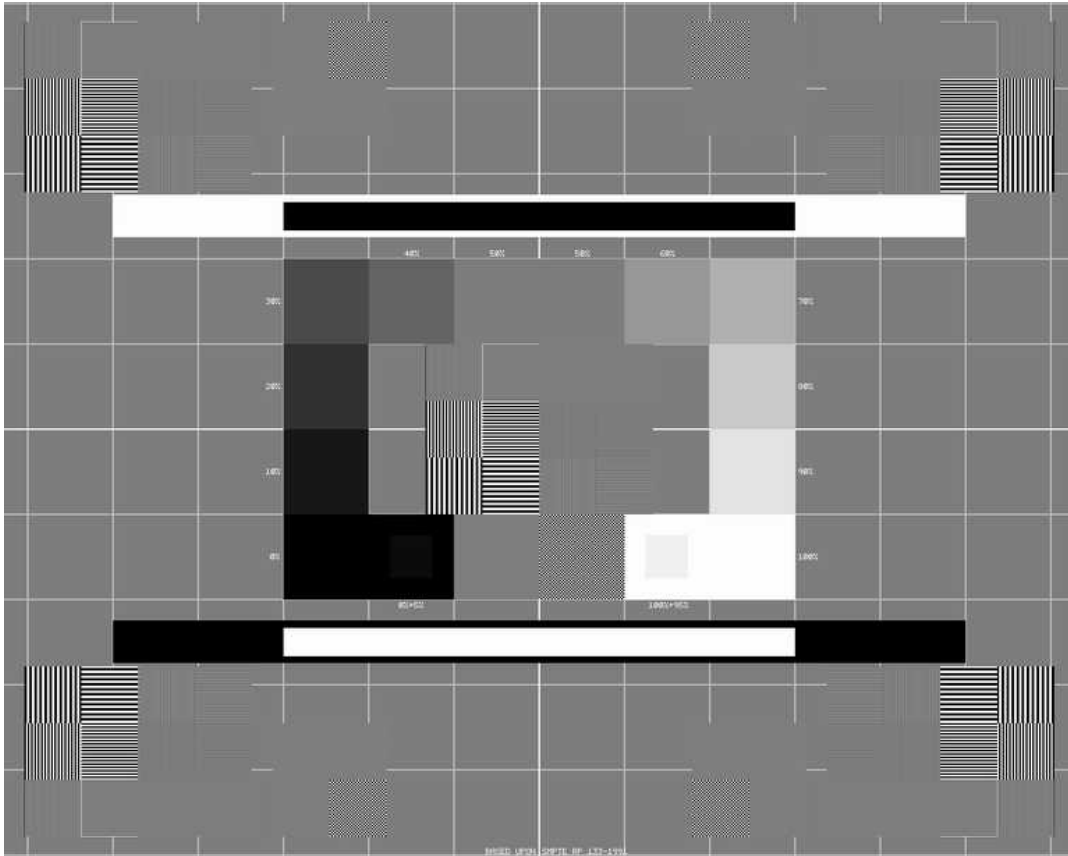
Insert a medium, select the medium from “Medium” dropdown list. Clicking “Import” button allows import of Setup file/Connectivity configuration file. After the process is completed, a message is shown to indicate file imported successfully.

Export System Log

Insert a medium, select the medium from “Medium” dropdown list. Click “Export” to export system Log.

SMPTE Test

System provides SMPTE test on the two monitors for the user. Click “SMPTE test” button, SMPTE test appeared on both screen.



Click any mouse button to return to normal screen.

Chapter9. Maintenance

9.1. Overview

This section describes routine performance checks that you can perform to ensure that the system is operating correctly. The performance checks listed are not intended to substitute for scheduled periodic maintenance. If problems are found during these checks, contact a qualified service engineer to troubleshoot and repair the system.

Prior to performing any of the performance checks in this section, it is important that potential hazards associated with these tasks are understood. Review the Introduction and Safety chapter of this manual before proceeding.

WARNING

Circuits inside the equipment use voltages that are capable of causing serious injury or death from electrical shock. Do not remove the covers or perform any type of service task, except as specifically instructed here.

Move the system into a safe operating area prior to beginning these checks and observe all radiation safety precautions. The performance checks should be performed as often as equipment is in use and circumstances warrant. The frequency of performance checks should be increased when the system is used extensively. In addition, circumstances such as accidents during transport or exposure to excessive fluids may warrant that performance checks be performed to verify operation of the equipment.

In addition to performance checks, safe cleaning practices are included and a description of periodic maintenance that should be performed. All periodic maintenance should be performed by a GE HUALUN medical systems Co. Ltd. representative or a qualified service engineer.

The system should have an expected service life of 10 years. Periodic maintenance of the system is required to maintain the system lifecycle. The first periodic maintenance shall be implemented after first 13 months of use. Then it is suggested to do once per year by a GE qualified service engineer.

9.2. Performance Checks

9.2.1. Mechanical Performance Check

C-Arm

1. Check the Wig-Wag brake operation. Check Wig-Wag for ease of movement (side to side swing) without excessive play.
2. Check the Horizontal Cross-Arm brake operation. Extend and retract the Horizontal Cross-Arm. Check for ease of movement without excessive play.
3. Check the lateral rotation brake operation. Check the rotation for ease of movement without excessive play.
4. Check the C-Arm orbital rotation brake operation. Check orbital rotation of the C-Arm for ease of movement without excessive play.
5. Check the operation of the brake release pedals.
6. Check for proper operation of the steering handle and control of the rear wheels. Check for ease of movement without excessive play.

Workstation

1. Check for proper control of the Workstation wheels. Check for ease of movement without excessive play.
2. Check the operation of brake pedal or wheel brake. Details please refer to section 3.2.9 "Workstation Brakes".
3. Check movement of the monitor arm for ease of movement without excessive play. Make sure that the monitors do not move after being positioned.

9.2.2. Electrical Performance Check

1. Inspect the high voltage cables (Bundle cable between mainframe base and C-Arm) for signs of wear and abrasion.
2. Inspect the footswitch and handswitch cables for signs of wear and abrasion.
3. Inspect the interconnect cable and power cable for sign of wear or abrasion, loose, missing or bending connector pins, and worn strain reliefs.
4. Turn the system on and verify that the system successfully completes the power-up sequence.
5. Raise and lower the vertical column by pressing the keys on the C-Arm.

WARNING

Some devices such as high voltage cables and electrolytic capacitors can retain a dangerous static charge for long periods after power has been removed. Do not touch these components unless power has been completely removed and they have discharged.

9.2.3. Emergency Switch Performance Check

This check is performed as a matter of routine operator maintenance to ensure that the Emergency Switch safety feature is functioning properly. Perform this check as often as possible in conditions such as removal from long-term storage and exposure to fluids.

1. Press and hold the vertical column control key and press Emergency Switch while motion is occurring.
2. Verify that mechanical motion stops and the X-Ray Off indicator lights up.
3. Press X-Ray switch and verify that X-Rays are disabled.
4. Press the power on/off switch to power off the system.
5. Rotate the Emergency Switch upward and then reboot the system.



Images will be lost when the system is restarted unless you have saved them. If you have saved images you can access them through the Workstation's Image Directory function.

9.2.4. Fluoro Mode Performance Check



This procedure produces X-Rays. Take the appropriate precautions.

1. Position the C-Arm so that the tube is directly above the image intensifier.
2. Place a suitable test object on the image intensifier.
3. Verify that auto mode is selected. The LED on the top left corner of the "Auto Technique" key on the control panel will be illuminated.
4. Press the left X-Ray switch on footswitch and handswitch. Verify that a digitized fluoro image of the object appears on the left monitor and that it remains there after the exposure is terminated.
5. Verify operation of the field size selection keys: NORM, MAG1 and MAG2.
6. Press the footswitch or handswitch to verify operation of the motorized collimation controls: shutter rotation, iris collimation, and shutter open/close.

9.2.5. Film Mode Performance Check

WARNING

This procedure produces X-Rays. Take the appropriate precautions.

1. Press the "Film" key on the C-Arm control panel to select film mode.
2. Install the film cassette holder on the image intensifier and load a film cassette.
3. Press the footswitch and handswitch and hold on until the exposure indicator light is off. The system will give continuous audible beeps during the exposure. If the X-Ray switch is released early during exposure, the system will give intermittent beeps to warn that the exposure is interrupted.
4. Develop the film and inspect the exposure.

9.3. Cleaning and Disinfections

WARNING

Before cleaning and disinfecting the equipment, make sure the system is turned off and the power cord is unplugged to avoid electrical shock.

Biohazard may be present on surface of the equipment. Wear rubber gloves during cleaning.

CAUTION

The system is not waterproof. Be careful not to spill or splash liquids where they can enter electronic assemblies.

9.3.1. Cleaning

Make sure the system is turned off and unplugged before cleaning. Clean the covers and panels periodically with a damp cloth. Use a mild detergent, if necessary, to remove scuffs, stains and dust. Do not use any solvents that may damage or discolor paint finishes or plastic components.

Be careful not to drip liquids where they can enter electronic assemblies through panel or cover seams.

WARNING

Water, soap, or other liquids, if allowed to drip into the equipment, can cause electrical short circuits leading to electric shock and fire hazards.

If liquids should accidentally spill into the system, do not connect power cord to power supply or turn the system on until the liquids have dried or evaporated completely.

Enameled parts and aluminium surface should only be wiped with a damp cloth and water, and then rubbed down with a soft dry woolen cloth. Never use corrosive cleaning agents, solvent or abrasive detergents or polishes. If you are uncertain of the nature of a cleaning agent, do not use it.

CAUTION

Always turn the system off and disconnect power before cleaning. Clean the external connector area with dry cloths only. Do not use liquid cleaners on this area.

Chrome parts should be cleaned only with a dry woolen cloth. Do not use abrasive polishes. To preserve the finish, use non-abrasive wax.

LCD monitor can be cleaned with little wet soft cloth or special cleanser. Do not use cleansers, which contain alcohol or acetone. Do not spray the liquid directly to the screen surface.

9.3.2. Disinfections

All parts of the equipment, including accessories and connecting cables, can be disinfected by wiping them with a cloth dampened with disinfectant, such as 75% alcohol. Never use corrosive or solvent disinfectants since it could cause damage to the equipment. If you are in doubt of the nature of a disinfecting agent, do not use it.



Flammable or potentially explosive disinfecting sprays must not be used since the mixed gas could ignite and cause personal injury and /or damage to equipment.



Disinfecting medical equipment rooms by means of sprays is not recommended since the sprays can penetrate the equipment, which causes electrical short circuit or corrosion.

If the nonflammable, non-explosive spray disinfectants are necessary to disinfect the rooms, the equipment must be switched off and allowed to cool first, which prevents the disinfectant from spraying into the equipment due to convection. Plastic sheeting must be used to cover the equipment thoroughly, following which disinfectant spraying can take place.

Once all traces of the disinfectant spray have dispersed, the plastic sheeting can be removed and the equipment itself can be disinfected in the recommended way.

If a spray has been used, the operator must make sure that all traces of the vapor have been dispersed before switching the equipment on again.

Disinfection techniques for both the equipment and the room must comply with all applicable laws and regulations that are in force within the jurisdiction in which the equipment is located.

9.4. Storage

9.4.1. Temporary Storage (less than 60 days)

To prepare the C-Arm for storage, move all mechanical assemblies into their most compact position, set all locks and brakes and remove all power.

Cover the C-Arm with a dust cover. Refer to the *Technical Reference* chapter for the range of environmental conditions in which the C-Arm can be safely stored.

9.4.2. Long Term Storage or Shipment (60 days or more)

To prepare the C-Arm for long-term storage or shipment, observe the following recommendations:

- Move all mechanical assemblies into their most compact positions, lock all the brakes and handles and remove all power. Coil the interconnect cable and the power cable.
- Wrap the image intensifier, X-Ray tube assembly, high voltage cable, and the control panel housing with bubble wrap. Make sure the monitors are secured in the monitor latch. Wrap the monitors with bubble wrap.
- Pack all accessories such as cassette holders and store them with the system.
- Cover the C-Arm and accessories. Attach each to a solid supportive shipping base and enclose in a protective container adequate for shipment or storage. Refer to the *Technical Reference* chapter for the range of environmental conditions in which the C-Arm can be safely stored.

WARNING

UPS battery should be disconnected when the system is not likely to be used for some time and before transportation. Battery can be left in the UPS.

9.5. Maintenance Schedule

9.5.1. User Routine Checks Items

The user of the equipment must follow the user routine check items in the table below. It is for the operator of the equipment to ensure that all checks have been satisfactorily met before attempting to use the equipment.

It is the operator's responsibility to provide these checks:

Item	Description	Frequency
All accessories	Availability and integrity	Daily
Power supply check	Ensure correct functioning	Daily
Indicators and buzzer test	Ensure correct functioning	Daily
X-Ray control check	Ensure correct functioning	Daily
Vertical movement check	Ensure correct functioning	Daily
Brakes and castors	Ensure correct functioning	Daily
Cabling	Inspect for kinks and / or cracks	Daily
Inscriptions and labels	Check for legibility	Annually
Handswitch and footswitch	Check functionality	Daily

German X-Ray ordinance Roentgenverordnung (R V) 16 and Qualitäts-sicherungs-Richtlinie (QS-RL) 3.1.3.8 require the SMPTE test pattern to be displayed on the system monitors for testing per Feb 2001 DIN V 6868-57. Clicking "SMPTE Test" button on Maintenance Setup screen, the operator can perform SMPTE test on the monitors.



Before using the equipment, verify that system operation is proper according to the check items above. System that is not fully functional may result in delays in medical treatment.

9.5.2. Periodic Maintenance

Periodic Maintenance should be performed by a GE HUALUN medical systems Co. Ltd. representative or a qualified service engineer.

The following periodic maintenance schedule is recommended.

Item	Description	Frequency
Earth (Ground)	Check maximum earth of whole system	Annually
Power supplies	Check earth leakage current	Annually
	Check internally- generated DC voltages	Annually
UPS	Check no leakage	Annually
	Check the appearance is normal	Annually
	Check the cable connection	Annually
	Check the UPS life, check the need for replacement	Annually
PCB's and racks	Ensure secure fitting and check for dust and corrosion	Annually
Motorized movement	Check electrical and mechanical movement	Annually
Bearings	Check movement freedom	Annually
Controls and Indicators	Check accuracy and functioning of:	
	All controls	Annually
	All visible / audible indicators	Annually
	Movement	Annually
	Dose / Dose rate / DAP	Annually
Alignment	Collimator alignment and field limitation	Annually
	Beam alignment and centering	Annually
Mechanical	Counterweight fasteners and cables	Annually
	All mechanical stops	Annually
	Wheels and wheel alignment	Annually
	Brakes and locks	Annually

Chapter10. Display Messages

10.1. Overview

This chapter describes messages that appear on the right Workstation monitor during system operation. Messages may indicate any of the following:

- Error Messages
- Warning Messages
- Information

When the error message appears, an error code will also be displayed on the kV, mA display field on the control panel, if the Mainframe is connected with Workstation.

10.2. Error Recovery Steps

Perform the following error recovery procedures if you encounter problems during start-up or operation:

1. Some messages can be cleared by pressing "OK" button or taking exposure, but will periodically re-display while the condition exists.
2. If an error message occurs, power off the system. Wait for one minute, and then reboot the system. If this fails to restore normal operation then proceed with step 3.
3. Press the power on/off button to power off the system, disconnect the power cord, and call for service. Refer to chapter 1 for communication center telephone numbers. Do not continue using the system.

WARNING

System may fail at any time. Ignoring error and warning messages may result in equipment damage and personal injury.

10.3. Messages

Each message contains the information of error description, how user will be affected and action suggested. For the detailed message list, please refer to Service Manual.

Chapter11. Labels and Symbols

11.1. Overview

This chapter describes labels and symbols that are located on your C-Arm and Workstation.

Two types of labels are described: warning labels and regulatory certification labels. Warning labels define potential hazards and advise against misuse that might result in personal injury. Familiarize yourself with these labels and their meanings in order to ensure a safe environment for both the patient and yourself. Regulatory labels indicate that the system meets the requirements of specific governmental, medical and industrial organizations.

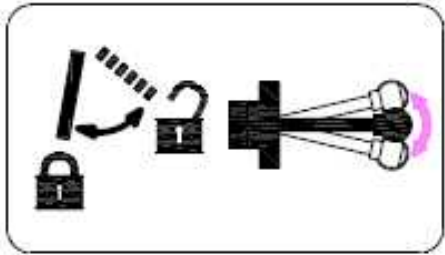
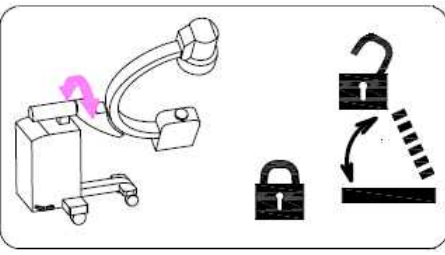
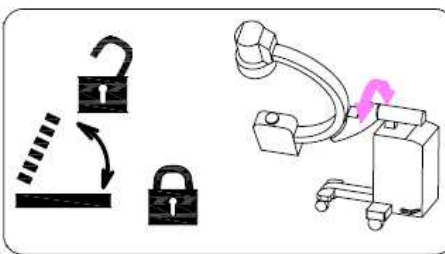
Symbols are provided to visually represent concepts.

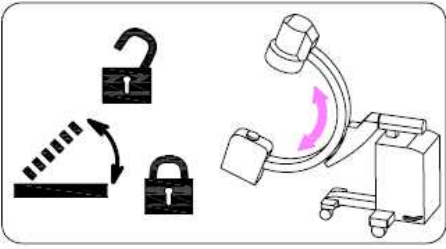
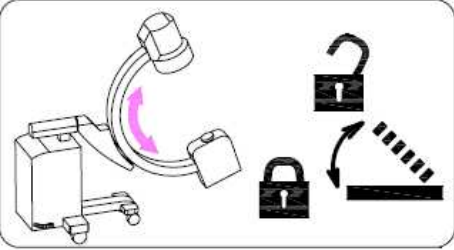
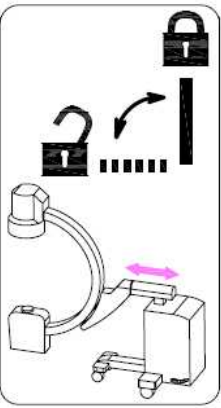
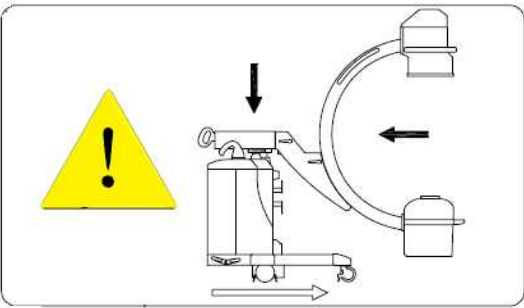


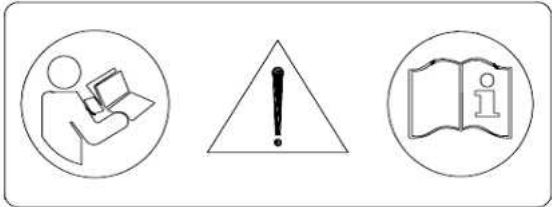
This manual is supplied only in English label, some labels may have a foreign language equivalent that will appear on your system.

11.2. Labels and Locations

11.2.1. Labels

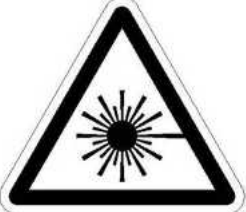
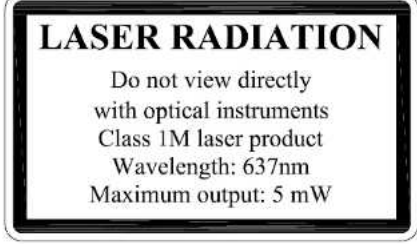
Item	Label	Quantity	Description
1		1	This label indicates the position of the C-Arm Wig-Wag movement handle. When the handle is rotated to dashed line position, the Wig-Wag brake is released. When the handle is in solid line position, the Wig-Wag brake is locked.
2		1	This label indicates the position of the C-Arm rotation movement brake handle. When the handle is rotated to the dashed line position, the brake is released. When the handle is in solid line position, the brake is locked.
3		1	This symbol indicates the position of the C-Arm rotation movement brake handle. When the handle is rotated to the dashed line position, the brake is released. When the handle is in solid line position, the brake is locked.

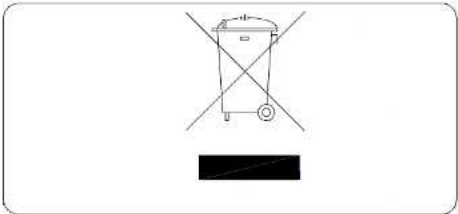
4		1	This label indicates the position of the C-Arm orbital movement brake handle. When the handle is rotated to the dashed line position, the brake is released. When the handle is in solid line position, the brake is locked.
5		1	This symbol indicates the position of the C-Arm orbital movement brake handle. When the handle is rotated to the dashed line position, the brake is released. When the handle is in solid line position, the brake is locked.
6		1	This label indicates the position of the C-Arm horizontal movement handle. When the handle is rotated to the dashed line position, the brake is released. When the handle is in solid line position, the brake is locked.
7		1	This label indicates that the Mobile C-Arm's mechanical subassemblies position should be placed in their most compact position before transporting or moving the system. Refer to <i>chapter 3 Mechanical Positioning</i> for details.
8	WARNING: Place all mechanical assemblies in their most compact (transport) position and lock brake handles prior to moving the equipment.	1	

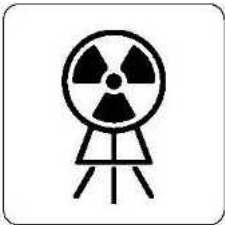
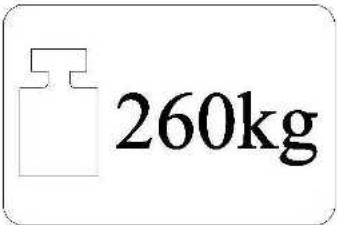
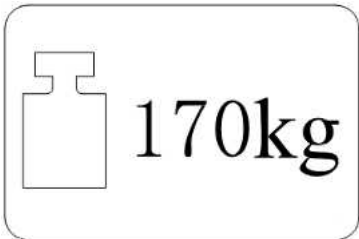
9		1	Note label. Pay attention and keep your feet away from this area.
10		5	This label is attached below Workstation handles. It is to note that the Workstation handles are not allowed to use to lift this system; This label is attached by printer cabinet. It is to note that the printer may slide when moving the Workstation. This label is attached on the control panel. It is to note that the operator manual should be read when you operate the control panel. This label is attached by UPS shut down button. Only when UPS can't shut down automatically after system power off process complete, the user should press the UPS shut down button to shut off UPS manually.
11		2	Monoblock heat label.
12	 To avoid data missing! Shut down: 1. Wait at least 1 minute before shutting down the system when any operation of the system is performed. 2. Press the power off key. 3. Wait for the system to shut down. Restart: 1. Wait at least 1 minute. 2. Press the power on key to restart.	1	This label on the control panel of the C-Arm indicates the procedure of shutting down and restart to avoid data missing.

13		1	Potential equalization terminal label.
14		8	Note label. Pay attention and keep your hands/fingers away from this area.
15		2	This label indicates that you should not touch the connector and the patient at the same time. The connector must be connected to the appointed device only. This port is only used for user to store patient information and image by USB. Any other equipment is forbidden to be connected strictly.
16		1	This label on the footswitch means that the degree of protection of the footswitch is IPX8. The electrical mechanism within the footswitch is protected from the effects of continuous immersion in water.
17		1	This label indicates the location of Emergency Switch.

18	Mobile C-arm X-ray Product REF Model Number: Brivo OEC 715 SN Serial Number: M Manufactured: Input: ~100/110/120 200/220/230/240 V, 50Hz /60Hz Maximum Line Impedance: 0.3 Ω 0.6 Ω 0.6 Ω Current(momentary): 30A 28A 25A Current(continuous): 20A 12A 10A Operation Mode: Continuous operation Loading Mode: Intermittent loading GE Hualun Medical Systems, Co. Ltd No. 1,Yong Chang North Road, Beijing Economic & Technological Development Zone Beijing 100176, China   CLASS I Rx only-U.S.A.	1	System rating plate indicates manufacture information and input power requirements.
19	X-ray Generator REF Model Number : Brivo OEC 2.2A X-ray Generator Part Number: 5085004 M Manufactured: SN Serial Number: Tube Manufacturer : Lohmann Tube Type: 110/3 DF Rated Voltage: 110kV Small Focal Spot (IEC 336): 0.6 $\times$ 1.4 Large Focal Spot (IEC 336): 1.4 GE HUALUN MEDICAL SYSTEMS CO. Ltd No.1 Yong Chang North Road, Beijing Economic Technological Development Zone Beijing 100176, China This component is certified to comply with 21CFR Subchapter J. CLASS I 	1	X-ray Generator rating plate.

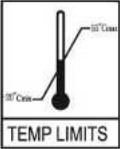
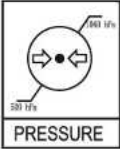
20		1	Collimator, Beam Limiting Device label.
21		1	Laser warning label. (This label will only exist on the laser aimer option.)
22		1	This label warns the operator or the patient not to look at the laser beams directly. (This label will only exist on the laser aimer option.)

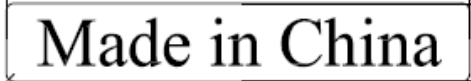
23		1	This label indicates the equipment was tested by a Notified Body and was found to be in compliance with the requirements of all relevant directives and standards in effect within the European Union at the time of manufacture.
24		1	This label indicates that waste electronic and electrical equipment must not be disposed of as unsorted municipal waste and must be collected separately. Please contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.
25		2	This is the label of branding.
26		N/A	Not used.
27		1	This label indicates the system was tested by ITS authorized body and was found to be in compliance with the requirements of all relevant directives and standards at the time of manufacture.

28		1	This label can be found on the X-Ray generator. It indicates the location of the X-Ray source and the controls used to produce ionizing X-Radiation. Use appropriate precautions and protective equipment when using the system and at all times when X-Rays are present.
29		1	USB warning label.
30		1	C-Arm weight label.
31		1	Workstation weight label.

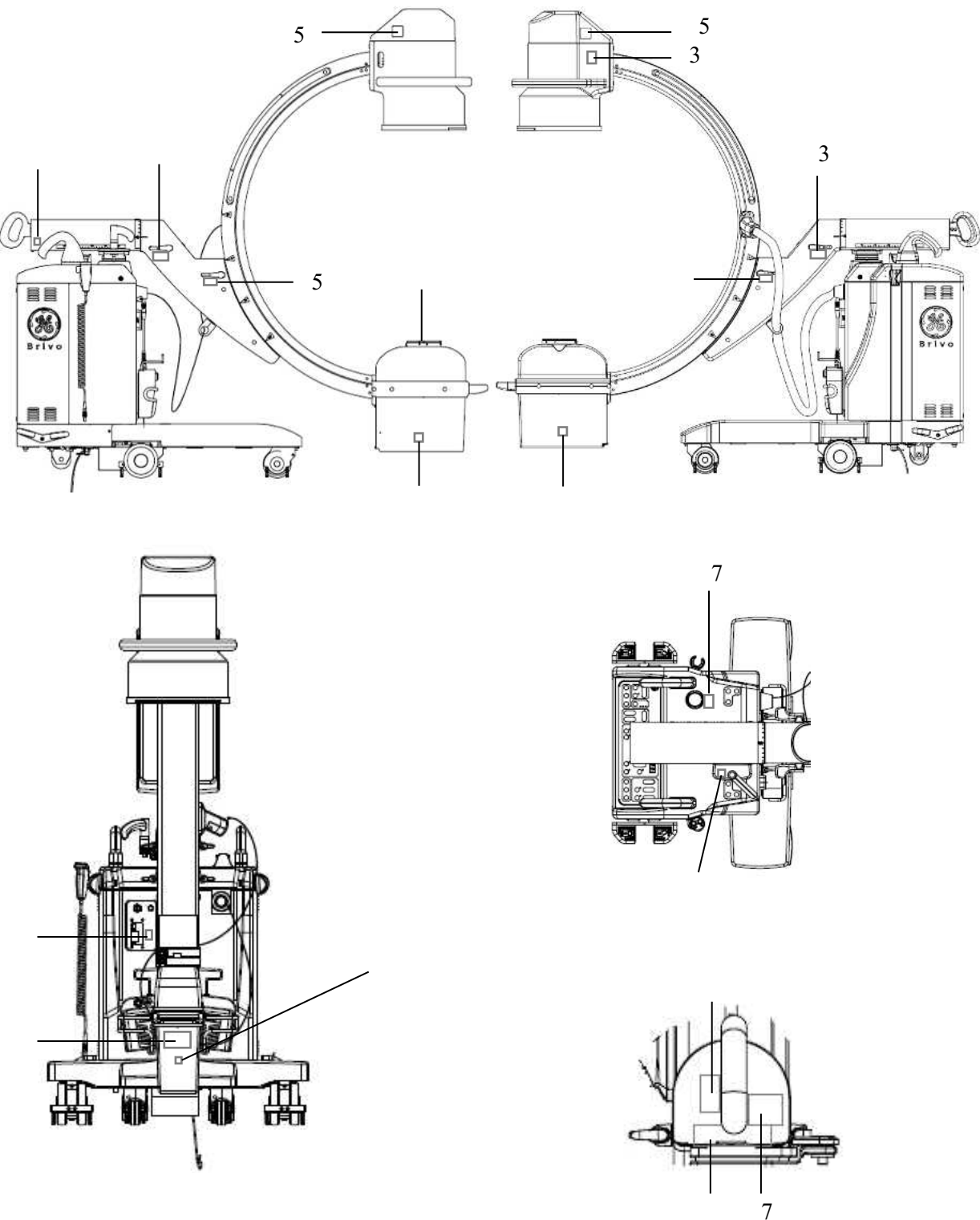
32		1	Image Intensifier label for Brivo OEC 715/785.
		1	Image Intensifier label for Brivo OEC 865.
		1	Image Intensifier label for Brivo OEC 715/785/865.
33		2	Hg label is attached on the monitor back.

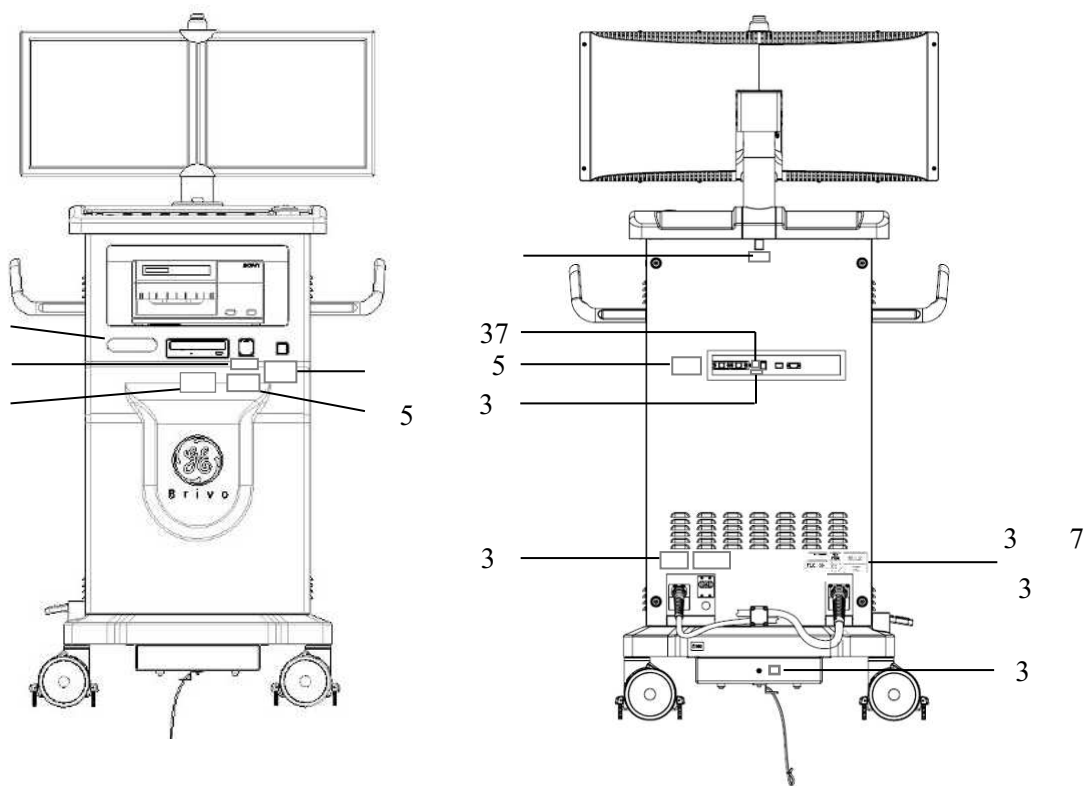
34	REMOVABLE FILTER 3.5mm Al Equivalent Filtration : 0.1mm Cu PN: 5085234 GE Hualun Medical Systems, Co.Ltd	1	Removable filter, 3.5 mm Al equivalent filtration 0.1 mm Cu.												
35	Removable Grid PN: 5085099 GE Hualun Medical Systems, Co.Ltd	1	Removable Grid.												
36	Skin Spacer PN: 5085235 GE Hualun Medical Systems, Co. Ltd	1	Skin spacer.												
37		1	WIFI label. Only available when you choose Wi-Fi Internet Adapter option for some configuration.												
38	Contains: IC:4054A-09400125 FCC ID : PY309400125	1	IC/FCC ID label.												
39	The Circuit Breaker or other over current protector used in power supply for this product shall meet below requirements:<table><tr><th>Voltage</th><th>Rated Tripping Current (Long term)</th><th>Rated Tripping Current (Short term,5 seconds)</th></tr><tr><td>220/230/240V</td><td>>10A</td><td>>25A</td></tr><tr><td>200V</td><td>>12A</td><td>>30A</td></tr><tr><td>100/110/120V</td><td>>20A</td><td>>30A</td></tr></table>	Voltage	Rated Tripping Current (Long term)	Rated Tripping Current (Short term,5 seconds)	220/230/240V	>10A	>25A	200V	>12A	>30A	100/110/120V	>20A	>30A	1	The Circuit Breaker or other over current protector used in power supply for this product shall meet the requirements.
Voltage	Rated Tripping Current (Long term)	Rated Tripping Current (Short term,5 seconds)													
220/230/240V	>10A	>25A													
200V	>12A	>30A													
100/110/120V	>20A	>30A													
40	  WARNING This X-ray unit may be dangerous to patient and operator unless safe exposure factors, operating instructions and maintenance schedules are observed.	1	This warning label is on the Workstation front cover.												

41	      	2	Packing label.
42		2	Brivo OEC 715 Product Brand name.
		2	Brivo OEC 785 Product Brand name.
		2	Brivo OEC 865 Product Brand name.
43	Component: SPOT FILM DEVICE REF Model Number: Cassette Holder 9" SN Serial Number: Manufactured: GE HUALUN MEDICAL SYSTEMS CO. Ltd No.1 Yong Chang North Road, Beijing Economic Technological Development Zone Beijing 100176,China This component is certified to comply with 21CFR Subchapter J.	1	Component, Spot Film Device label.

44	UDI: Sample Text  (01)00001234567895(11)150700(21)AAAAAYY00001AA	1	Unique Device Identifier (UDI) Every medical device has a unique marking for identification. It is attached on the mainframe base cover. Note that this is only an example of a UDI marking. The characters used in the UDI marking represent specific identifiers. Device identifier: ● (01) GS1 global trade item number (GTIN) of the device. ● 00001234567895 Global trade item number. Production identifiers: ● (11) GS1 application identifier for the manufacturing date of the device. ● 150700 Manufacturing date: year-month-day (YYMMDD) with DD always equal to 00. ● (21) GS1 application identifier for the serial number of the device. AAAAAYY00001AA Serial number.
45		1	This label indicates Country of Origin.

11.2.2. Locations of the Labels





11.3. Symbols

Be familiar with the following symbols that may appear on equipment and schematics, so you can safely maintain and operate the system:

Name	Symbol	Description
Dangerous Voltage		This symbol identifies areas where hazardous voltages may be present. Use appropriate safety precautions.
Earth Ground Connection		This symbol identifies an earth ground connection that is necessary to maintain grounding throughout the system.
X-Ray Source		This X-Ray Source symbol can be found on the X-Ray generator. It indicates the location of the X-Ray source and the controls used to produce ionizing X-Radiation. Use appropriate precautions and protective equipment when using the system and at all times when X-Rays are present.
Radioscopy		This symbol is used to identify controls that will produce ionizing X-Radiation when activated. Always use appropriate precautions when generating X-Rays.
Attention		Attention, refer to accompanying documentation for information. You can refer to this manual for the operation instructions.
Type B		This symbol indicates Type B applied parts. User is protected against electric shock by protective earth grounding of all accessible conductive parts or the use of non-conductive parts.
Ionization Radiation		This symbol means that there is ionization radiation.

Chapter12. Technical Reference

12.1. Overview

The policy of GE Healthcare is one of continual product development and improvement.

For this reason, GE Healthcare reserves the right to change the operating characteristics and specifications of newer products at any time, without prior notice, and without incurring any obligation relating to previously manufactured items.

The specifications listed are limited to general performance and physical data. Specifications of optional equipment provided by other manufacturers are given in the applicable manuals provided with those options.

12.2. System Specifications

12.2.1. Classification Type

- Class I equipment (as defined by IEC60601-1)
- Type B protection against electric shock (IEC60601-1)
- Ordinary protection against ingress of water
Degree of Protection:
 System: IPX0
 Footswitch: IPX8
- Operation Mode:
 Continuous operation
- Loading Mode:
 Intermittent loading
- All parts of the equipment, including accessories and connecting cables, can be disinfected by wiping them with a cloth dampened with disinfectant, such as 75% alcohol.

12.2.2. Environmental Requirements

Operating Conditions

Temperature:	+10° C +40° C (+50F +104F)
Humidity:	20% 80% Relative, non-condensing
Atmospheric Pressure:	700hPa 1060hPa

Transportation and Storage

Temperature:	-20° C +55° C (-4F +131F)
Humidity:	10% 95% Relative, non-condensing
Atmospheric Pressure:	500hPa 1060hPa

12.2.3. Power Requirements

Voltage: AC 100V/110V/120V 200V 220V/230V/240V

Frequency: 50Hz/60Hz

Rated Momentary Line Current (shorter than 5s):

- 30A (100V/110V/120V)
- 28A (200V)
- 25A (220V/230V/240V)

Rated Continuous Line Current:

- 20A (100V/110V/120V)
- 12A (200V)
- 10A (220V/230V/240V)

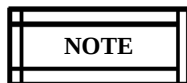
Maximum line impedance: $0.3\ \Omega$ $0.6\ \Omega$ $0.6\ \Omega$

Max Line Regulation (Percent): 15% at maximum radiographic exposure.

Line regulation is based on measurements made at the input (primary winding) of the isolation transformer under standby conditions and at maximum radiographic exposure by the following equation:

Line regulation $= 100 (V_n - V_1)/V_1$

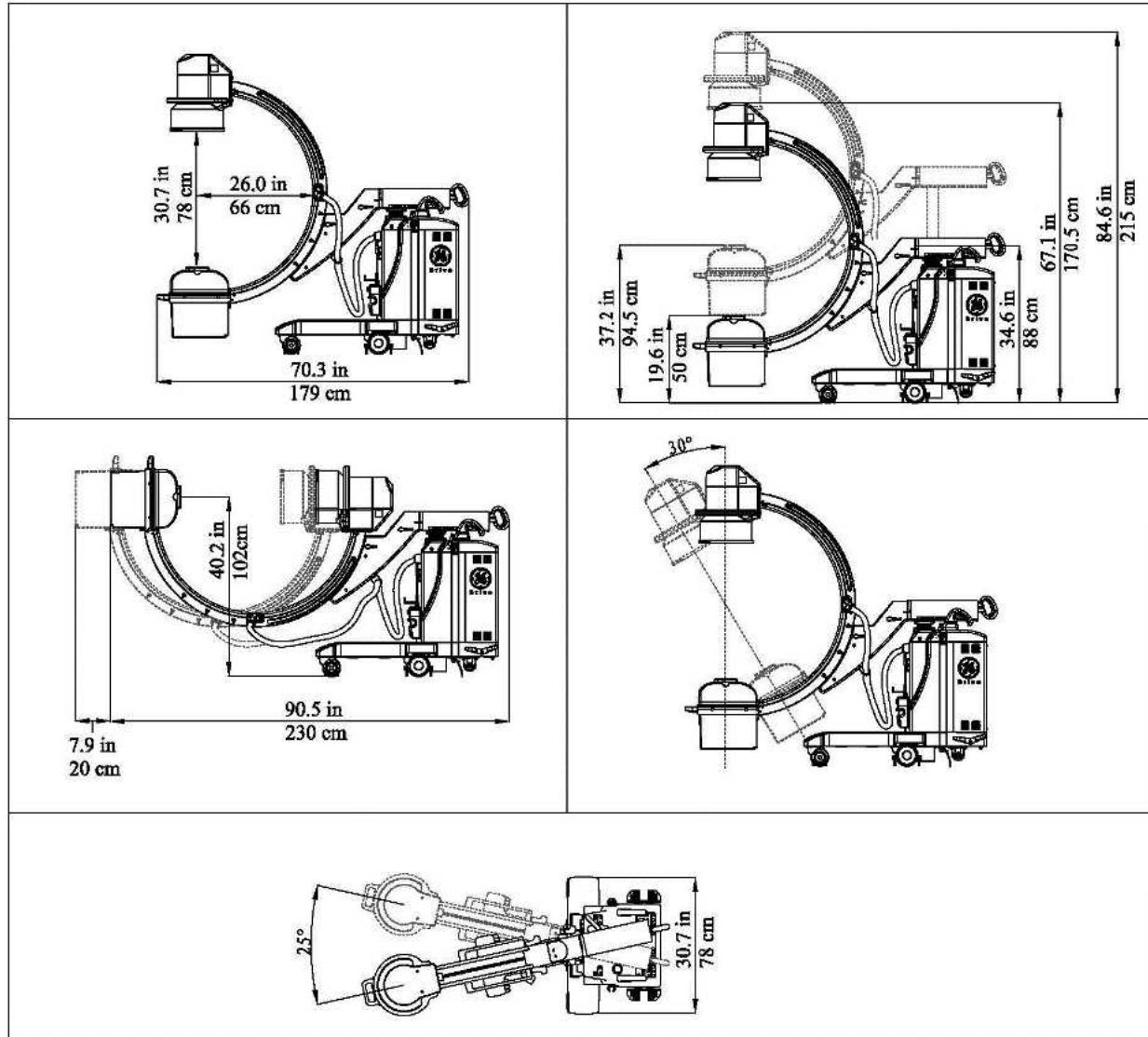
Where V_n = no load voltage (standby), and V_1 = loaded voltage (maximum radiographic exposure).



16A power supply is required in China. And 20 A is required in Japan.

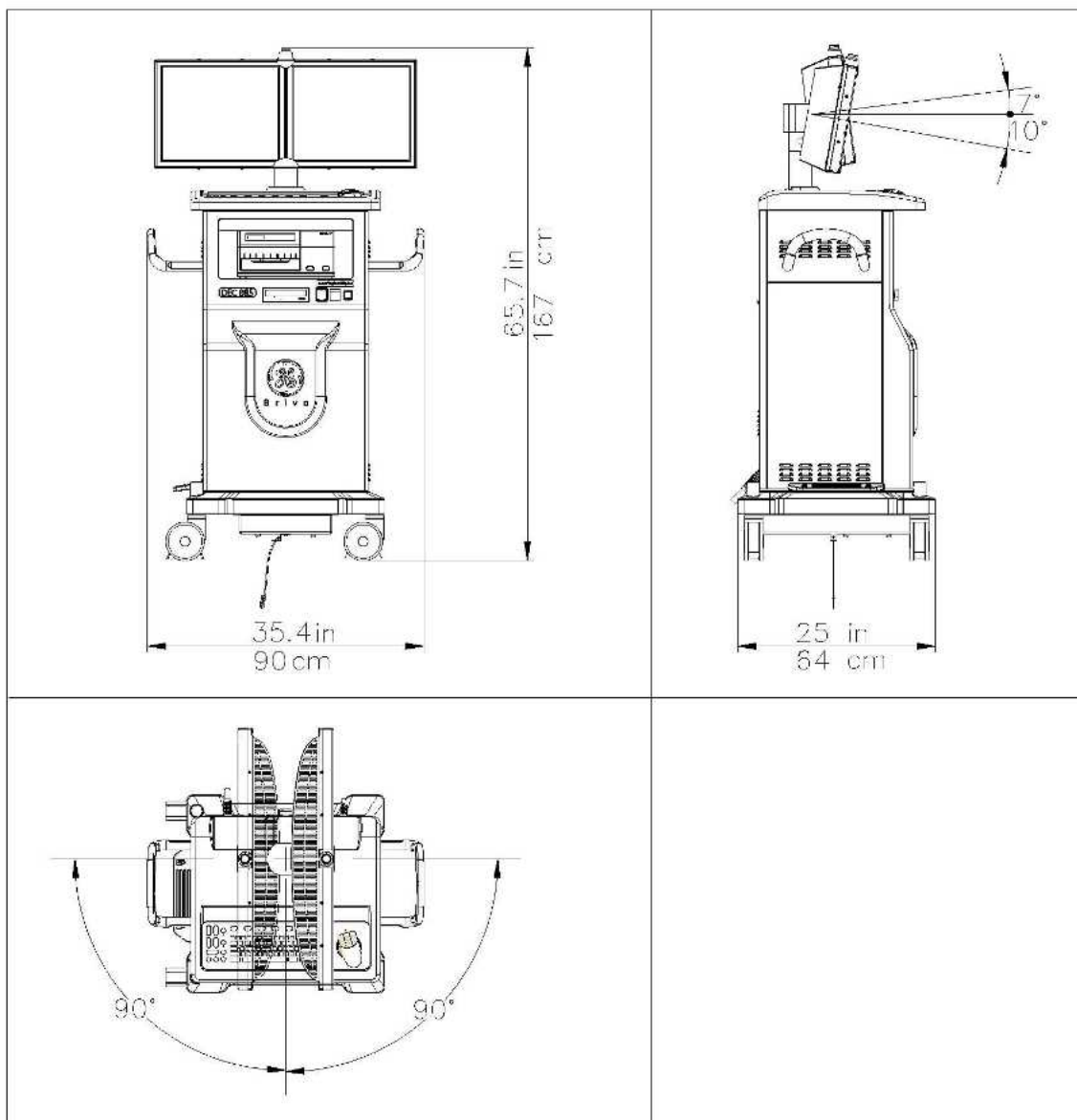
12.2.4. Physical Specifications

C-Arm



SID (Source Image Distance)	39.3" (100cm)
Horizontal travel	8" (20cm)
Vertical travel	17.5" (44.5cm)
Orbital rotation	120° (90° underscan /30° over scan)
Lateral rotation	± 205°
Wig-Wag	25° (± 12.5°)
Weight	260±10kg

Workstation



Weight: 170±10kg

12.3. Specification of Key Components

12.3.1. X-Ray Generator

- **Nominal electric power**
2.0kW (100kV, 20mA, 0.1s)
- **Maximum electric power**

Normal Fluoro	440W (110kV, 4mA)
High Level Fluoro	660W (110kV, 6mA; 55kV, 12mA)
Low Dose	220W (110kV, 2mA)
Digital Spot	1.76 kW (110kV, 16mA) 200V, 220V, 230V, 240V; 1.1 kW (110kV, 10mA) 100V, 110V, 120V
Film	2.2 kW (110kV, 20mA) 200V, 220V, 230V, 240V; 1.1 kW (110kV, 10mA) 100V, 110V, 120V.

- **High frequency generator** 40kHz
- **Voltage** 40kV 110kV

Normal Fluoro	When the X-Ray tube current reaches its highest value of 4mA, the obtainable highest X-Ray tube voltage is 110kV.
High Level Fluoro	When the X-Ray tube current reaches its highest value of 12mA, the obtainable highest X-Ray tube voltage is 55kV.
Low Dose	When the X-Ray tube current reaches its highest value of 2mA, the obtainable highest X-Ray tube voltage is 110kV.
Digital Spot	When the X-Ray tube current reaches its highest value of 16mA, the obtainable highest X-Ray tube voltage is 110kV 200V, 220V, 230V, 240V. When the X-Ray tube current reaches its highest value of 10mA, the obtainable highest X-Ray tube voltage is 110kV 100V, 110V, 120V.
Film	20mA, stationary 200V, 220V, 230V, 240V, the obtainable highest X-Ray tube voltage is 110kV; 10mA, stationary 100V, 110V, 120V, the obtainable highest X-Ray tube voltage is 110kV.

- **Current**

Normal Fluoro	0.1 mA 4 mA; when the X-Ray tube voltage reaches its highest value of 110kV, the obtainable highest X-Ray tube current is 4mA.
High Level Fluoro	0.2 mA 12 mA; when the X-Ray tube voltage reaches its highest value of 110kV, the obtainable highest X-Ray tube current is 6mA.
Low Dose	0.1 mA 2 mA; when the X-Ray tube voltage reaches its highest value of 110kV, the obtainable highest X-Ray tube current is 2mA.
Digital Spot	0.2 mA 16 mA; when the X-Ray tube voltage reaches its highest value of 110kV, the obtainable highest X-Ray tube current is 16mA 200V, 220V, 230V, 240V. 0.2 mA 10 mA; when the X-Ray tube voltage reaches its highest value of

	110kV, the obtainable highest X-Ray tube current is 10mA 100V, 110V, 120V.
Film	1mAs 80mAs 200V, 220V, 230V, 240V; current: 20mA, stationary; the obtainable highest X-Ray tube voltage is 110kV; 1mAs 40mAs 100V, 110V, 120V; current: 10mA, stationary; the obtainable highest X-Ray tube voltage is 110kV.

- **Film**

Voltage: 40 110kV;

mAs: 1.0, 1.3, 1.6, 2.0, 2.5, 3.2, 4.0, 5.0, 6.3, 8.0, 10, 13, 16, 20, 25, 32, 40, 50, 63, 80 mAs 200V, 220V, 230V, 240V;

mAs: 1.0, 1.3, 1.6, 2.0, 2.5, 3.2, 4.0, 5.0, 6.3, 8.0, 10, 13, 16, 20, 25, 32, 40 mAs 100V, 110V, 120V.

- **Pulse Fluoro** 1/2/4/8pps

- The lowest Current Time Product or the combination of Loading Factors: 40kVp, 1mAs.

- **Fluoroscopy Duty Cycle**

80 kVp 1.0 mA continuous

- **Film Mode Duty Cycle**

110 kVp 20 mA, 4 seconds (80 mAs)

Once per 5 min (1.3%) continuous

Once per 2min (3.3%) for 10 exposures

- **kVp Accuracy** 7%

- **mA Accuracy** 10%

- **mAs Accuracy** (10% + 0.2mAs)

- **Linearity Film mode linearity** 0.18

- **Reproducibility Coefficient of Variation (COV)** 0.045

- **Measurement Basis for Technique Factors**

kVp - The peak value of high voltage generator output in the interval after a 20 mS delay period to the end of the exposure.

mA - The time average of the current flow into the high voltage cable/X-ray tube assembly, beginning at the point where kVp crosses the 35 kVp level.

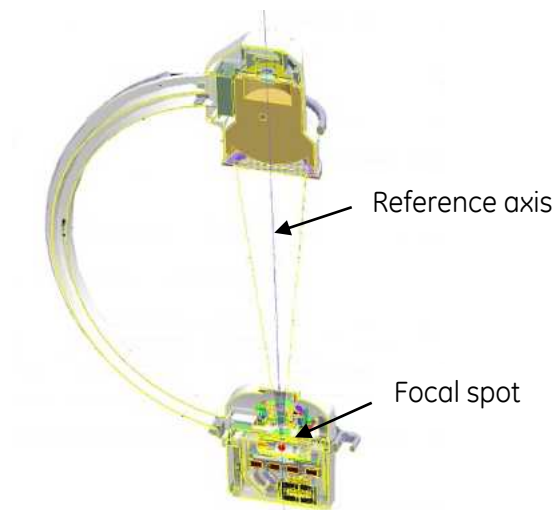
Time - Measurement of exposure time begins when the kVp crosses the 35 kVp level (80% of kVp selected).

mAs - The time integral of mA as defined above.

12.3.2. X-Ray Source Assembly

Inherent Filter	2.55 mm Al 50KV with LOHMANN tube
Removable filter	3.5 mm Al 50kV
Maximum continuous heat dissipation rate	177W
Tube assembly dimension (length×Width×Height)	327mm 132 mm 238 mm
Tube assembly weight	17.5 kg
Maximum heat capacity	900,000HU
Maximum cooling rate	12,500HU/min
Cooling method	Natural cooling
Maximum symmetrical radiation field on the X-axis	100cm from the spot: Fluoro: 235 mm (diagonal) Film: 240mm (diameter)
Leakage radiation	Dose rate is less than 1.0mGy/h, averaged over any area of 100 cm ² of which no principal linear dimension exceeds 20 cm at 1 m from the focal spot under conditions of loading.
Leakage radiation test condition	110 kV, 2mA

Location of the focal spot and the reference axis



The reference axis is from the center of the focal spot to the center of the image intensifier. Focal spot's tolerances on the reference axis is zero.

X-Ray tube

Nominal high voltage	110 kV
Nominal input power	2.2 kW
Maximum anode heat capacity	53 kJ
Target material	Tungsten
Target angle	10°
Small focal spot (IEC336) (Normal Fluoro, High Level Fluoro, Digital Spot)	0.6 × 1.4 (IEC336)
Large focal spot (IEC336) (Radiography)	1.4 (IEC336)
Anode	Stationary
Typical filament specification	LOHMANN X-Ray tube: 4.2V, 3.5A
Maximum filament current	4.6A
Permanent Filtration	LOHMANN X-Ray tube: 0.55mm Al 50kV
Weight	0.65 kg
Manufacturer and model	LOHMANN, 110/3DF
Maximum voltage between anode and ground	55kV
Maximum voltage between cathode and ground	-55kV
X-Ray radiation angle	10 degree

Bracket (between the tube assembly and the collimator)

Filter	0.8 mm Al
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Collimator

Filtration	0 mm Al
Input power	DC 5V, 1W
Electrical connection mode	through 3mm, 12pins connector
Mechanical connection mode	cast aluminum bracket, screw fitting

- Iris collimator**

Fluoroscopy:

Nominal diameter: 23/15/11cm (9"/6"/4.5")

X-Ray beam continuously adjustable to an area of less than 5 × 5 cm at the image receptor

Film:

Nominal diameter circle: 23cm (9")

- **Shutter collimator**

Full Range Rotation Angle: 360°

Operation: Controlled on panel

The total filtration

3.35 mm Al /4.3 mm Al without removable filter

6.85 mm Al with removable filter

12.3.3. Grid

Type	Circular
Focus distance	100 cm

12.3.4. Image Intensifier

Mode	Normal	Mag1	Mag2
Input field size	9"(23 cm)	6"(15cm)	4.5"(11cm)
Central resolution (23XZ4ST/BS)	44Lp/cm	50Lp/cm	54Lp/cm
Central resolution (TH9438HXH249VR19 ST)	52Lp/cm	58Lp/cm	68Lp/cm
Central resolution (E5830SD-P10A)	52Lp/cm	58Lp/cm	68Lp/cm

12.3.5. Camera

Image resolution	1K x 1K, 12 bit
Frame Rate	25 frame/second

12.3.6. Others

Monitor

Type	Brivo OEC 715/785: color LCD Brivo OEC 865 right monitor: mono LCD touchscreen Brivo OEC 865 left monitor: mono LCD
Size	19"
Resolution	1280X1024

UPS

Output Capacity	550 VA / 330 W
Input Voltage	220V 230V AC
Frequency	50/60Hz

Image Storage

Capacity	100,000 images 1K x 1K image resolution
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Digital Processor

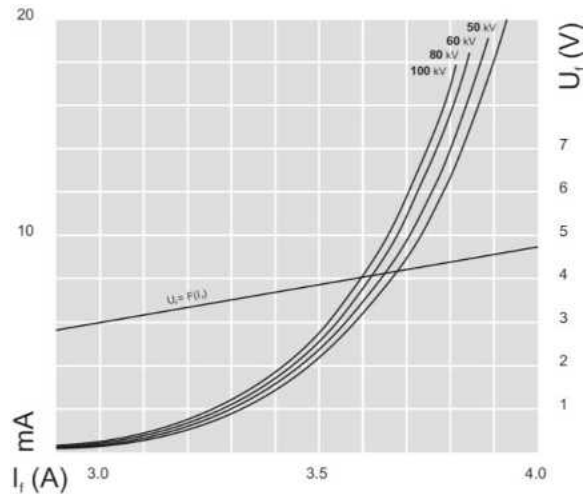
Function	Negate, Edge Enhancement, Auto Brightness/Contrast, Manual Brightness/Contrast Adjustment, Rotation/Reversal/Invert, Smart Metal, Zoom/Roam, Swap, Recursive Noise Reduction and Last Image Hold
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12.4. Tube Rating Chart

The following charts and data describe tube characteristics when operated with three-phase full-wave rectification, a reasonable approximation to the high frequency generator with minimal ripple.

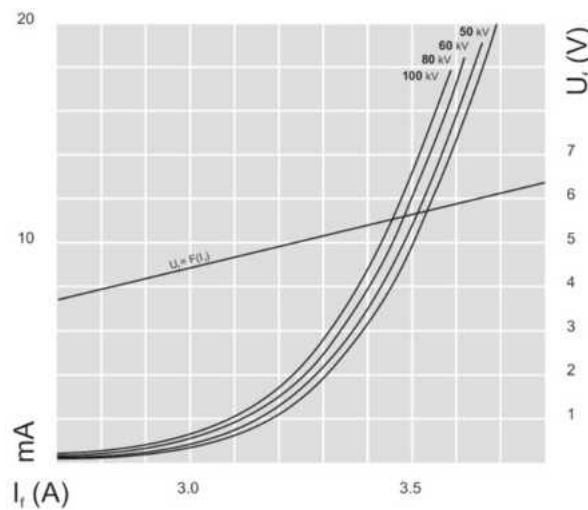
12.4.1. Filament Emission Characteristics

Small Focal Spot



LOHMANN

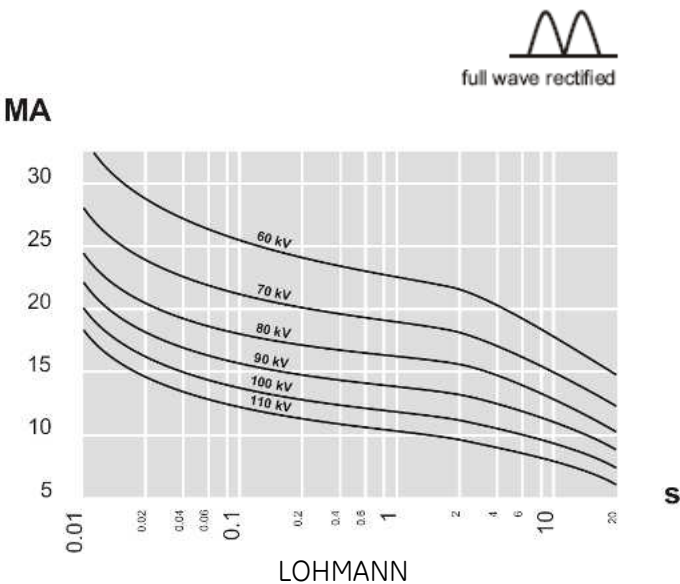
Large Focal Spot



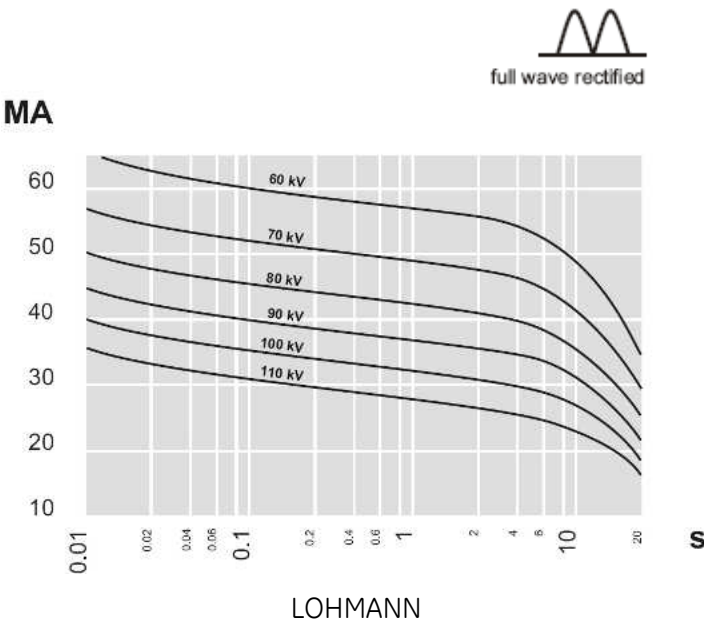
LOHMANN

12.4.2. Single Load Ratings

Small Focal Spot

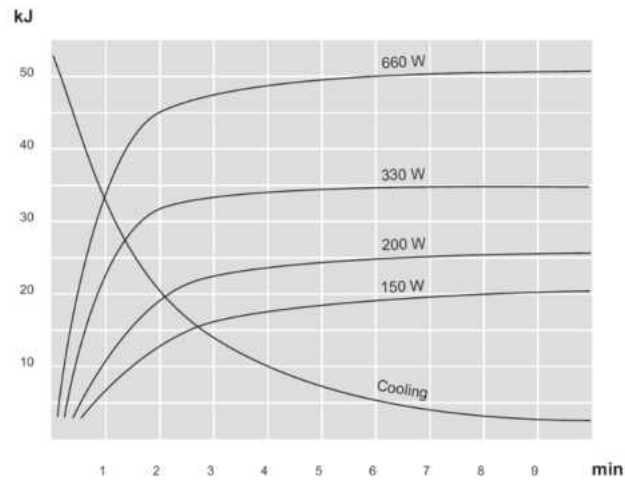


Large Focal Spot



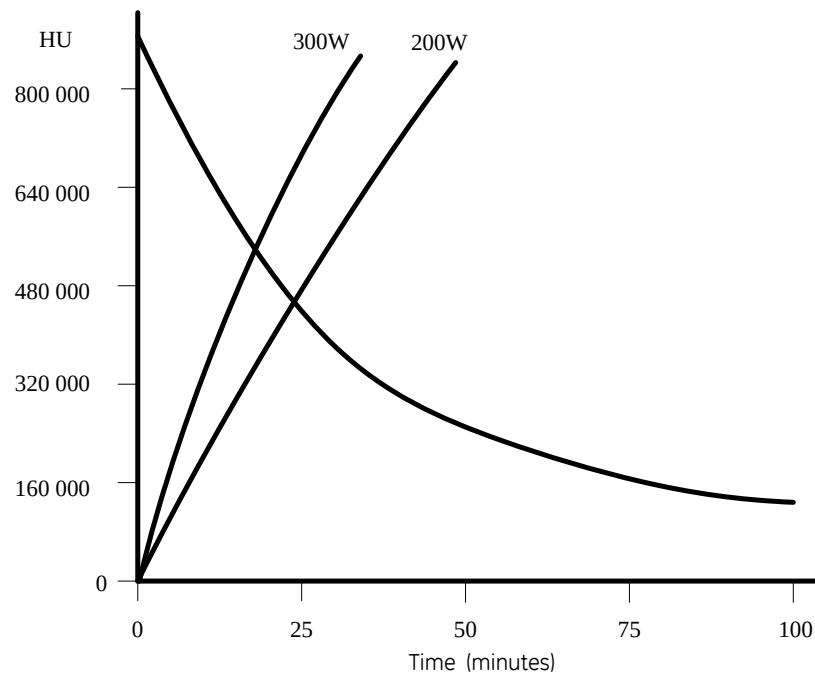
12.4.3. Thermal Characteristic

Anode Heating and cooling characteristics



LOHMANN

Tube Housing heating and cooling characteristic



12.5. Scatter Radiation

Isokerma Map Test

This test method is according to clause 203.13.4 in EN 60601-2-43:2010.

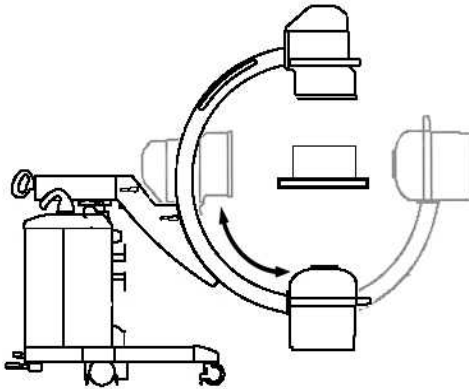
The measurement is performed at 2 planes at typical height: 1m & 1.5m from the ground.

The scatter radiation is measured in 2 layouts as vertical and lateral configuration.

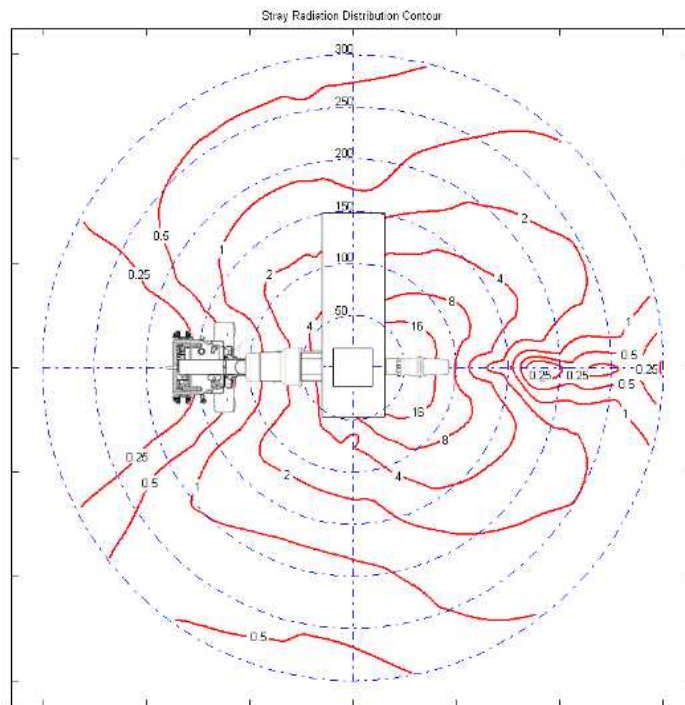
For the vertical configuration, the table is at height 1m. I.I. is above the tube.

For the lateral configuration, the table is at height 1m. The center of I.I. is at same altitude as the center of phantom. The distance from the surface of I.I. to center of phantom is 40cm.

The layout is as below:

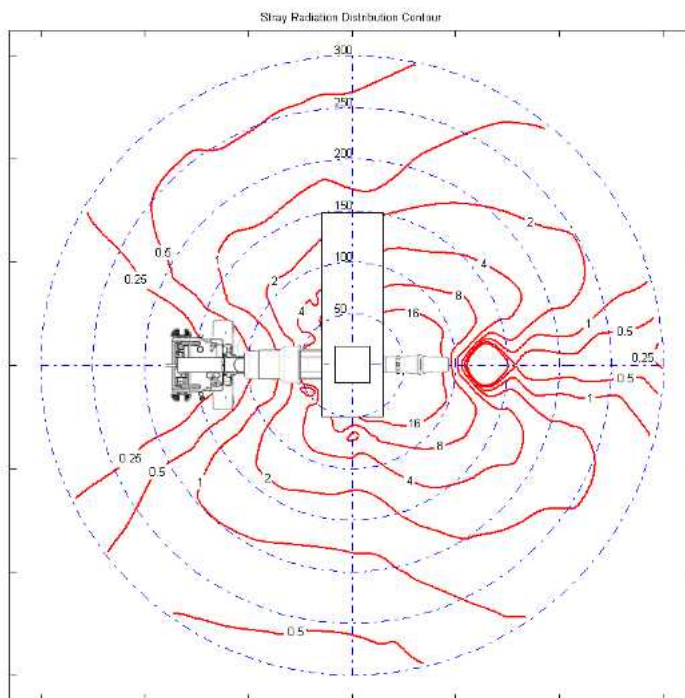


Lateral configuration, 1.5m



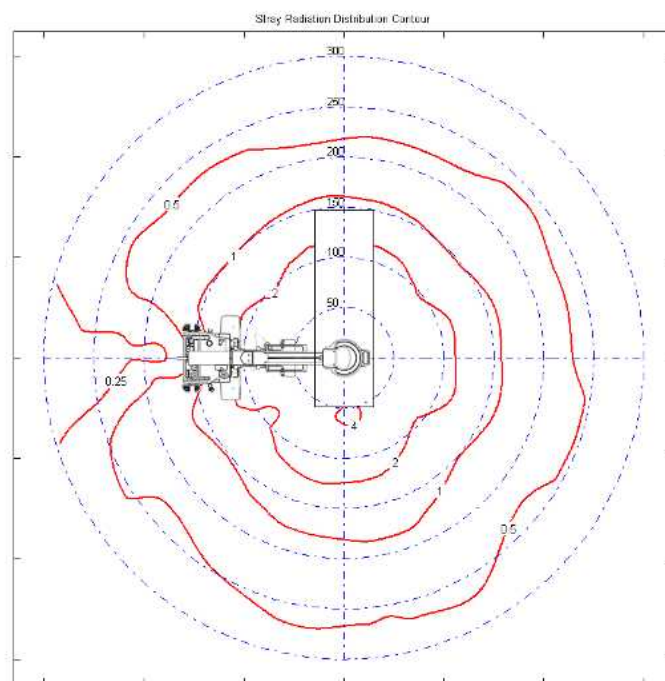
LOHMANN

Lateral configuration, 1m



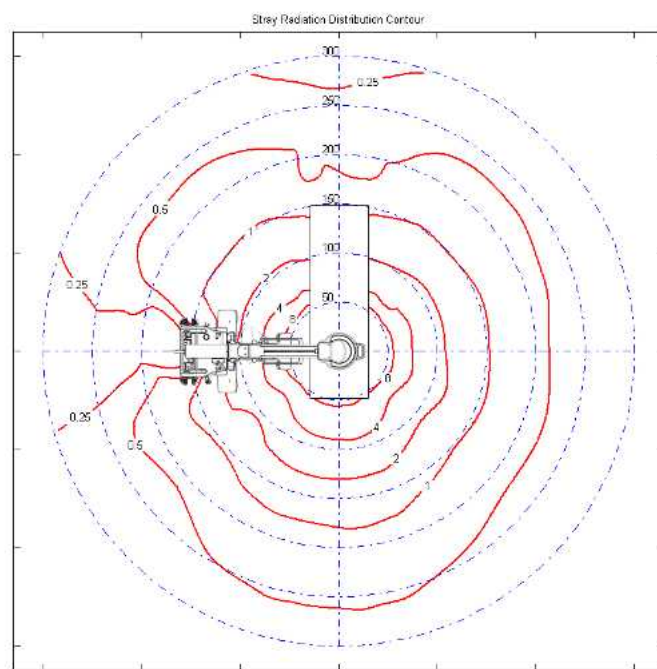
LOHMANN

Vertical configuration, 1.5m



LOHMANN

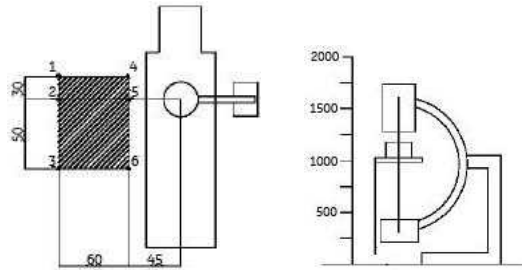
Vertical configuration, 1m



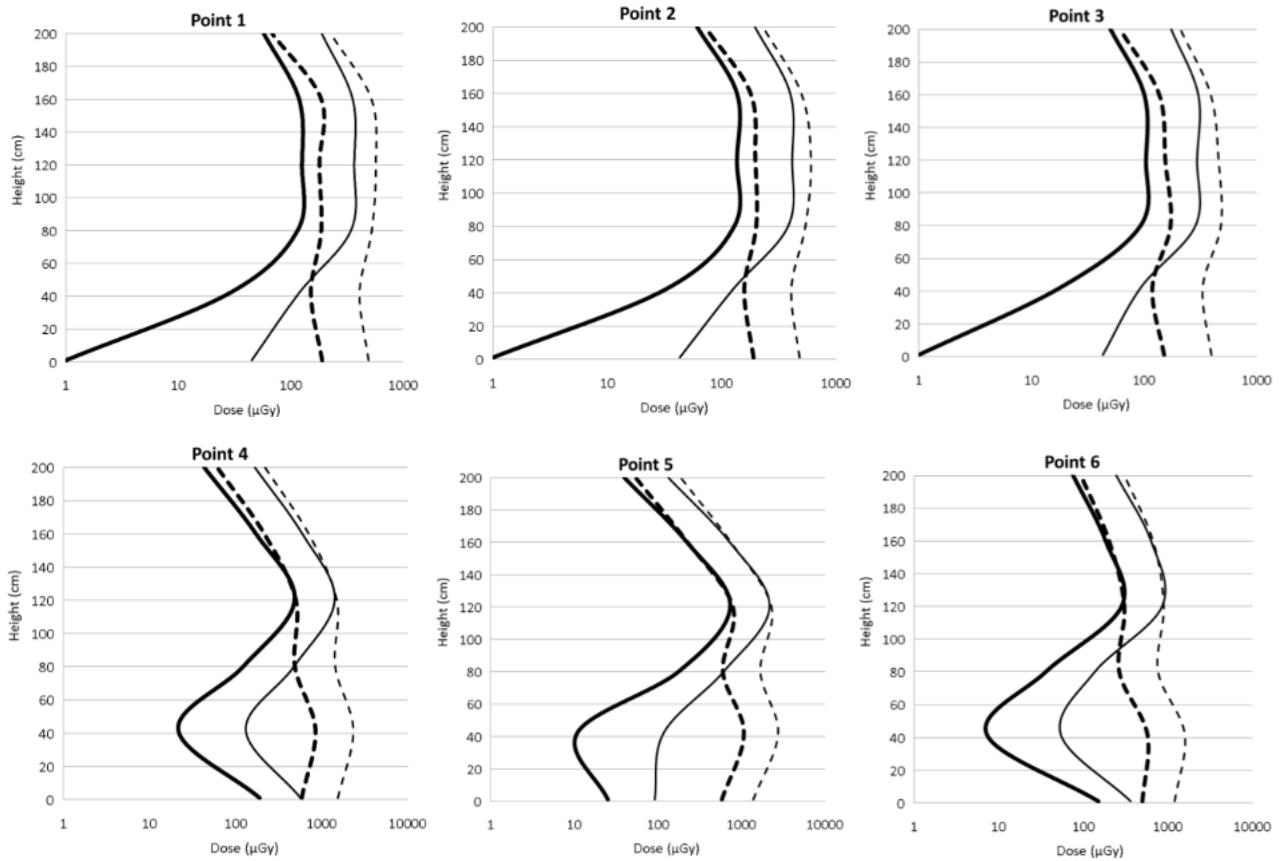
LOHMANN

Significant ones of Occupancy Test

This test method is according to clause 13.4 in IEC 60601-1-3:2008.



LOHMANN X-Ray tube



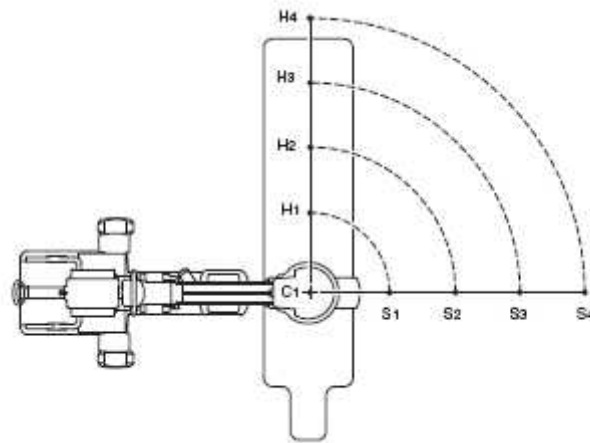
- 110kV, 4mA, without shielding
- 110kV, 4mA, with shielding
- 70kV, 3mA, without shielding
- 70kV, 3mA, with shielding

Horizontal & Vertical Plane Test

This test is performed at fluoroscopic mode in 70kV, 3mA, without shielding.

Horizontal Plane

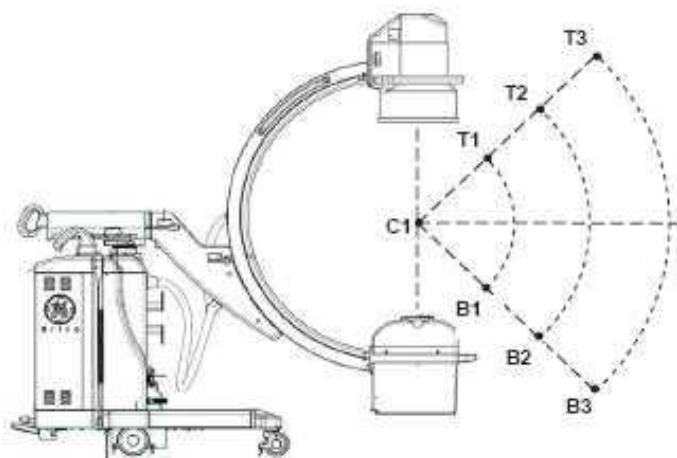
The following illustration shows measurement points in a plane perpendicular to the beam axis at a height indicated by reference point C1. C1 is the 30-inch SSD, and is illustrated in the Vertical Plane illustration, below. The measurement points are 25 cm apart. The table following the illustration shows expected scatter radiation at each point.



Measurement Point	Dose Rate LOHMANN X-ray tube
H1	3.32 mGy/hr
H2	0.67 mGy/hr
H3	0.28 mGy/hr
H4	0.19 mGy/hr
S1	3.60 mGy/hr
S2	0.81 mGy/hr
S3	0.36 mGy/hr
S4	0.19 mGy/hr

Vertical Plane

The following illustration shows measurement points in a plane through the beam axis at a height indicated by reference point C1, which represents the 30-inch SSD. The measurement points are 25 cm apart. The table following the illustration shows expected scatter radiation at these points.



Measurement Point	Dose Rate LOHMANN X-ray tube
T1	1.56 mGy/hr
T2	0.60 mGy/hr
T3	0.28 mGv/hr
C1	0.01 Gv/min
B1	Too close to phantom for measurement
B2	2.03 mGv/hr
B3	0.76 mGv/hr

12.6. Room Interface (RIF)

Room Interface (RIF) is an interface interactive with operation room or some peripheral equipment. The lamps in operation rooms such as Room In Use indicator or X-Ray On indicator can be connected to system to warn user, injector can be controlled by room interface if necessary, also the exposure can be restrained of door status in case of excessive radiation to user, all functions in RIF are not mandatory when operating.

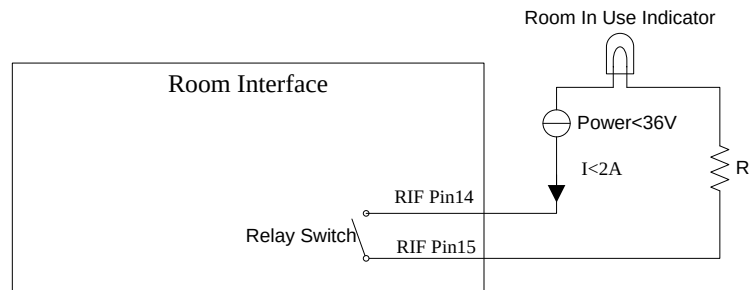
System provides a HDB15 connector as room interface output/Input to connect each indicators, on the back of workstation.



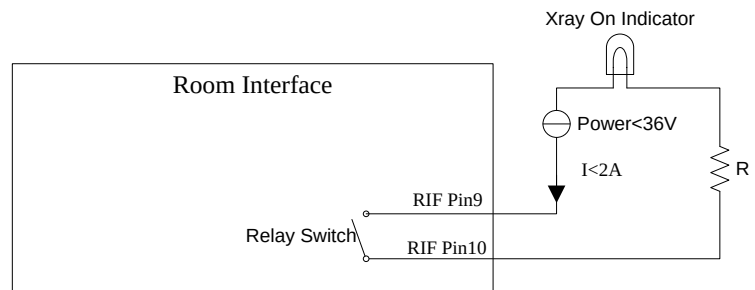
For Room Interface (RIF) connection, the rules listed below shall be complied with:

1. All equipment and power connect to RIF shall be UL/IEC60601 certified.

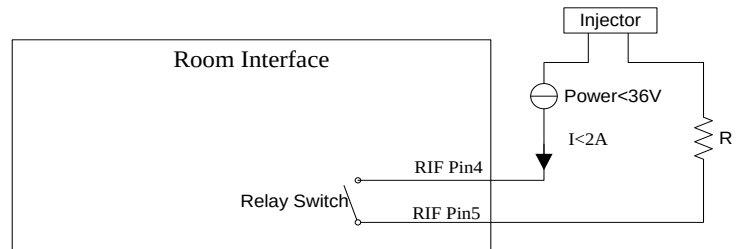
2. For "Room in Use Indicator", the power shall be less than DC 36V, and current shall be limited within 2A. The connection diagram is recommended as below.



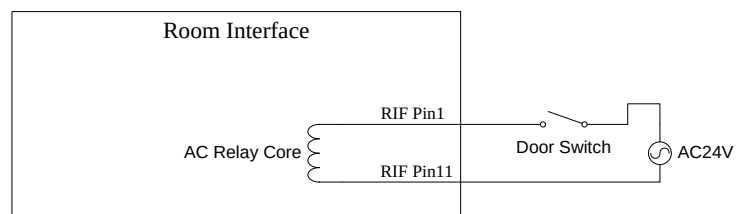
3. For "X-Ray on Indicator", the power shall be less than DC 36V, and current shall be limited within 2A. The connection diagram is recommended as below.



4. For "Injector in Use Indicator", the power shall be less than DC 36V, and current shall be limited within 2A. The connection diagram is recommended as below.



5. For "Door Open Detector", an AC 24V with minimum power 24W is needed to drive an AC relay in internal circuit, the door switch status will be detected by system. The connection diagram is recommended as below.



12.7. Options

12.7.1. Thermal/ Film Printers

	UP-D897	UP-990AD	UP-971AD/UP-991AD	UP-D898MD
Power requirements	100 to 240V AC, 50/60Hz	100 to 120V AC, 220 to 240V AC, 50/60 Hz	100 to 240V AC, 50/60Hz	100 to 240V AC, 50/60Hz
Dimensions (W × H × D)	154 × 88 × 240 mm	316 × 132.5 × 305 mm	316 × 132.5 × 265 mm	154 × 88 × 240 mm
Mass	2.6 kg	Approx. 8 kg	Approx. 7 kg	2.5 kg
Resolution	325 dpi	325 dpi	325 dpi	325 dpi
Gradation	8 bits (256 levels) processing	256 levels processing	256 levels processing	8 bits (256 levels) processing
Picture size	320 × 100 mm (max.)	200 × 150 mm	600 × 200 mm (max.)	320 × 100 mm (max.)
Printing speed	Approx. 2 seconds/image (960 × 1280 dots) (printing at the high speed) Approx. 3.3 second/image (960 × 1280 dots) (printing at the normal speed)	8 seconds/Image	8 seconds/Image	Approx. 1.9 seconds/image (960 × 1280 dots) (printing at the high speed) Approx. 3.3 second/image (960 × 1280 dots) (printing at the normal speed)
Picture elements	4096 × 1280 dots (max.)	3414 × 2560 dots	7680 × 2560 dots (max.)	4096 × 1280 dots (max.)
Interface	Hi-Speed USB (USB2.0)	Hi-speed USB	Hi-Speed USB (USB2.0)	Hi-Speed USB (USB2.0)

For detailed information, please refer to user manual of printer.

12.7.2. Laser Aimer

The laser Aiming Device produces a laser beam, which is used to indicate the central axis of the X-Ray beam. An opaque cross-hair located on the laser aimer provides a reference point in the X-Ray image that can be used in conjunction with the laser beam to obtain an accurate placement angle during alignment procedures.

Brivo OEC 715/785/865 Mobile C-Arm X-Ray Product provides two laser aimer options, which are attached to the tube and Image Intensifier.

Laser Aimer, which is attached to the tube, is incorporated with the tube cover. Press the laser aimer's ON switch to generate the laser beam.



Perform the following procedure to install and operate the laser aimer on the Image Intensifier. The C-Arm can be placed in any orientation that is convenient for attaching the laser aimer. There are no alignment marks or notches on the laser aimer that need to match any points on the image intensifier.



When the laser aimer is attached to the C-Arm, releasing the orbital or rotational brake could cause the C-Arm to move. Make sure to hold the C-Arm to control its movement whenever a brake is released.

1. Place the laser aimer's attachment handle in the open position.
2. Move the laser aimer flush with the Image Intensifier, but slightly off center.
3. Center the laser aimer with the relation to the Image Intensifier. As you center the laser aimer, the attachment hooks catch on the edge of the image intensifier.
4. Turn the attachment handle inward until the handle is completely closed and the laser aimer is securely attached to the image intensifier. Insert the metal bolt to the hole of the handle to lock.



Do not grasp the Laser Aimer's handle to position the C-Arm or to move the system. Pulling the handle may cause the Laser Aimer to release from the image intensifier and fall.

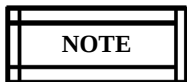
5. Use the image intensifier handle to position the C-Arm or to move the system to the desired location.

6. After attaching the laser aimer, place the C-Arm in the position it will be used during the procedure.
7. Turn the laser aimer ON and adjust the beam alignment knobs until the laser beam is centered on the crossing of the cross-hair on the tube cover.



If the laser is ON, do not look directly into the laser aimer's aperture.

8. Using fluoroscopy, position the cross-hair in the fluoro image, at the center of the area of interest.



The location where the laser beam marks the patient corresponds to the center of the area of interest, indicated by the cross-hair in the fluoro image.

If you don't want the cross hair to show on the image, remove the laser cover from the laser aimer.

1. Switch OFF the power to the laser aimer.
2. Grasp the two knobs on the laser cover.
3. Press down and rotate the cover (approximately 10 degrees counterclockwise) until the knobs align with the notches on the housing.
4. Lift off the cover and resume laser aimer operation.



Regardless of this procedure to adjust or operate the equipment may result harmful radiation.

12.7.3. 30cm Skin Spacer

Skin Spacer is able to install on the tube cover and removable filter. Install the skin spacer as the following steps:

1. Put the skin spacer on the X-Ray tube cover or removable filter. Make sure the skin spacer's hole superpose with the screws.
2. Rotate the skin spacer clockwise to lock it onto the cover.



12.7.4. Film Cassette Holder

Film cassette holder dimension: 33.1cm x35.0cm x5.2cm for 9" Image Intensifier.

Film cassette dimensions: 10" x 12" (24 cm x 30cm) for 9" Image Intensifier.

If user need install the film cassette holder, please make sure the lock handle is locked into place.



12.7.5. Video Distributor

The video distributor provides a means of viewing the images displayed on the left monitor on an external monitor. There are two kinds of video distributor provided as option. The weight of them is $2.3 \pm 0.2\text{kg}$ and $1.0 \pm 0.2\text{kg}$.

The DVI video connector in the Workstation console provides video from the Workstation to external monitor. Connect the video cable to DVI interface and the external video output will be the same as the video displayed on the left monitor.

12.7.6. Removable Filter

In some countries or regions, removable filter is required to be used in pediatric or other applications, please use this option as requirement of local regulations. After use of the removable filter, it will reduce the dose radiation for the patient and technician. Meantime, for the same parts imaging, the kVp and mA may increase and tube sustainable exposure time may be shortened.

The removable filter includes two sensors and a 3.5 mm Al filter.

12.7.7. Removable Grid

In some countries or regions, removable grid is required to be used in pediatric or other applications, please use this option as requirement of local regulations. When user conduct imaging without a removable grid, system can obtain images with lower dose, but it may cause image blurring, image quality may be not very good.

If user need install the removable grid, please make sure it is locked into place.

12.7.8. Wi-Fi Internet Adapter

When using the wireless network, operator should open the Wi-Fi Internet Adapter on the back Workstation and then setup the wireless network. Refer to section 6.3.3 “DICOM Setup” for details.

A Wi-Fi Internet Adapter converts wire network interface into wireless network interface. The WiFi adaptor complies with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. It does not exceed the Class B limits for radio-noise emissions form digital apparatus as set out in the Radio Interface Regulations of the Canadian Department of Communications. It complies with the Essential Requirements of the R&TTE Directive of the European Union (1999/5/EC).

12.8. Material Safety Data Sheets

Manufacturer’s Material Safety Data Sheets are available from the manufacture upon request. Contact the following manufacturers with regard to the materials listed:

Oil Sinkiang Kelamayi Lubricating Oil Plant

Address: Jinlong Town Youlong Road No. 8, Kelamayi City , Sinkiang Uigur Municipality

Post Code: 834003

Tel: 0990-6831426

12.9. Material Recycling

Parts or accessories at end-of-life

The disposing of parts and accessories must be in accordance with national regulations or local regulations for waste processing. All materials and components that could have risks to the environment must be removed from the end-of-life parts and accessories (For examples: transformer oil, etc...).

Please consult your local GEMS representative before discarding these products.

Packing materials

The materials used to pack our equipment are recycled. They must be collected and processed in accordance with the local regulations.

12.10. Replacement Parts

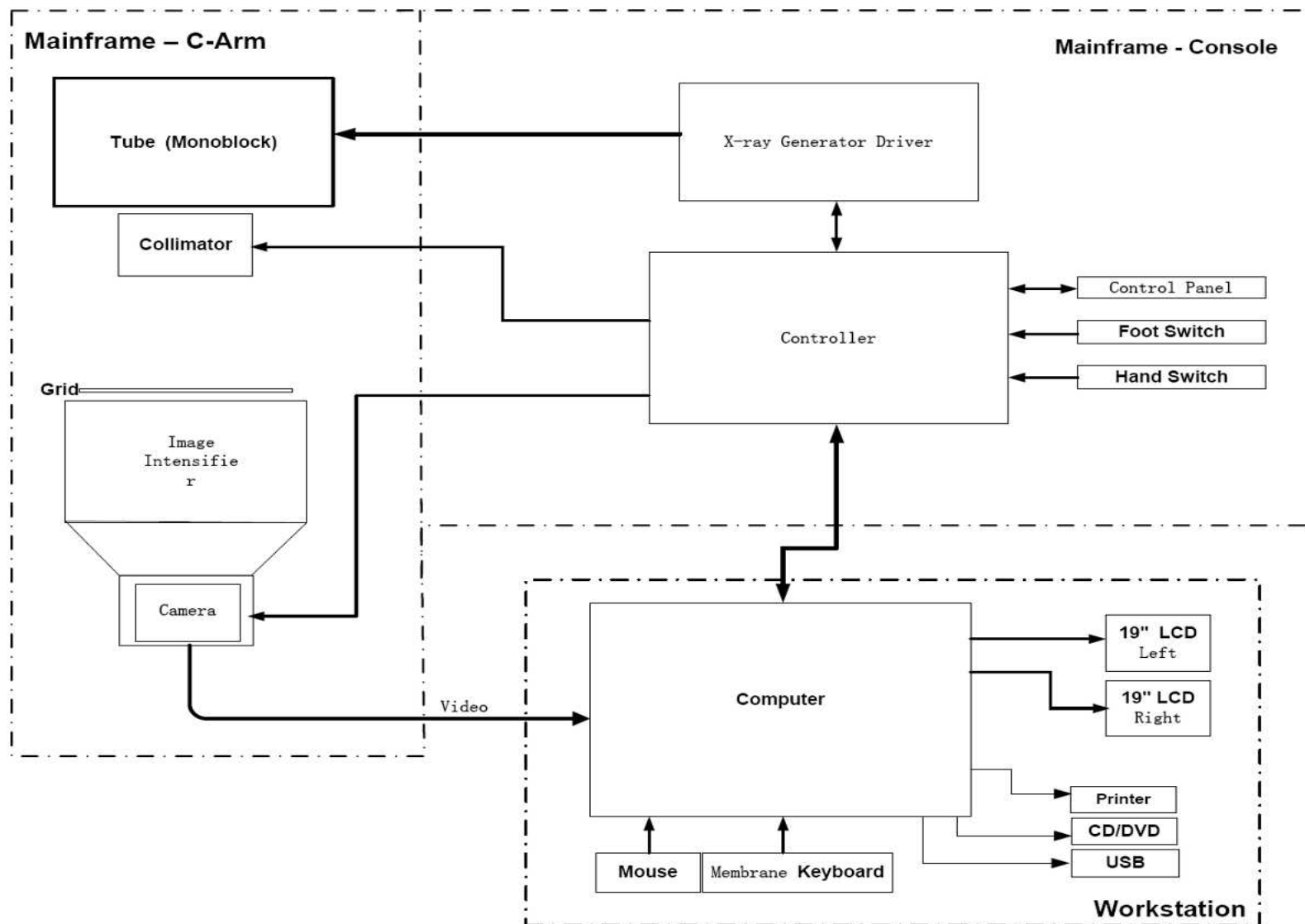
Parts as below can be replaced in the field. Call service to replace or maintain the parts.

- Circuit board and cable
- Collimator
- DC power supply, fuse, capacitor, filters, breaker and so on
- Power on indicator and exposure indicator
- LCD monitor
- Memory bank, mother board, hard disk of computer, UPS, mouse and membrane keyboard
- Image Intensifier
- Monoblock
- Footswitch, handswitch and emergency stop switch
- Handles, wheels and Covers

WARNING

Never make any modifications or adjustments to the equipment unless directed by a qualified GE Healthcare representative.

12.11. System Block



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For service issues, contact the service representative or service office in your country.

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